

# Welcome

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## About Battery

First, read the document [About Battery.pdf](#) in the unzipped folder.

If you did not download the zip file, please download it and unzip it via link below.

[https://github.com/Freenove/Freenove\\_Big\\_Hexapod\\_Robot\\_Kit\\_for\\_Raspberry\\_Pi/archive/master.zip](https://github.com/Freenove/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/archive/master.zip)

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- Product quality issues
- Product use and build issues
- Questions regarding the technology employed in our products for learning and education
- Your input and opinions are always welcome
- We also encourage your ideas and suggestions for new products and product improvements

For any of the above, you may send us an email to:

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- Keep this product out of the reach of children under 6 years old.
- This product should be used only when there is adult supervision present as young children lack necessary judgment regarding safety and the consequences of product misuse.
- This product contains small parts and parts, which are sharp. This product contains electrically conductive parts. Use caution with electrically conductive parts near or around power supplies, batteries and powered (live) circuits.
- When the product is turned ON, activated or tested, some parts will move or rotate. To avoid injuries to hands and fingers, keep them away from any moving parts!
- It is possible that an improperly connected or shorted circuit may cause overheating. Should this happen, immediately disconnect the power supply or remove the batteries and do not touch anything until it cools down! When everything is safe and cool, review the product tutorial to identify the cause.
- Only operate the product in accordance with the instructions and guidelines of this tutorial, otherwise parts may be damaged or you could be injured.
- Store the product in a cool dry place and avoid exposing the product to direct sunlight.
- After use, always turn the power OFF and remove or unplug the batteries before storing.

## About Freenove

Freenove provides open source electronic products and services worldwide.

Freenove is committed to assist customers in their education of robotics, programming and electronic circuits so that they may transform their creative ideas into prototypes and new and innovative products. To this end, our services include but are not limited to:

- Educational and Entertaining Project Kits for Robots, Smart Cars and Drones
- Educational Kits to Learn Robotic Software Systems for Arduino, Raspberry Pi and micro: bit
- Electronic Component Assortments, Electronic Modules and Specialized Tools
- **Product Development and Customization Services**

You can find more about Freenove and get our latest news and updates through our website:

<http://www.freenove.com>

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 [support@freenove.com](mailto:support@freenove.com)

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## Machinery Parts

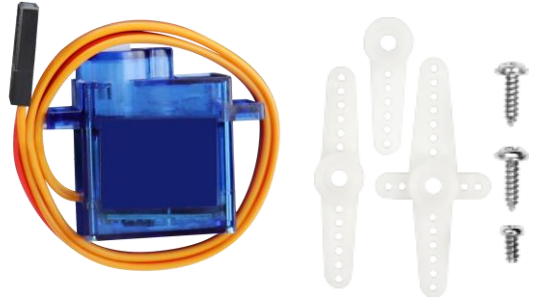
|                                                                                                                                                  |                                                                                                                                                  |                                                                                                                                                  |                                                                                                                                                   |                                                                                                                                                |                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p><b>M3*55</b><br/>Copper Standoff</p> <p>x8<br/>Freenove</p> |  <p><b>M3*30</b><br/>Copper Standoff</p> <p>x6<br/>Freenove</p> |  <p><b>M3*20</b><br/>Copper Standoff</p> <p>x6<br/>Freenove</p> |  <p><b>M2.5*8</b><br/>Copper Standoff</p> <p>x6<br/>Freenove</p> |  <p><b>M1.8*8</b><br/>Screw</p> <p>x80<br/>Freenove</p>     |  <p><b>M1.4*6</b><br/>Screw</p> <p>x12<br/>Freenove</p>     |
|  <p><b>M4*14</b><br/>Screw</p> <p>x74<br/>Freenove</p>          |  <p><b>M3*12</b><br/>Screw</p> <p>x40<br/>Freenove</p>          |  <p><b>M2.5*8</b><br/>Screw</p> <p>x10<br/>Freenove</p>         |  <p><b>M2*16</b><br/>Screw</p> <p>x8<br/>Freenove</p>            |  <p><b>M3*10</b><br/>Sunk Screw</p> <p>x10<br/>Freenove</p> |  <p><b>M4</b><br/>Non-slip nuts</p> <p>x74<br/>Freenove</p> |
|  <p><b>M3</b><br/>Nut</p> <p>x20<br/>Freenove</p>               |  <p><b>M2</b><br/>Nut</p> <p>x8<br/>Freenove</p>                |                                                                                                                                                  |                                                                                                                                                   |                                                                                                                                                |                                                                                                                                                |

## Transmission Parts

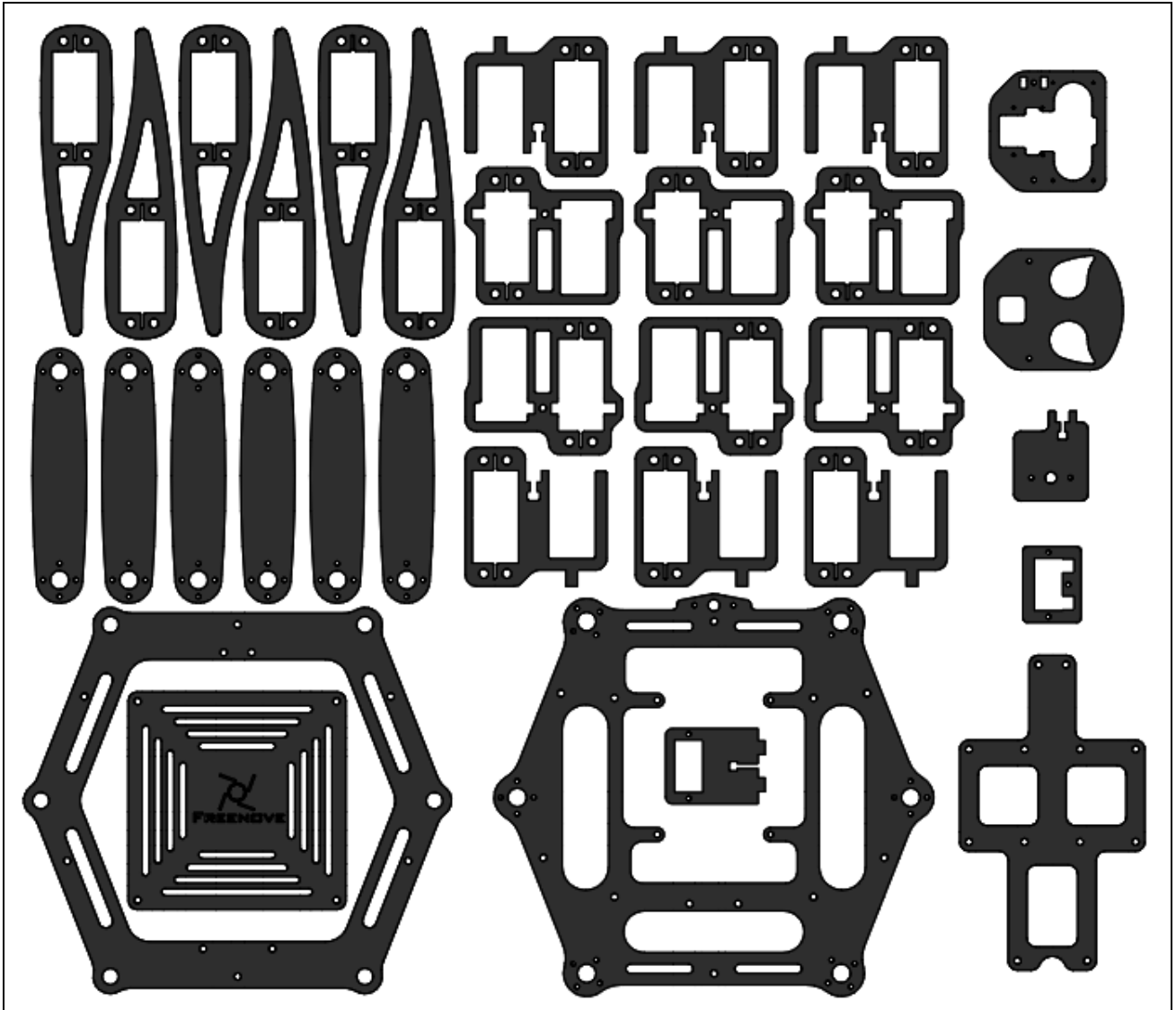
Servo package 1 x18



Servo package 2 x2

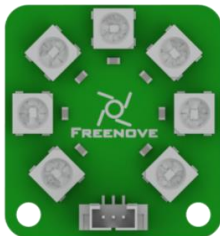
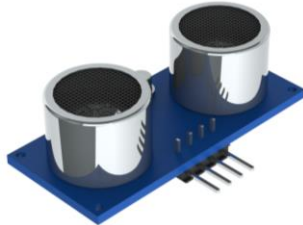
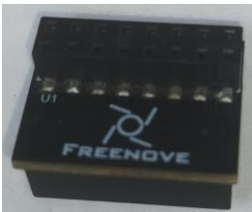


## Acrylic Parts



## Electronic Parts

Please note: There are two models of connection boards, remember their version numbers.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                   |                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| LED module x1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Camera x1                                                                         | HC-SR04 Ultrasonic Module x1                                                        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |  |
| Connection board (PCB_V1.0)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                   | Connection board (PCB_V2.0)                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                   |   |
| <p>Note:</p> <p>The PCB Version 2 is not on sale yet. They are current at our warehouse. The only distinction between Version 1 and Version 2 of the adapter board lies in the modification of the LED control pin. Version 1 employs GPIO18, whereas Version 2 has switched to GPIO10.</p> <p>Due to significant changes in the Raspberry Pi 5, the library rpi-ws281x-python is unable to function properly. This library was designed to work with GPIO18, and the lack of detailed documentation on the RP1 chip from the Raspberry Pi authorities has hindered the resolution of this issue.</p> <p>Consequently, we have introduced an alternative approach, utilizing the SPI MOSI pin for communication with the LED lights. Although this restored the functionality of the LEDs, it also meant that SPI-related pins (SCK, MISO) could not be used for other purposes. This serves as an interim measure, and once the rpi-ws281x-python library resumes normal functionality on pi 5, Version 2 is expected to be phased out.</p> <p>If you want to achieve the function of V2, you can connect GPIO10 to GPIO18 with a jumper wire, as shown below.</p>  |                                                                                   |                                                                                     |
| Jumper Wire F/F(4) x1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                   |                                                                                     |



1.25mm 3Pin LED cable (same direction) x1



FPC 25cm 15Pin camera cable (reversed direction) x1



FPC soft line x1



**Please note that this cable only come with the purchase of the Raspberry Pi 5 version.**

Battery holder x2



## Tools

Cross screwdriver (2mm) x1



Cross screwdriver (4.5mm) x1



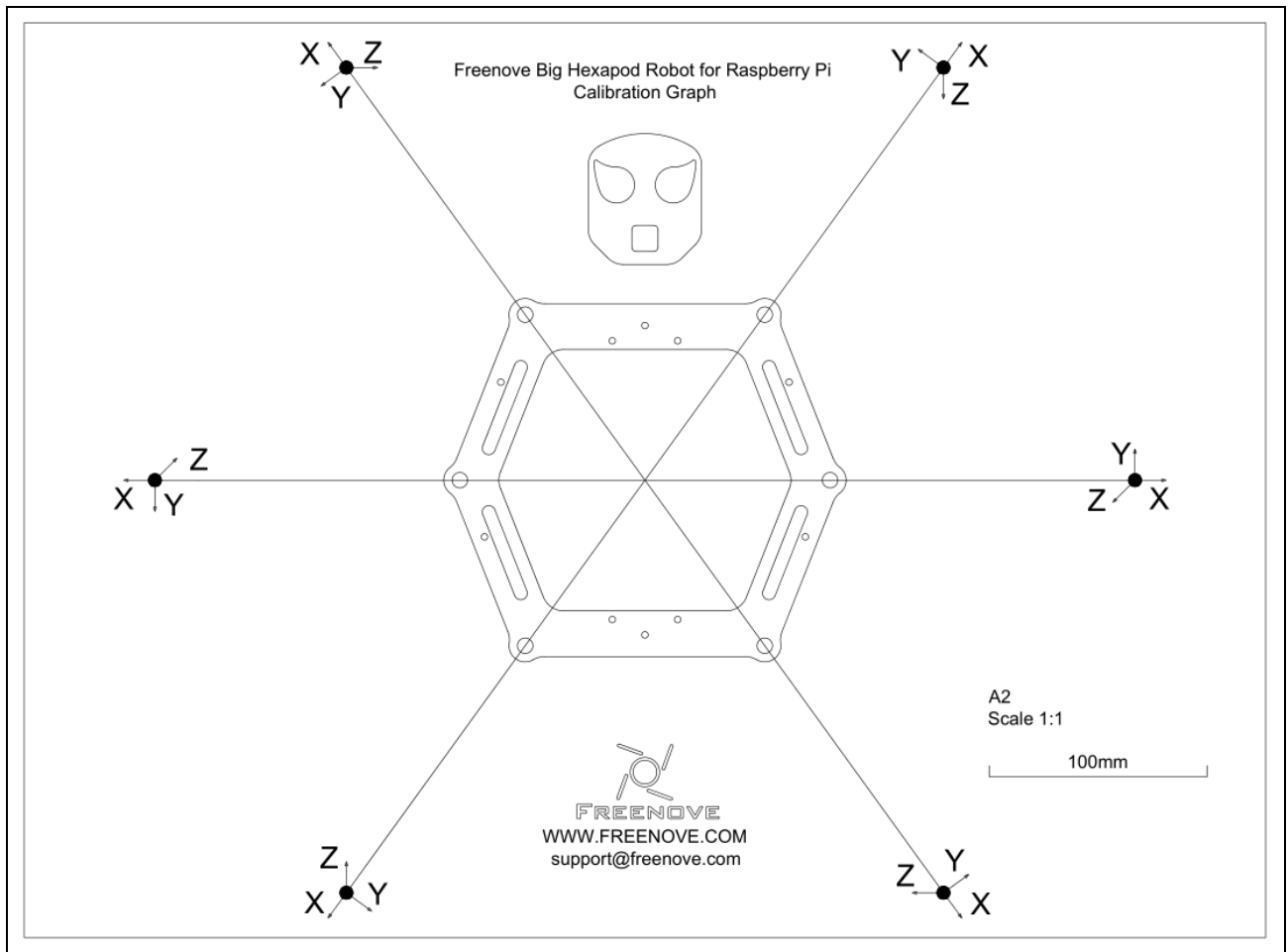
Cable Tidy x150cm



Multifunctional Spanner x1



## Calibration Graph

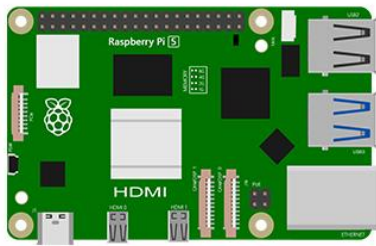


## Required but NOT Contained Parts

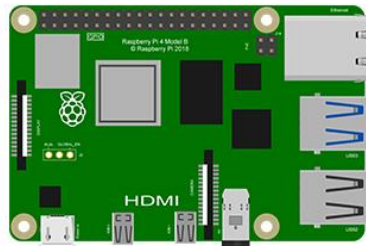
Four 18650 lithium batteries without protection board. Discharge current is greater than 15A



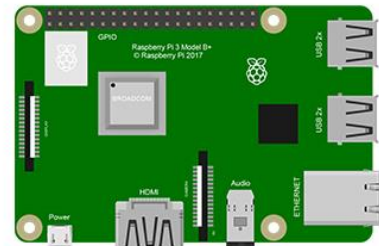
Raspberry Pi (Recommended model: Raspberry 5 / 4B / 3B+) x1



**Raspberry Pi 5**



**Raspberry Pi 4 Model B**



**Raspberry Pi 3 Model B+**

## Preface

Welcome to use this robot. Following this tutorial, you can make a very cool robot with many functions.

This kit is based on Raspberry Pi, a popular control panel, so you can share and exchange your experience and design ideas with many enthusiasts all over the world. The parts in this kit include all electronic components, modules, and mechanical components required for making the smart car. And all of them are packaged individually. There are detailed assembly and commissioning instructions in this book.

And if you encounter any problems, please feel free to contact us for fast and free technical support.

[support@freenove.com](mailto:support@freenove.com)

The contents in this book can help enthusiasts with little technical knowledge to make a robot. If you are very interested in Raspberry Pi, and want to learn how to program and build the circuit, please visit our website [www.freenove.com](http://www.freenove.com) or contact us to buy the kits designed for beginners:

Freenove Basic\LCD1602\Super\Ultrasonic\RFID\Ultimate Starter Kit for Raspberry Pi

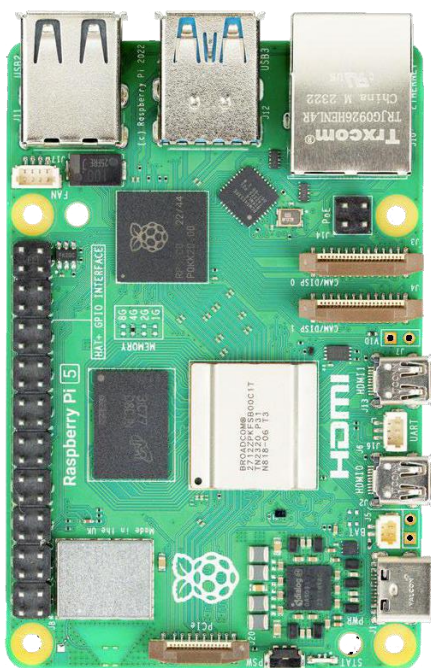
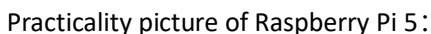
# Raspberry Pi Introduction

Raspberry Pi (called RPi, RPI, RasPi, the text these words will be used alternately later), a micro-computer with size of a card, quickly swept the world since its debut. It is widely used in desktop workstation, media center, smart home, robots, and even the servers, etc. It can do almost anything, which continues to attract fans to explore it. Raspberry Pi used to be running with Linux system and along with the release of windows 10 IoT. We can also run it with Windows. Raspberry Pi (with interfaces USB, network, HDMI, camera, audio, display and GPIO), as a microcomputer, can be running in command line mode and desktop system mode. Additionally, it is easy to operate just like Arduino, and you can even directly operate the GPIO of CPU.

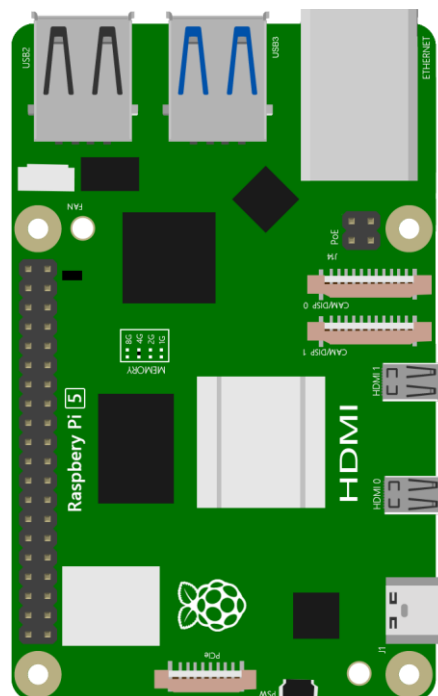
So far, at this writing, Raspberry Pi has advanced to its fifth generation product offering. Version changes are accompanied by increases in upgrades in hardware and capabilities.

The A type and B type versions of the first generation products have been discontinued due to various reasons. What is most important is that other popular and currently available versions are consistent in the order and number of pins and their assigned designation of function, making compatibility of peripheral devices greatly enhanced between versions.

Below are the raspberry pi pictures and model pictures supported by this product. They have 40 pins.



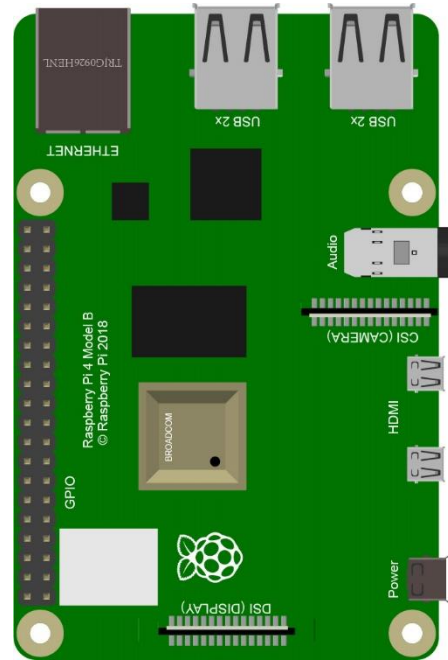
### Model diagram of Raspberry Pi 5:



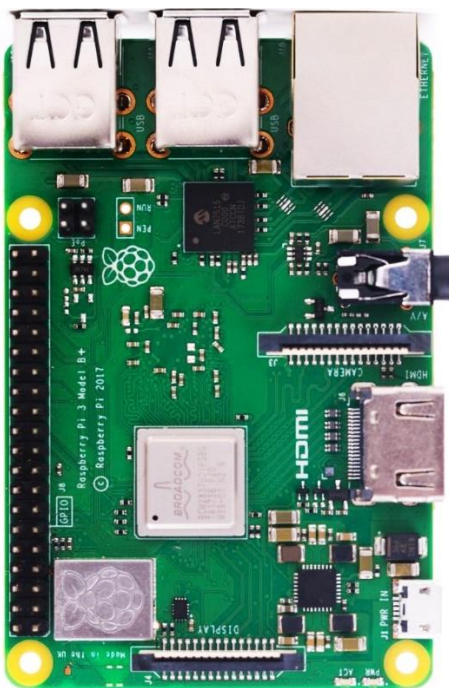
Actual image of Raspberry Pi 4 Model B :



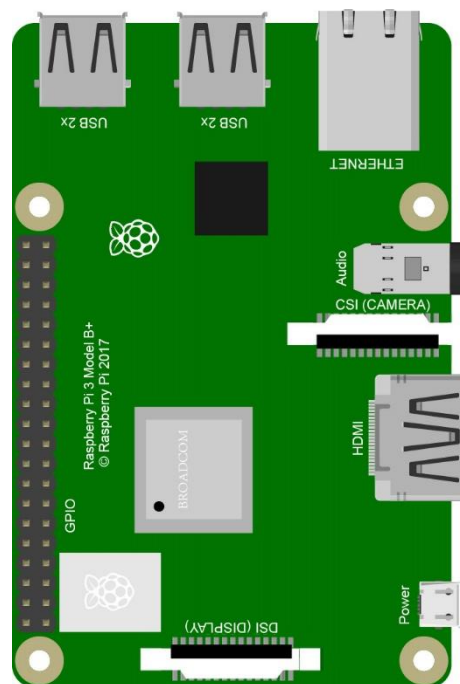
CAD image of Raspberry Pi 4 Model B :



Actual image of Raspberry Pi 3 Model B+ :



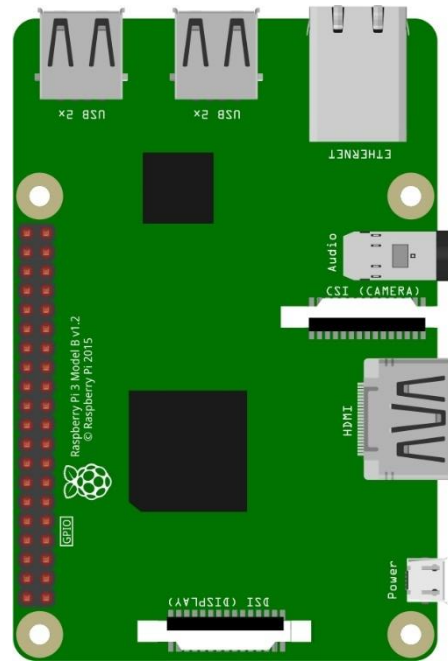
CAD image of Raspberry Pi 3 Model B+ :



Actual image of Raspberry Pi 3 Model B:



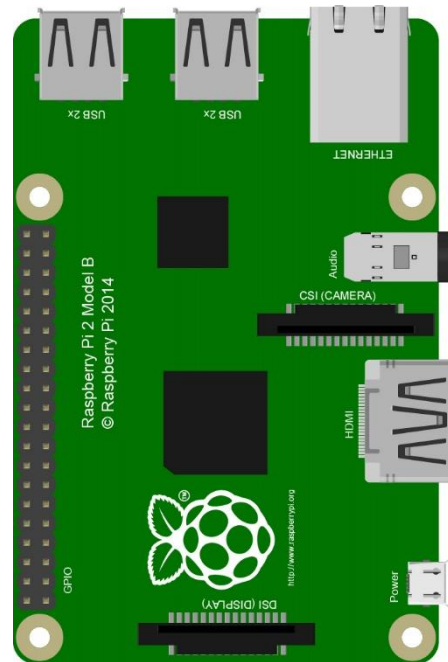
CAD image of Raspberry Pi 3 Model B:



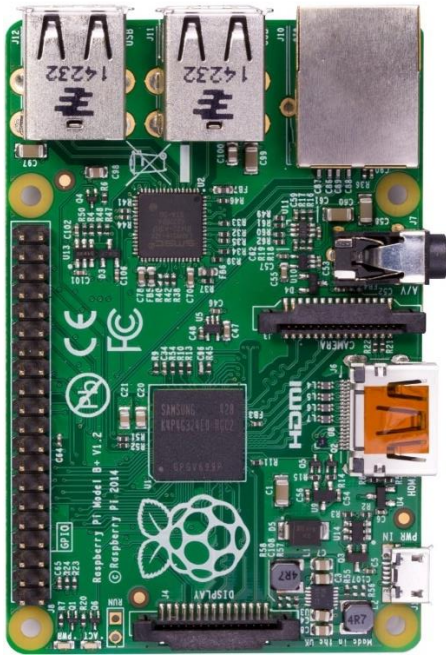
Actual image of Raspberry Pi 2 Model B:



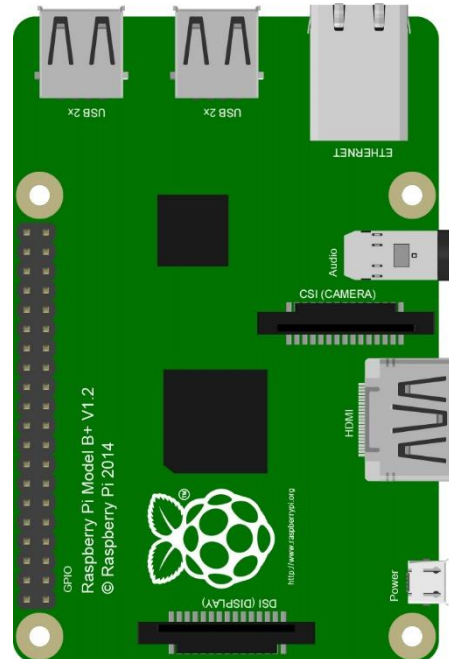
CAD image of Raspberry Pi 2 Model B:



Actual image of Raspberry Pi 1 Model B+:



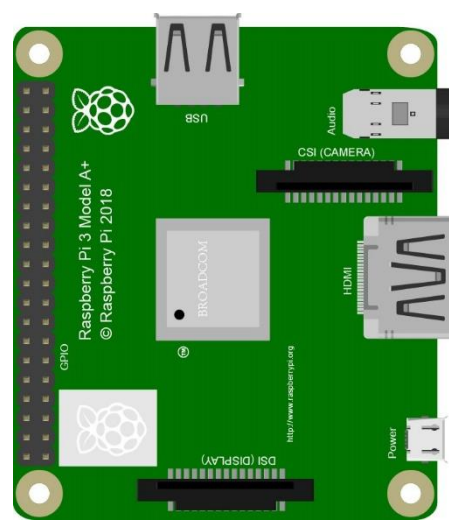
CAD image of Raspberry Pi 1 Model B+:



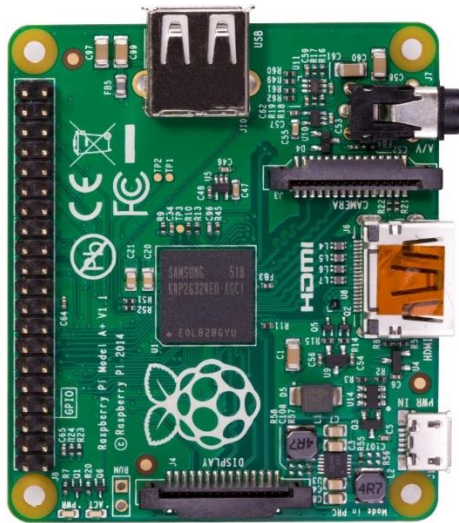
Actual image of Raspberry Pi 3 Model A+:



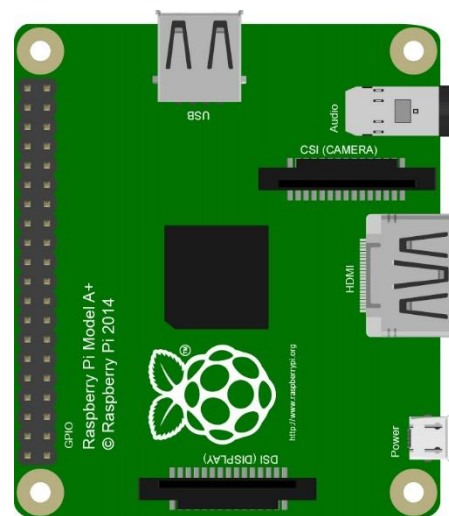
CAD image of Raspberry Pi 3 Model A+:



Actual image of Raspberry Pi 1 Model A+:



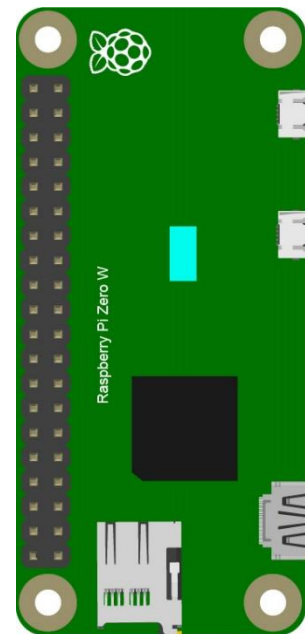
CAD image of Raspberry Pi 1 Model A+:



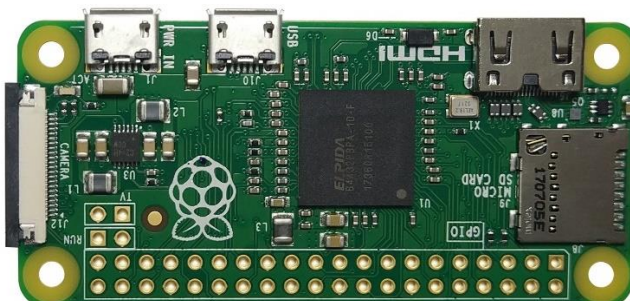
Actual image of Raspberry Pi Zero W:



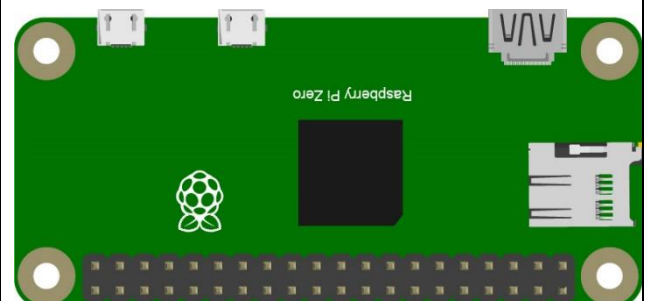
CAD image of Raspberry Pi Zero W:



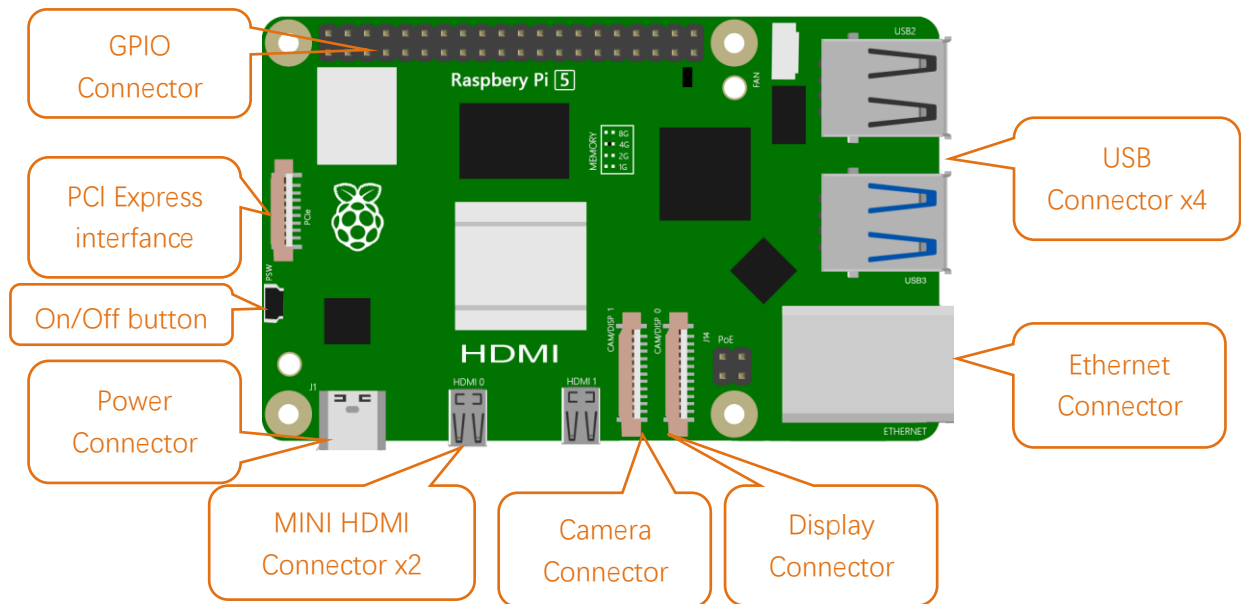
Actual image of Raspberry Pi Zero:



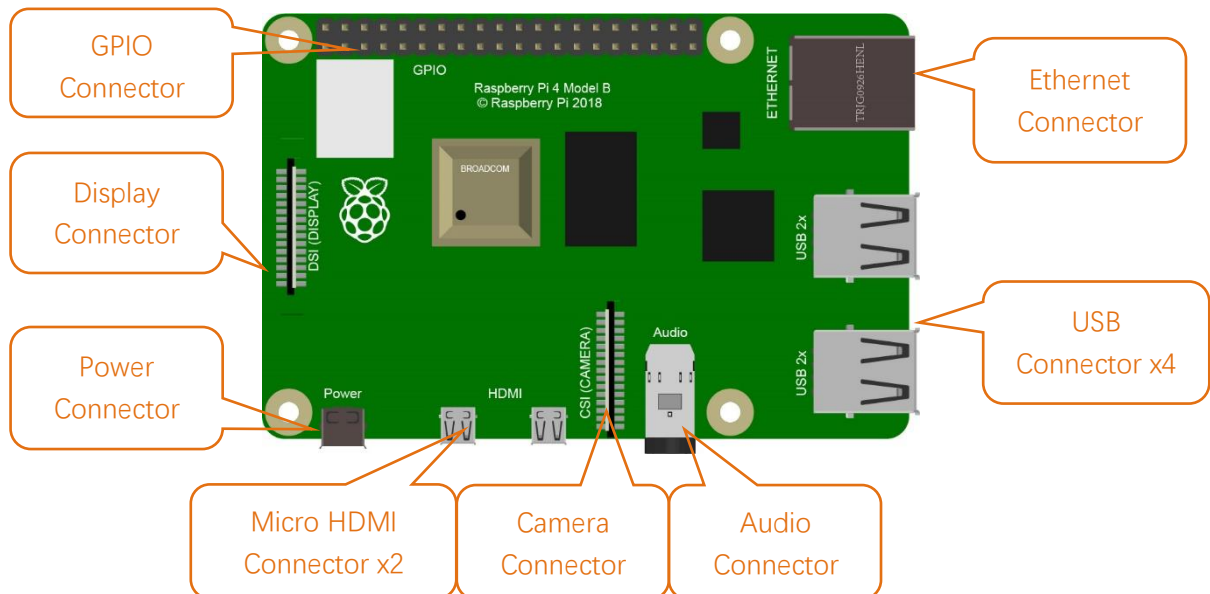
CAD image of Raspberry Pi Zero:



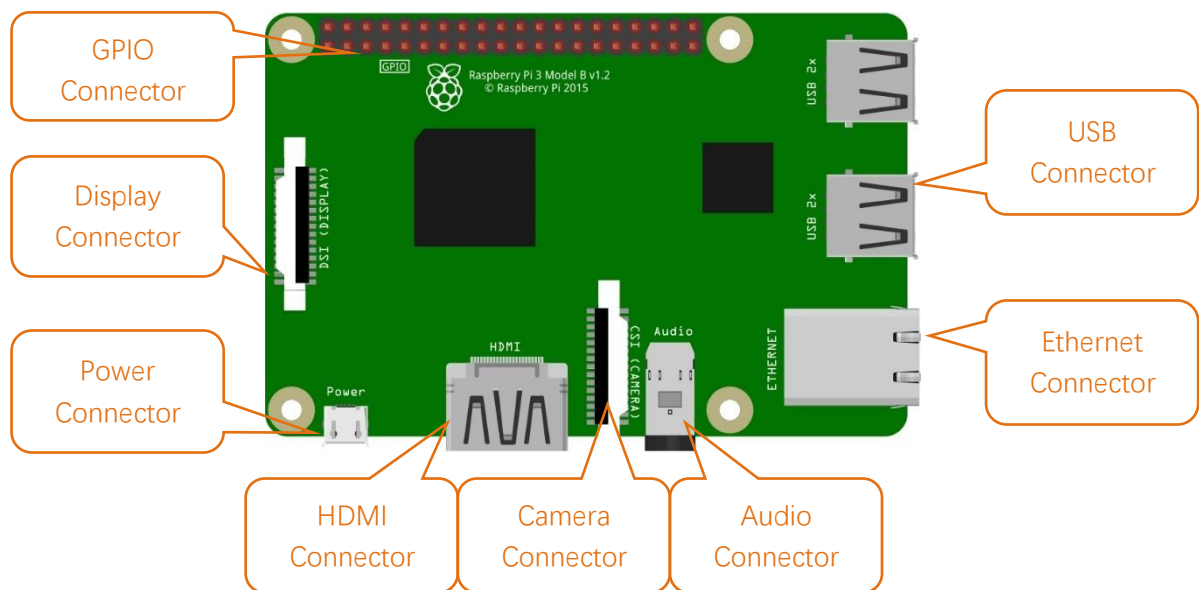
Below are the raspberry pi pictures and model pictures supported by this product. They have 40 pins.  
Hardware interface diagram of RPi 5 is shown below:



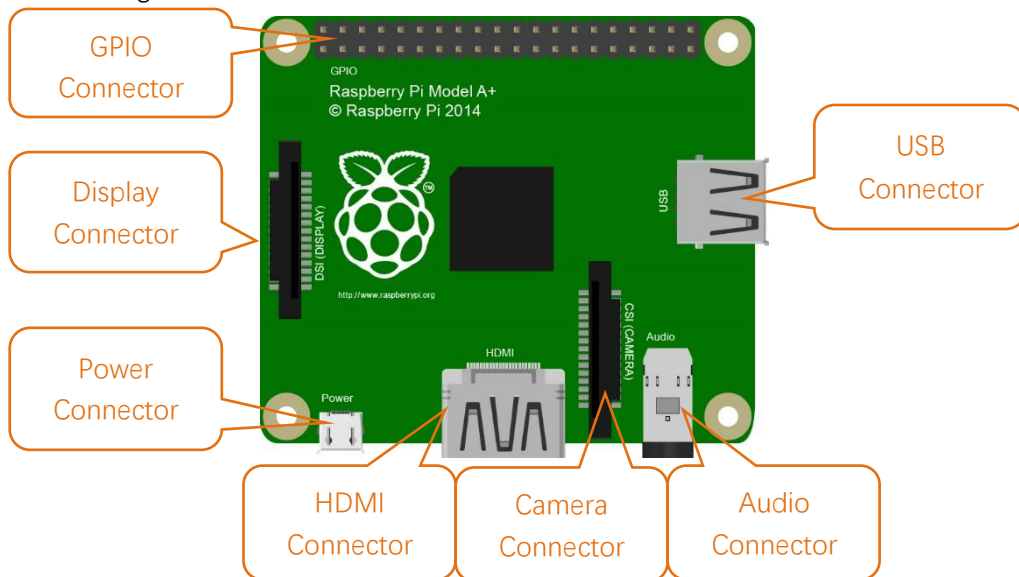
Hardware interface diagram of RPi 4B is shown below:



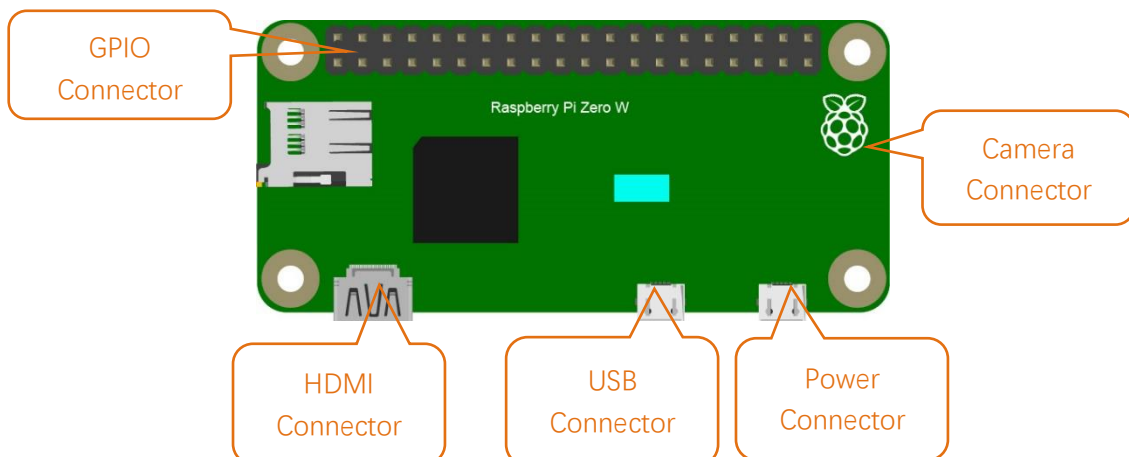
Hardware interface diagram of RPi 3B+/3B/2B/1B+:



Hardware interface diagram of RPi 3A+/A+:



Hardware interface diagram of RPi Zero/Zero W:



## GPIO

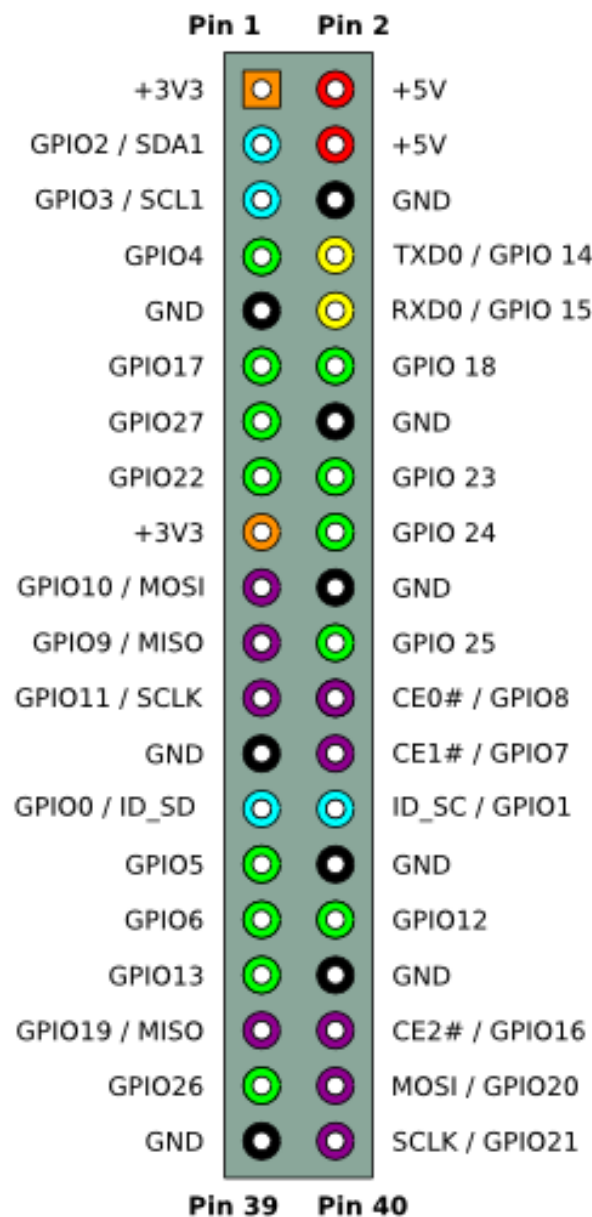
GPIO: General purpose input/output. We will introduce the specific feature of the pins on the Raspberry Pi and what you can do with them. You can use them for all sorts of purposes. Most of them can be used as either inputs or outputs, depending on your program.

When programming the GPIO pins there are 3 different ways to refer to them: GPIO numbering, physical numbering, WiringPi GPIO Numbering.

### BCM GPIO Numbering

Raspberry Pi CPU use BCM2835/BCM2836/BCM2837 of Broadcom. GPIO pin number is set by chip manufacturer. These are the GPIO pins as that computer recognizes. The numbers are unordered and don't make any sense to humans. You will need a printed reference or a reference board that fits over the pins.

Each pin is defined as below:



For more details about pin definition of GPIO, please refer to <http://pinout.xyz/>

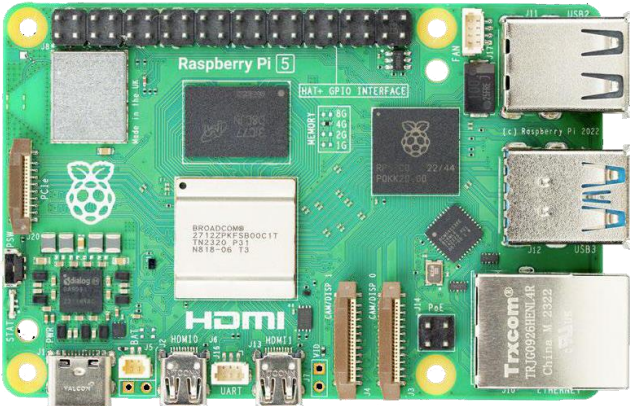
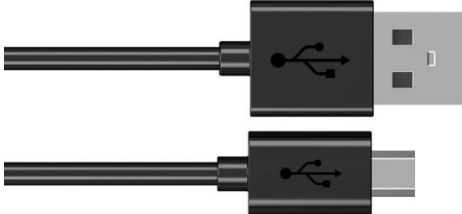
# Chapter 0 Raspberry Pi Preparation

## Install a System

Firstly, install a system for your RPi.

## Component List

### Required Components

|                                                                                                                                   |                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Raspberry Pi 5 / 4B / 3B+ (Recommended)</p>  | <p>5V/3A Power Adapter. Different versions of Raspberry Pi have different power requirements.</p>  |
| <p>Micro USB Cable x1</p>                      | <p>Micro SD Card (TF Card) x1, Card Reader x1</p>                                                 |

This robot also supports the following versions of the Raspberry Pi, but **additional accessories** need to be prepared by yourself.

| Raspberry                      | Additional accessories                                                                                                                                          |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Raspberry Pi Zero W</b>     | Camera cable(>25cm) for zero w, 15 Pin 1.0mm Pitch to 22 Pin 0.5mm<br><a href="https://www.amazon.com/dp/B076Q595HJ/">https://www.amazon.com/dp/B076Q595HJ/</a> |
| <b>Raspberry Pi Zero 1.3</b>   | wireless network adapter,<br>Camera cable(>25cm) for zero w, 15 Pin 1.0mm Pitch to 22 Pin 0.5mm,<br>OTG cable (USB Type micro B to USB Type A)                  |
| <b>Raspberry Pi 2 Model B</b>  | wireless network adapter                                                                                                                                        |
| <b>Raspberry Pi 1 Model A+</b> | wireless network adapter                                                                                                                                        |
| <b>Raspberry Pi 1 Model B+</b> | wireless network adapter                                                                                                                                        |

Power requirements of various versions of Raspberry Pi are shown in following table:

| Product                 | Recommended PSU current capacity | Maximum total USB peripheral current draw          | Typical bare-board active current consumption |
|-------------------------|----------------------------------|----------------------------------------------------|-----------------------------------------------|
| Raspberry Pi 1 Model A  | 700mA                            | 500mA                                              | 200mA                                         |
| Raspberry Pi 1 Model B  | 1.2A                             | 500mA                                              | 500mA                                         |
| Raspberry Pi 1 Model A+ | 700mA                            | 500mA                                              | 180mA                                         |
| Raspberry Pi 1 Model B+ | 1.8A                             | 1.2A                                               | 330mA                                         |
| Raspberry Pi 2 Model B  | 1.8A                             | 1.2A                                               | 350mA                                         |
| Raspberry Pi 3 Model B  | 2.5A                             | 1.2A                                               | 400mA                                         |
| Raspberry Pi 3 Model A+ | 2.5A                             | Limited by PSU, board, and connector ratings only. | 350mA                                         |
| Raspberry Pi 3 Model B+ | 2.5A                             | 1.2A                                               | 500mA                                         |
| Raspberry Pi 4 Model B  | 3.0A                             | 1.2A                                               | 600mA                                         |
| Raspberry Pi 5          | 5.0A                             | 1.6A (600mA if using a 3A power supply)            | 800mA                                         |
| Raspberry Pi 400        | 3.0A                             | 1.2A                                               | 800mA                                         |
| Raspberry Pi Zero       | 1.2A                             | Limited by PSU, board, and connector ratings only  | 100mA                                         |
| Raspberry Pi Zero W     | 1.2A                             | Limited by PSU, board, and connector ratings only. | 150mA                                         |
| Raspberry Pi Zero 2 W   | 2A                               | Limited by PSU, board, and connector ratings only. | 350mA                                         |

For more details, please refer to <https://www.raspberrypi.org/help/fags/#powerReqs>

In addition, RPi also needs an Ethernet network cable used to connect it to a WAN (Wide Area Network).

The Raspberry Pi 5 provides 1.6A of power to downstream USB peripherals when connected to a power supply capable of 5A at +5V (25W). When connected to any other compatible power supply, the Raspberry Pi 5 restricts downstream USB devices to 600mA of power.

## Optional Components

Under normal circumstances, there are two ways to login to Raspberry Pi: 1) Using a stand-alone monitor. 2) Using a remote desktop or laptop computer monitor “sharing” the PC monitor with your RPi.

## Required Accessories for Monitor

If you choose to use an independent monitor, mouse and keyboard, you also need the following accessories:

1. A display with a HDMI interface
2. A Mouse and a Keyboard with an USB interface

As to Pi Zero and Pi Zero W, you also need the following accessories:

1. A Mini-HDMI to HDMI Adapter and Cable.
2. A Micro-USB to USB-A Adapter and Cable (Micro USB OTG Cable).
3. A USB HUB.
4. USB to Ethernet Interface or USB Wi-Fi receiver.

For different Raspberry Pi Modules, the optional items may vary slightly but they all aim to convert the interfaces to Raspberry Pi standards.

|                                                          | Pi Zero                                    | Pi A+ | Pi Zero W            | Pi 3A+ | Pi B+/2B             | Pi 3B/3B+            | Pi 4B | Pi 5 |
|----------------------------------------------------------|--------------------------------------------|-------|----------------------|--------|----------------------|----------------------|-------|------|
| Monitor                                                  | Yes (All)                                  |       |                      |        |                      |                      |       |      |
| Mouse                                                    | Yes (All)                                  |       |                      |        |                      |                      |       |      |
| Keyboard                                                 | Yes (All)                                  |       |                      |        |                      |                      |       |      |
| Micro-HDMI to HDMI Adapter & Cable                       | Yes                                        | No    | Yes                  | No     | No                   | No                   | No    | No   |
| Micro-HDMI to HDMI Adapter & Cable                       | No                                         |       |                      |        |                      |                      | Yes   |      |
| Micro-USB to USB-A Adapter & Cable (Micro USB OTG Cable) | Yes                                        | No    | Yes                  | No     |                      |                      |       |      |
| USB HUB                                                  | Yes                                        | Yes   | Yes                  | Yes    | No                   | No                   | No    | No   |
| USB to Ethernet Interface                                | select one from two or select two from two |       | optional             |        | Internal Integration | Internal Integration |       |      |
| USB Wi-Fi Receiver                                       |                                            |       | Internal Integration |        | optional             |                      |       |      |

### Required Accessories for Remote Desktop

If you do not have an independent monitor, or if you want to use a remote desktop, you first need to login to Raspberry Pi through SSH, and then open the VNC or RDP service. This requires the following accessories.

|                                                          | Pi Zero | Pi Zero W | Pi A+ | Pi 3A+ | Pi B+/2B | Pi 3B/3B+/4B/5 |
|----------------------------------------------------------|---------|-----------|-------|--------|----------|----------------|
| Micro-USB to USB-A Adapter & Cable (Micro USB OTG Cable) | Yes     | Yes       | No    | NO     |          |                |
| USB to Ethernet interface                                | Yes     | Yes       | Yes   |        |          |                |

# Raspberry Pi OS

**Without Screen - Use Raspberry Pi - under Windows PC:** [https://www.youtube.com/watch?v=XpiT\\_ezb\\_7c](https://www.youtube.com/watch?v=XpiT_ezb_7c)

**With Screen - Use Raspberry Pi - under Windows PC:** <https://youtu.be/HEywFsFrj3I>

## Automatically Method

You can follow the official method to install the system for raspberry pi via visiting link below:

<https://projects.raspberrypi.org/en/projects/raspberry-pi-setting-up/2>

In this way, the system will be downloaded **automatically** via the application.

## Manually Method

After installing the Imager Tool in the **link above**. You can **also** download the system **manually** first.

Visit <https://www.raspberrypi.org/downloads/>

## Manually install an operating system image

Browse a range of operating systems provided by Raspberry Pi and by other organisations, and download them to install manually.

See all download options



## Operating system images

Many operating systems are available for Raspberry Pi, including Raspberry Pi OS, our official supported operating system, and operating systems from other organisations.

Raspberry Pi Imager is the quick and easy way to install an operating system to a microSD card ready to use with your Raspberry Pi. Alternatively, choose from the operating systems below, available to download and install manually.

Download:

- [Raspberry Pi OS](#)
- [Raspberry Pi OS \(64-bit\)](#)
- [Raspberry Pi OS \(Legacy\)](#)
- [Raspberry Pi OS \(Legacy, 64-bit\)](#)
- [Raspberry Pi Desktop](#)

## Raspberry Pi OS

Our recommended operating system for most users.

Compatible with:

All Raspberry Pi models

### Raspberry Pi OS with desktop

Release date: July 4th 2024  
System: 32-bit  
Kernel version: 6.6  
Debian version: 12 (bookworm)  
Size: 1,205MB  
[Show SHA256 file integrity hash:](#)  
[Release notes](#)

Download

[Download torrent](#)  
[Archive](#)

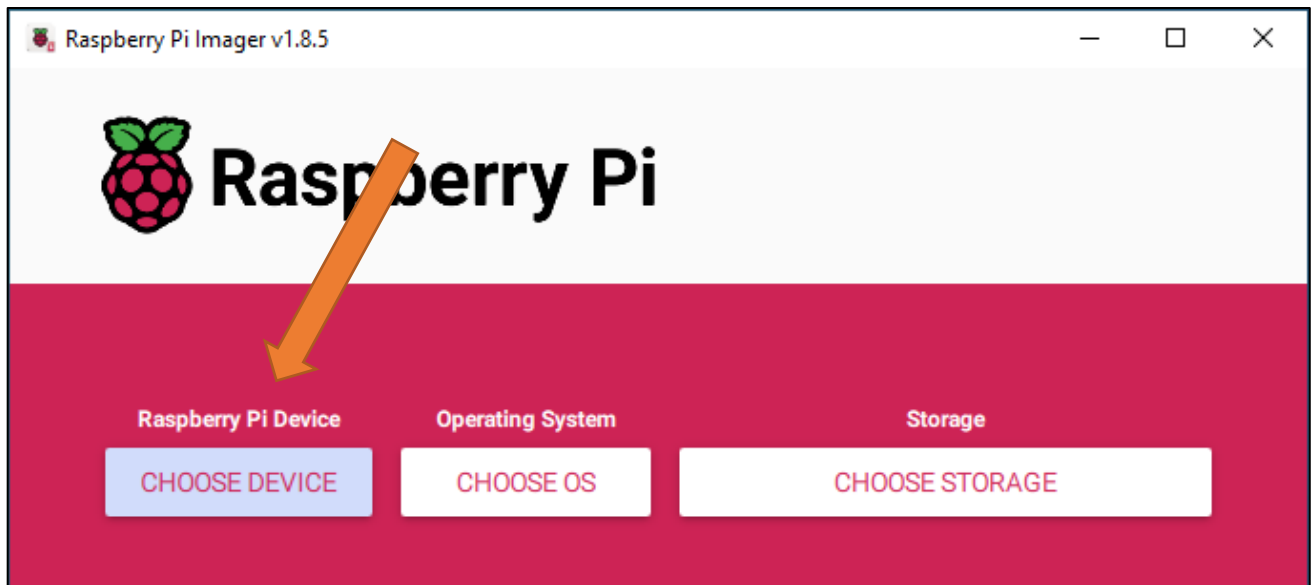
And then the zip file is downloaded.

### Write System to Micro SD Card

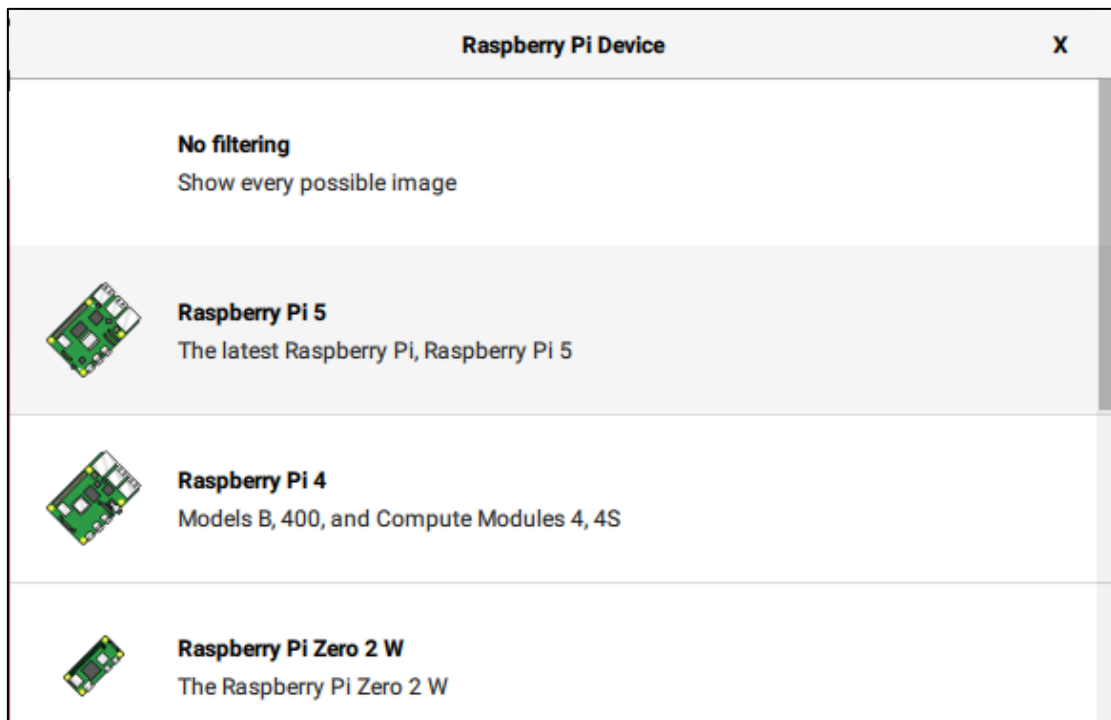
First, put your Micro **SD card** into card reader and connect it to USB port of PC.



Then open imager tool. Clicked Choose Device.

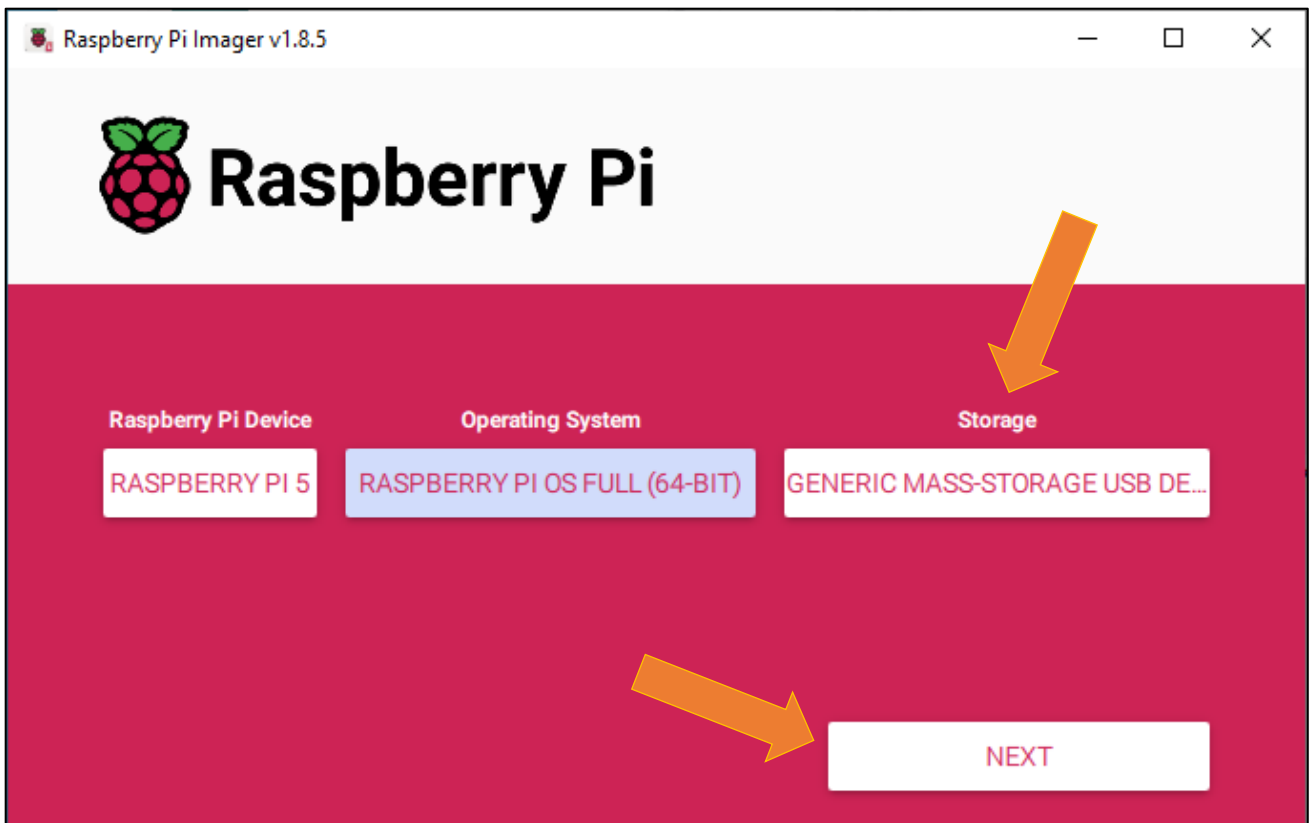


Select a Raspberry Pi Device based on your Raspberry Pi version. It will help us filter out the right version of the system for the Raspberry PI.

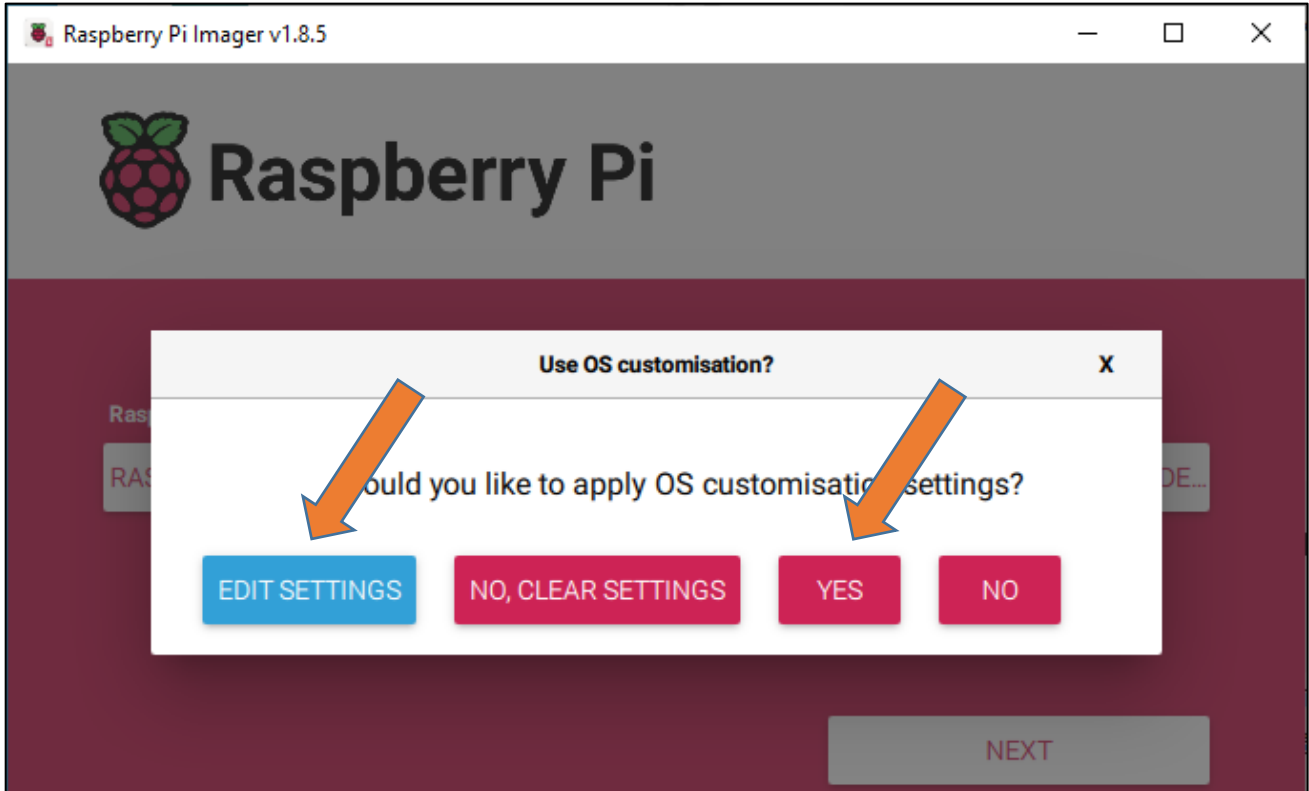




Choose the SD card. Then click “Next”.



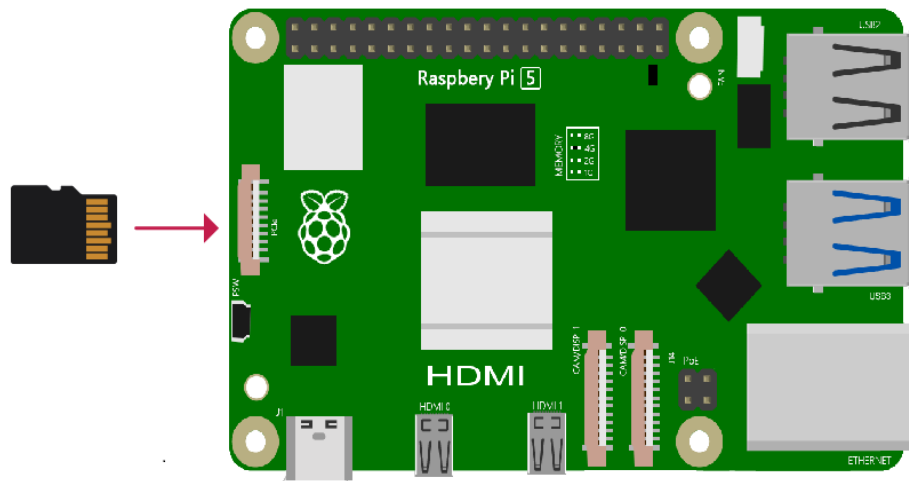
You can configure the Raspberry PI according to your needs.





Insert SD card

Then remove SD card from card reader and insert it into Raspberry Pi.

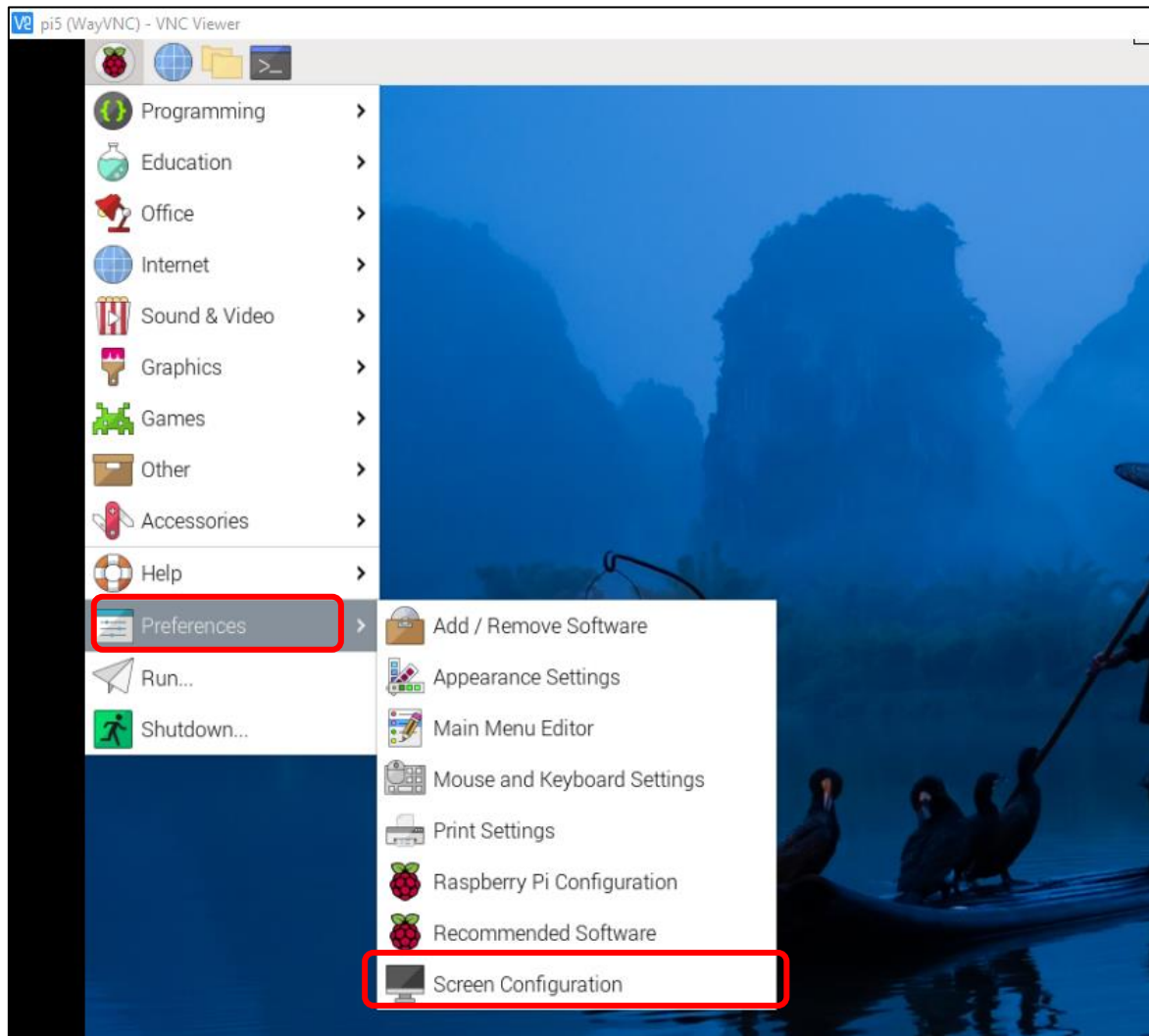


Connect to the power supply and wait for the Raspberry PI to turn on.

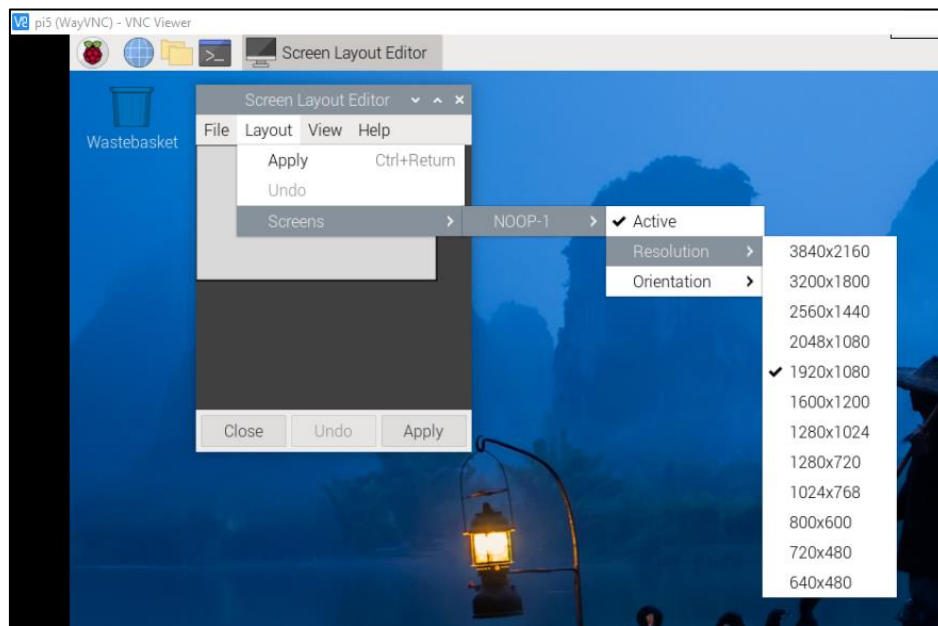


## Set Resolution

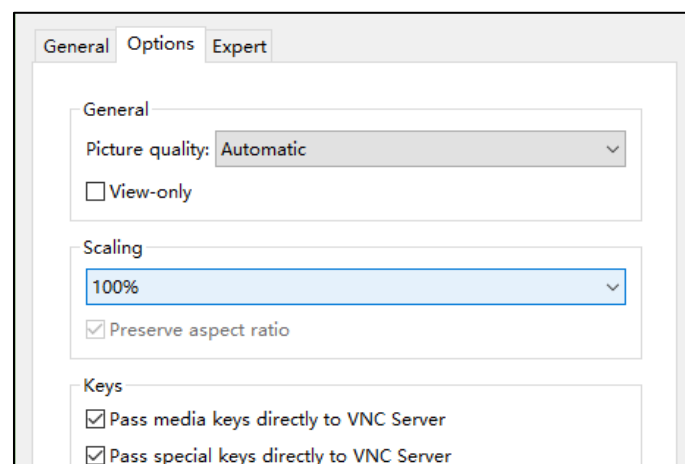
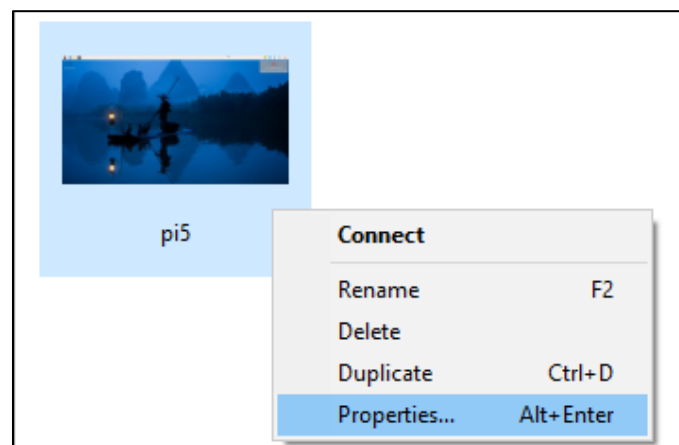
You can also set other resolutions.



If you don't know what resolution to set properly, you can try 1920x1080.

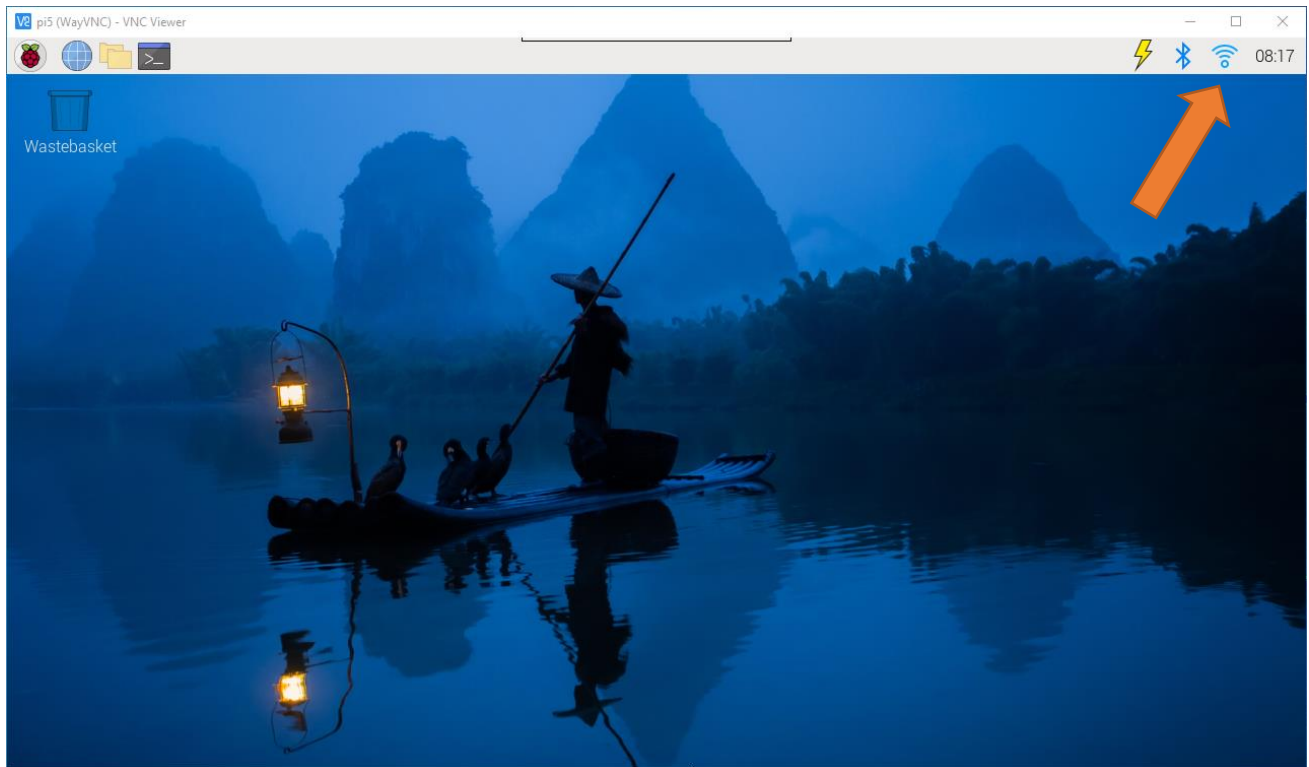


In addition, your VNC Viewer window may zoom your Raspberry Pi desktop. You can change it. On your VNC View control panel, click right key. And select Properties->Options label->Scaling. Then set proper scaling.



Here, you have logged in to Raspberry Pi successfully by using VNC Viewer and operated proper setting.

Raspberry Pi 5/4B/3B+/3B integrates a Wi-Fi adaptor. If you did not connect Pi to WiFi. You can connect it to wirelessly control the robot.





You can also use the IP address to log in Pi.

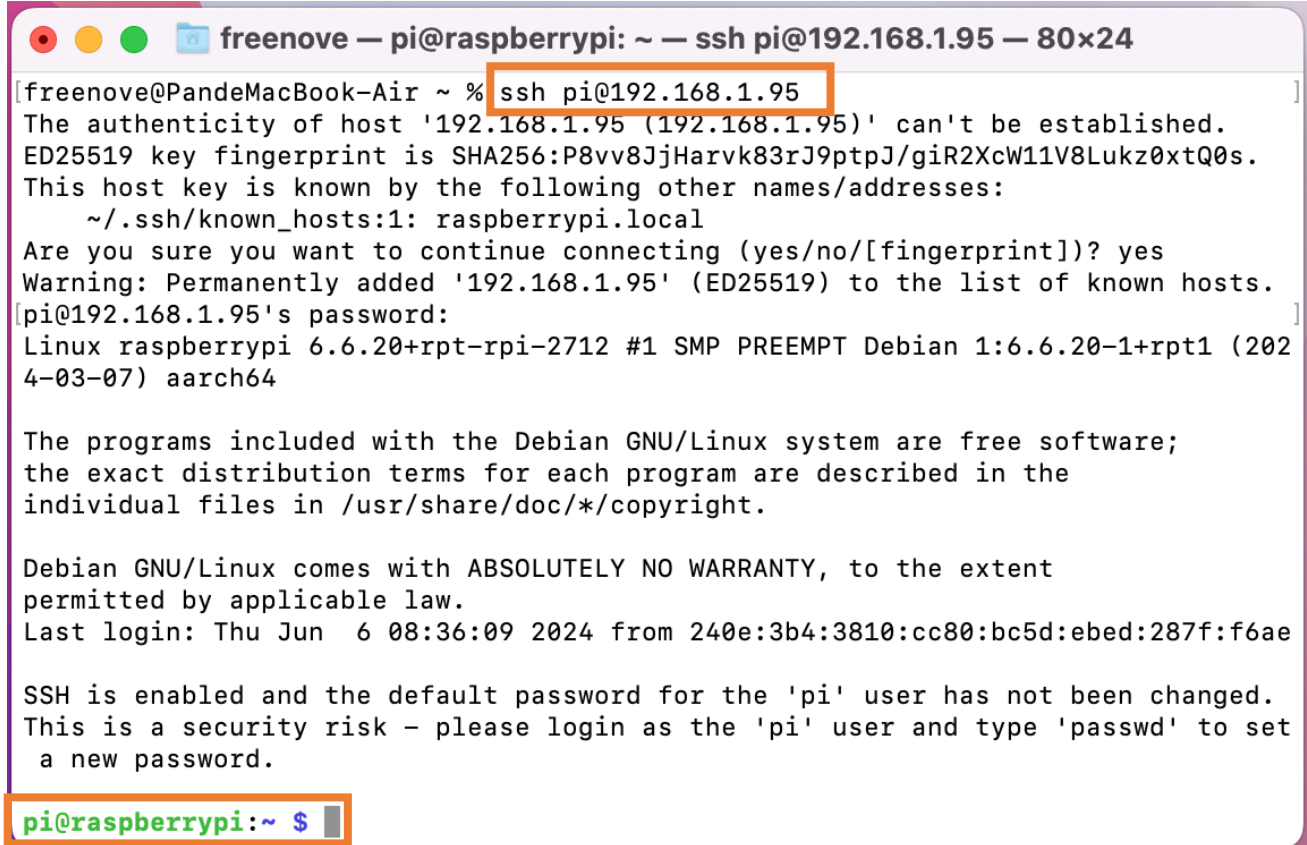
Enter **router** client to **inquiry IP address** named “raspberry pi”.

For example, I have inquired to **my RPi IP address**, and it is “192.168.1.95”.

Open the terminal and type following command.

```
ssh pi@192.168.1.95
```

When you see **pi@raspberrypi:~ \$**, you have logged in Pi successfully. Then you can skip to next section.



```
freenove — pi@raspberrypi: ~ — ssh pi@192.168.1.95 — 80x24
[freenove@PandeMacBook-Air ~ % ssh pi@192.168.1.95
The authenticity of host '192.168.1.95 (192.168.1.95)' can't be established.
ED25519 key fingerprint is SHA256:P8vv8JjHarvk83rJ9ptpJ/giR2XcW11V8Lukz0xtQ0s.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: raspberrypi.local
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.95' (ED25519) to the list of known hosts.
pi@192.168.1.95's password:
Linux raspberrypi 6.6.20+rpt-rpi-2712 #1 SMP PREEMPT Debian 1:6.6.20-1+rpt1 (202
4-03-07) aarch64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Thu Jun  6 08:36:09 2024 from 240e:3b4:3810:cc80:bc5d:ebd:287f:f6ae

SSH is enabled and the default password for the 'pi' user has not been changed.
This is a security risk - please login as the 'pi' user and type 'passwd' to set
a new password.

pi@raspberrypi:~ $
```

Then you can skip to [VNC Viewer](#).



### Enable VNC Viewer

Type the following command. And select Interface Options→P5 VNC → Enter→Yes→OK. Here Raspberry Pi may need be restarted, and choose ok. Then open VNC interface.

```
sudo raspi-config
```

SSH is enabled and the default password for the 'pi' user has not been changed. This is a security risk – please login as the 'pi' user and type 'passwd' to set a new password.

```
pi@raspberrypi:~ $ sudo raspi-config
```

#### Raspberry Pi Software Configuration Tool (raspi-config)

- |                            |                                             |
|----------------------------|---------------------------------------------|
| 1 System Options           | Configure system settings                   |
| 2 Display Options          | Configure display settings                  |
| <b>3 Interface Options</b> | <b>Configure connections to peripherals</b> |
| 4 Performance Options      | Configure performance settings              |
| 5 Localisation Options     | Configure language and regional settings    |
| 6 Advanced Options         | Configure advanced settings                 |
| 8 Update                   | Update this tool to the latest version      |
| 9 About raspi-config       | Information about this configuration tool   |

<Select>

<Finish>

#### Raspberry Pi Software Configuration Tool (raspi-config)

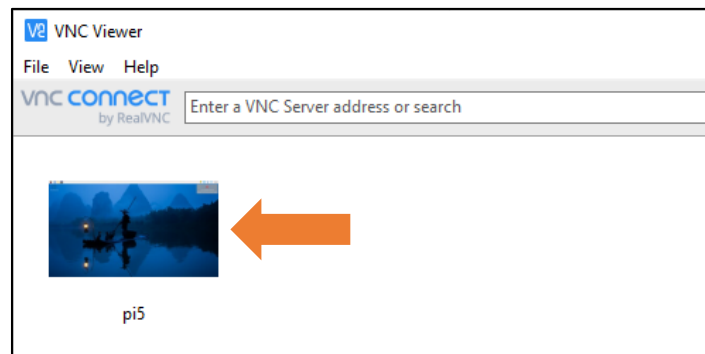
- |                |                                                        |
|----------------|--------------------------------------------------------|
| I1 SSH         | Enable/disable remote command line access using SSH    |
| <b>I2 VNC</b>  | <b>Enable/disable graphical remote desktop access</b>  |
| I3 SPI         | Enable/disable automatic loading of SPI kernel module  |
| I4 I2C         | Enable/disable automatic loading of I2C kernel module  |
| I5 Serial Port | Enable/disable shell messages on the serial connection |
| I6 1-Wire      | Enable/disable one-wire interface                      |
| I7 Remote GPIO | Enable/disable remote access to GPIO pins              |

<Select>

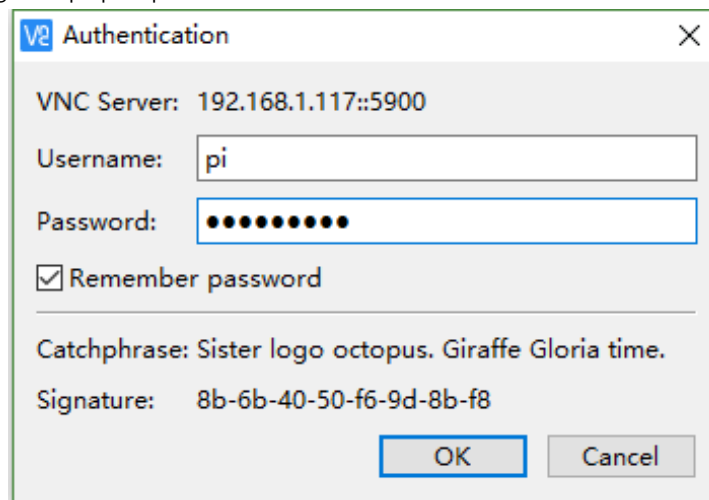
<Back>



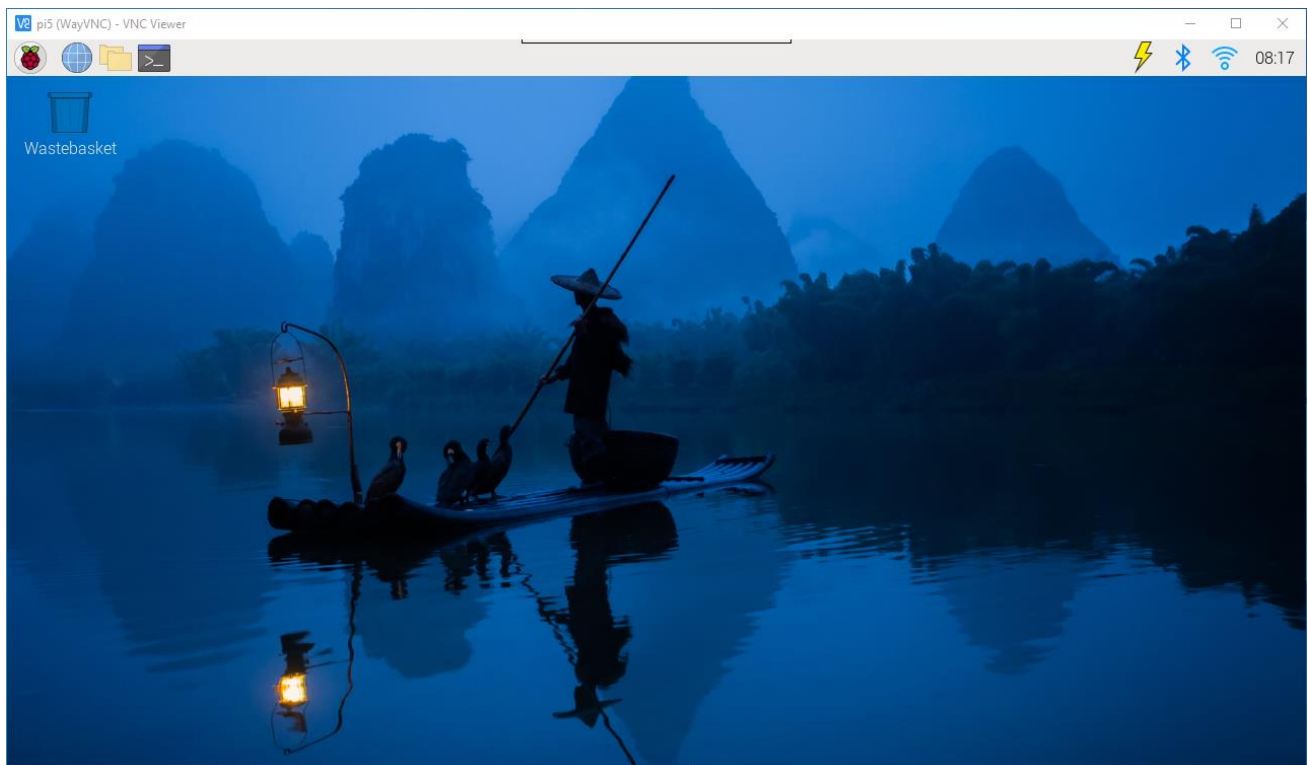
Then on the VNC Viewer panel, double-click new connection you just created,



and the following dialog box pops up.



Enter username: **pi** and Password: **raspberry**. And click OK.



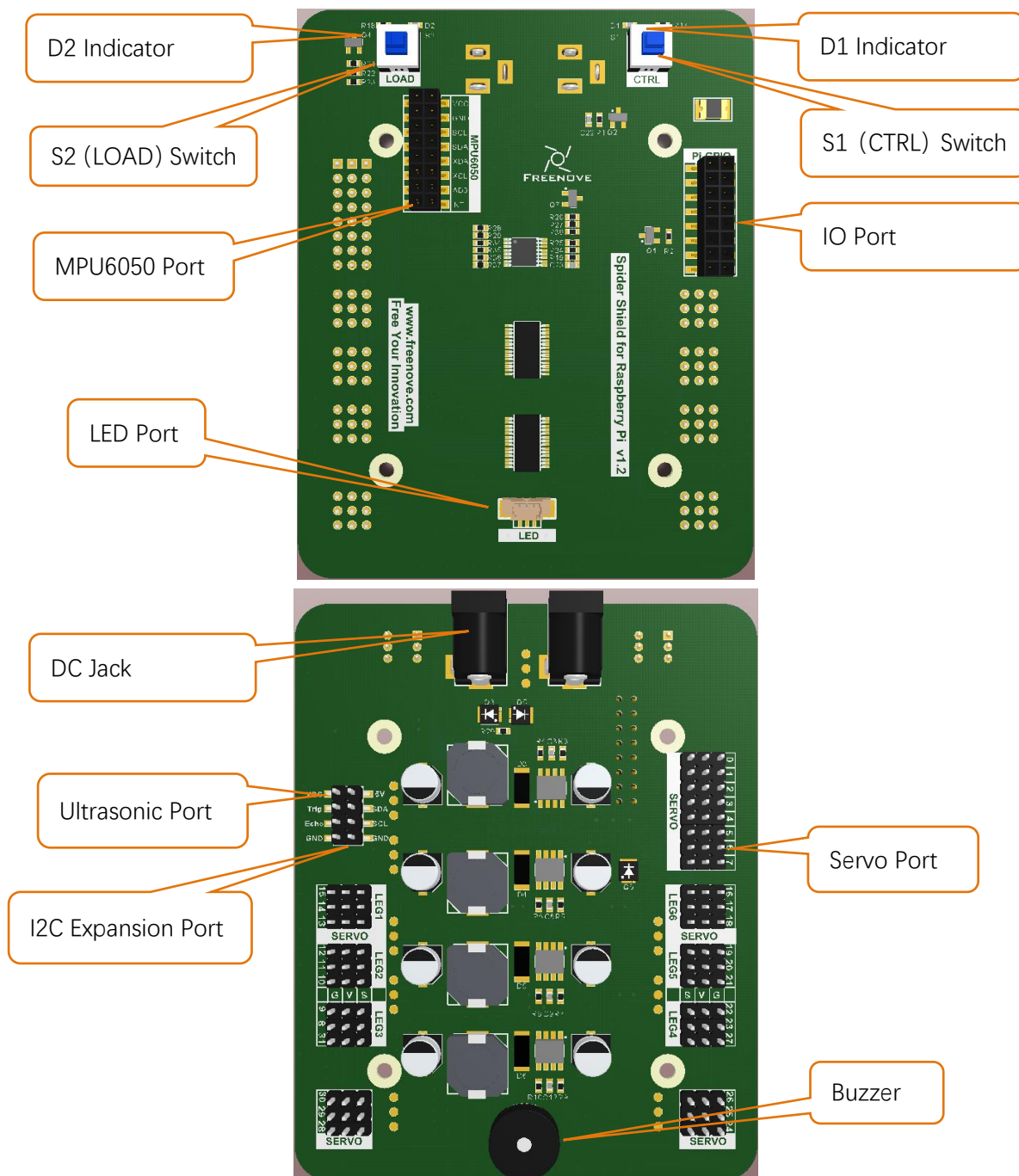
Here, you have logged in to Raspberry Pi successfully by using VNC Viewer

 [support@freenove.com](mailto:support@freenove.com)

# Spider Shield for Raspberry Pi

## Description of Robot Control Board

The control board is mainly connected the Raspberry Pi through the “PI-GPIO” port and the positioning holes on the board are suitable for the Raspberry Pi. The features and functions of the control panel are as follows.



- **Spider Shield board:** powered by two sets of batteries without protection plates, with each set containing two 18650 batteries and voltage ranging from 7V-8.4V.
- **DC socket:** Two DC sockets available, each connecting to one set of batteries, each containing two 18650 batteries.
- **S1(CTRL) Switch:** It mainly controls chips like PCA9685 and ADS7830 and the power supply of the Raspberry Pi. When it is pressed, D1 indicator will turn ON.
- **S2(LOAD) Switch:** It mainly controls the power supply of servos. When it is pressed, D2 indicator will turn ON.
- **LED port:** Used to connect the LED Module, controlled directly by RPi.
- **MPU6050 Module port:** Used to connect MPU6050, controlled directly by RPi.
- **PI-GPIO port:** Used to connect control board with RPi, facilitating RPi to control the robot
- **I2CExtension Board:** Used for the extension of other I2C devices.
- **Servo port:** 32 servo ports available.
- **Sonic module port:** The interface of HC-SR04 Ultrasonic Module is provided.
- **Buzzer:** Turning ON when it is at high level and OFF at low level, directly controlled by RPi

# Chapter 1 Installation of Python Libraries(Important)

In this chapter, we will do some necessary foundational preparation work: Start your Raspberry Pi and install some necessary libraries. And we will assemble the robot in the next chapter.

Caution: Do NOT install it without referring to the tutorial. Otherwise, it may cause installation errors and damage the components.

## Note:

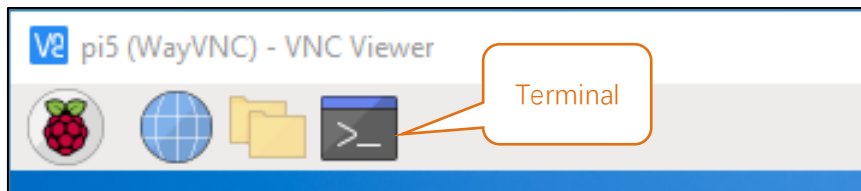
- 1, **Please use Raspberry Pi OS with Desktop**
- 2, The installation of libraries takes much time. **You can power Raspberry Pi with a power supply Cable.**
- 3, If you are using **remote desktop** to login Raspberry Pi, you need to use [VNC viewer](#).

You can watch this video while reading this chapter: <https://youtu.be/cVrwOoORItE>

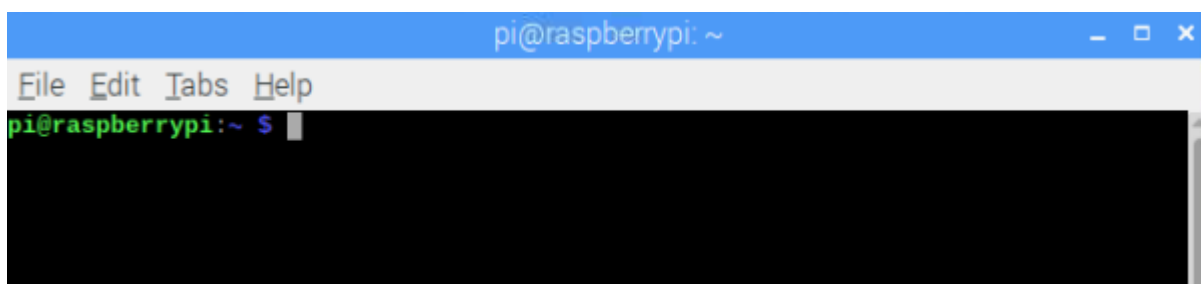
We will help you with any concerns via [support@freenove.com](mailto:support@freenove.com)

## Step 1 Obtain the Code

To download the code, you can power Raspberry Pi with a power supply cable **or** switch on S1 (Power Switch). Then open the Raspberry Pi and the terminal. You can open the terminal by clicking as shown below, or you can press "CTRL + ALT + T" on the desktop.



The terminal is shown below:



Enter the following commands in the terminal to obtain the robot's code and save it to the user's directory "Pi". (Note that here are two commands, please enter them one by one in order.)

```
cd ~  
git clone --depth 1  
https://github.com/Freenove/Freenove\_Big\_Hexapod\_Robot\_Kit\_for\_Raspberry\_Pi.git
```

Downloading need much time. Please wait with patience.

You can also find and download the code by visiting our official website (<http://www.freenove.com>) or our GitHub repository (<https://github.com/freenove>).

**Note:** all codes of the robot are written with Python3. If run with Python2, it may cause errors.

## Set Python3 as default python (Necessary)

First, execute python to check the default python on your Raspberry Pi. Press Ctrl-Z to exit.

```
pi@raspberrypi:~ $ python
```

If it is python3, you can skip this section.

**If it is python2, you need execute the following commands to set default python to python3.**

1. Enter directory /usr/bin

```
cd /usr/bin
```

2. Delete the original python link.

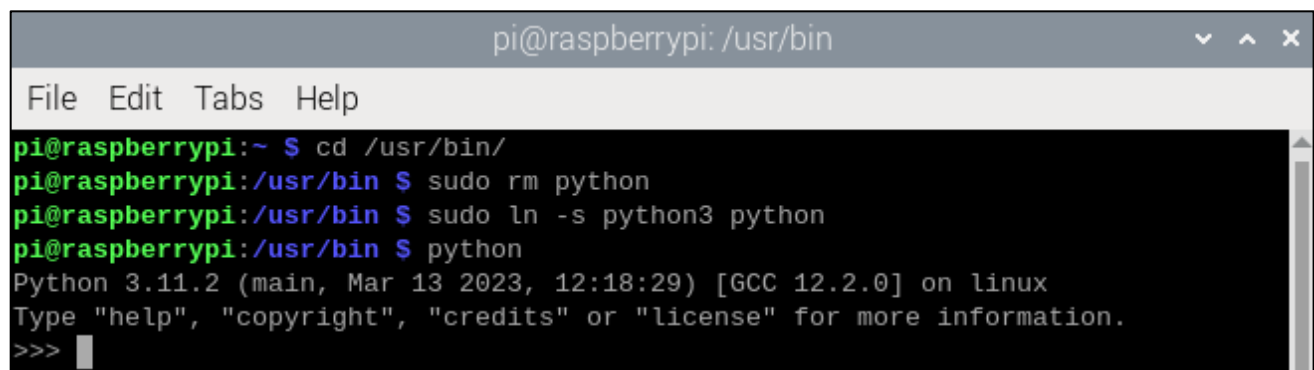
```
sudo rm python
```

3. Create new python links to python3.

```
sudo ln -s python3 python
```

4. Check python. Press Ctrl-Z to exit.

```
python
```



```
pi@raspberrypi: /usr/bin
File Edit Tabs Help
pi@raspberrypi:~ $ cd /usr/bin/
pi@raspberrypi:/usr/bin $ sudo rm python
pi@raspberrypi:/usr/bin $ sudo ln -s python3 python
pi@raspberrypi:/usr/bin $ python
Python 3.11.2 (main, Mar 13 2023, 12:18:29) [GCC 12.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> █
```

If you want to set python2 as default python in **other projects**, just repeat the commands above and change python3 to python2.

### Shortcut Key

Now, we will introduce several shortcuts that are very **useful** and **commonly used** in terminal.

1. **up and down arrow keys**. History commands can be quickly brought back by using up and down arrow keys, which are very useful when you need to reuse certain commands.

When you need to type command, pressing “↑” will bring back the previous command, and pressing “↓” will bring back the latter command.

2. **Tab key**. The Tab key can automatically complete the command/path you want to type. When there are multiple commands/paths conforming to the already typed letter, pressing Tab key once won't have any result. And pressing Tab key again will list all the eligible options. This command/path will be directly completed when there is only one eligible option.

As shown below, under the '~' directory, enter the Documents directory with the “cd” command. After typing “cd D”, press Tab key, then there is no response. Press Tab key again, then all the files/folders that begin with “D” is listed. Continue to type the character “oc”, then press the Tab key, and then “Documents” is completed automatically.

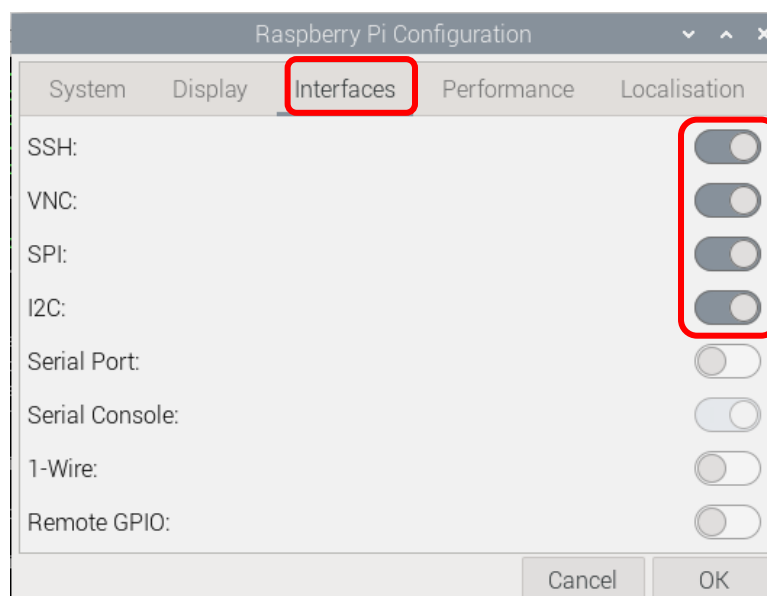
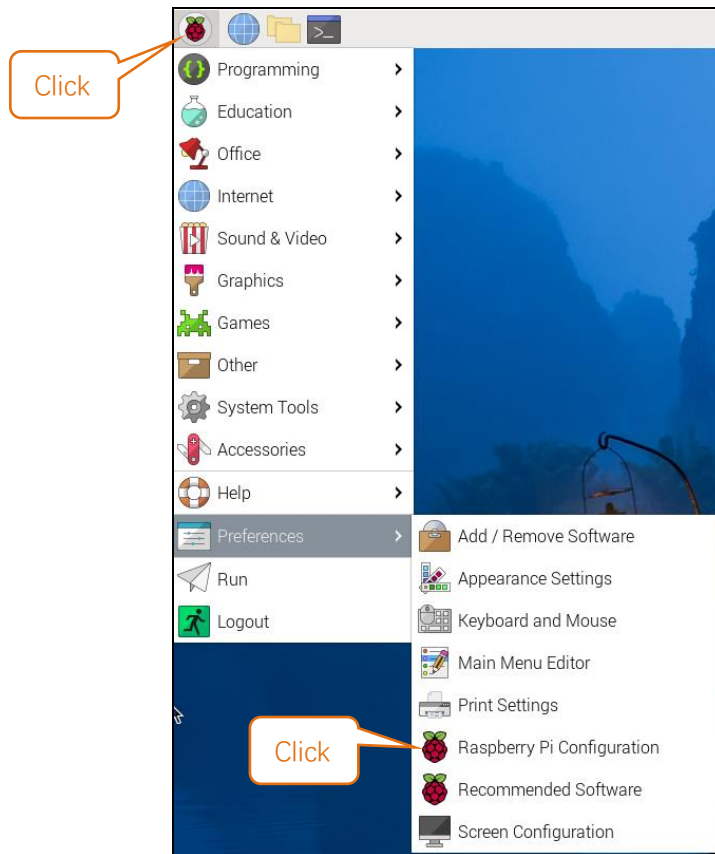
```
pi@raspberrypi:~ $ cd D
Desktop/  Documents/ Downloads/
pi@raspberrypi:~ $ cd Doc
```

```
pi@raspberrypi:~ $ cd D
Desktop/  Documents/ Downloads/
pi@raspberrypi:~ $ cd Documents/
```

## Step 2 Configuration

### Enable I2C, SPI and VNC

The I2C interface Raspberry Pi is disabled by default. You need to open it manually.



Note that if your connector version is V2.0, remember to turn on the SPI feature. If your connector version is V1.0, turn off the SPI feature.

## Set I2C Baud Rate

Open the terminal and enter the following command.

```
sudo nano /boot/firmware/config.txt
```

The default I2C Baud Rate is 100000. Now we change it to 400000, because this can speed up the response speed of the servos to make robot dog walk faster. If the baud rate is 100,000, the robot walks slowly.

Scrolling the middle of the mouse to find **dtparam=i2c\_arm=on**, and add **"i2c\_arm\_baudrate=400000"**.

```
# uncomment for composite PAL
#sdtv_mode=2

#uncomment to overclock the arm. 700 MHz is the default.
#arm_freq=800

# Uncomment some or all of these to enable the optional hardware interfaces
dtparam=i2c_arm=on,i2c_arm_baudrate=400000
#dtparam=i2s=on
#dtparam=spi=on

[ Wrote 67 lines ]
^G Get Help  ^O Write Out  ^W Where Is  ^K Cut Text  ^J Justify   ^C Cur Pos
^X Exit      ^R Read File  ^_ Replace   ^U Uncut Text ^T To Spell  ^_ Go To Line
```

Press "CTRL"+"O" and then "Enter" to save the modified content. Then press "CTRL"+"X" to exit editing.

After the modification is completed, reboot Raspberry Pi to make the change work. You can also reboot the Raspberry Pi after completing the step 3 below.

Type a command to check whether the I2C module is enabled:

```
lsmod | grep i2c
```

If I2C module has been enabled, the following content will show up (the numbers showing in your device may be different):

```
File Edit Tabs Help
pi@raspberrypi:~ $ lsmod | grep i2c
i2c_dev                16384  0
i2c_mux_pinctrl        12288  0
i2c_mux                12288  1 i2c_mux_pinctrl
i2c_brcmstb            12288  0
i2c_bcm2835            16384  1
pi@raspberrypi:~ $
```

### Install I2C-Tools

Type the command to install I2C-Tools.

```
sudo apt-get install i2c-tools
```

### Install python-smbus

Python-smbus is a module of the program Python, which contains some classes and methods to operate I2C.

Type the following command to install python-smbus:

```
sudo apt-get install python3-smbus
```

### Additional supplement

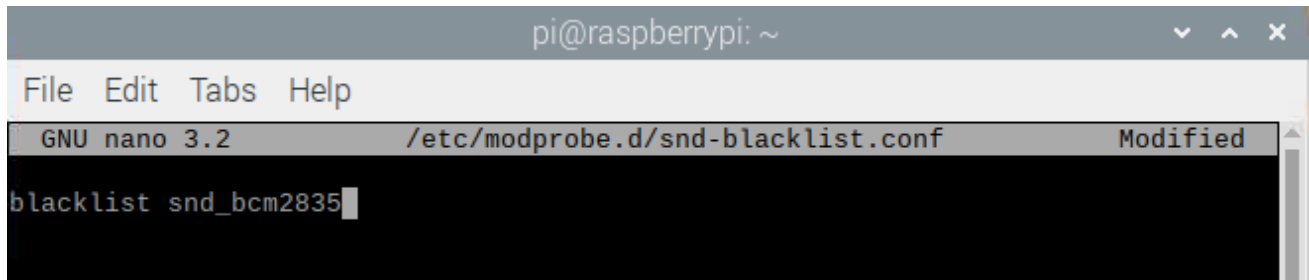
Raspberry Pi, other than 5, 4B and 400, needs to disable the audio module, otherwise the LED will not work properly.

1. Create a new snd-blacklist.conf and open it for editing

```
sudo nano /etc/modprobe.d/snd-blacklist.conf
```

Add following content: After adding the contents, you need to press Ctrl+O, Enter, Ctrl+Z.

```
blacklist snd_bcm2835
```



```
pi@raspberrypi: ~  
File Edit Tabs Help  
GNU nano 3.2 /etc/modprobe.d/snd-blacklist.conf Modified  
blacklist snd_bcm2835
```

2. We also need to edit config file.

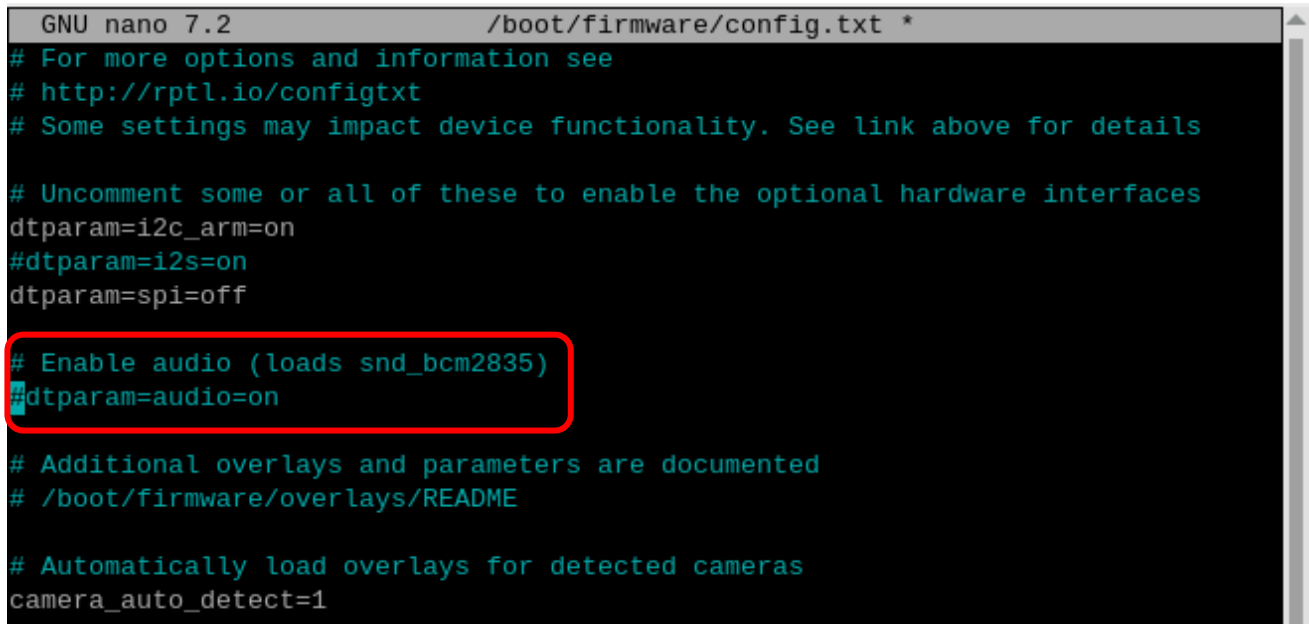
```
sudo nano /boot/firmware/config.txt
```

Find the contents of the following two lines (with Ctrl + W you can search):

```
# Enable audio (loads snd_bcm2835)  
dtparam=audio=on
```

Add # to comment out the second line. Press Ctrl+O, Enter, Ctrl+X.

```
# Enable audio (loads snd_bcm2835)  
# dtparam=audio=on
```



```
GNU nano 7.2 /boot/firmware/config.txt *  
# For more options and information see  
# http://rptl.io/configtxt  
# Some settings may impact device functionality. See link above for details  
  
# Uncomment some or all of these to enable the optional hardware interfaces  
dtparam=i2c_arm=on  
#dtparam=i2s=on  
dtparam=spi=off  
  
# Enable audio (loads snd_bcm2835)  
#dtparam=audio=on  
  
# Additional overlays and parameters are documented  
# /boot/firmware/overlays/README  
  
# Automatically load overlays for detected cameras  
camera_auto_detect=1
```

It will take effect after restarting, and you can restart after executing the next section.

If you want to restart the audio module, just restore the content modified in the above two steps.

## Step 3 Execute the Installed Program

- Execute following commands to enter directory of “setup.py”.

```
cd ~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code
```

- Run setup.py

```
sudo python setup.py
```

This program will automatically install the rpi\_ws281x, PyQt5 library, etc. Please **reboot** the Raspberry Pi after the installation is completed, as shown below.

If it is your first time to run the script, it will ask you to input the camera type. At this point, input the content according to the camera you have. (The one included in this kit is of the type ov5647).

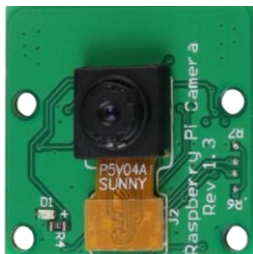
```
rpi_ws281x 4.3.4 is already the active version in easy-install.py

Installed /usr/local/lib/python3.11/dist-packages/rpi_ws281x-4.3.
Processing dependencies for rpi_ws281x==4.3.4
Finished processing dependencies for rpi_ws281x==4.3.4

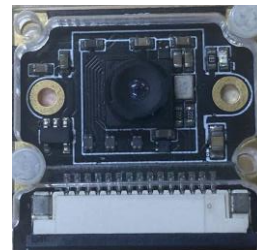
All libraries have been installed successfully.
Getting Raspberry Pi version...
Detected Raspberry Pi 5
Backing up /boot/firmware/config.txt.bak
Backup of /boot/firmware/config.txt created at /boot/firmware/con
Updated /boot/firmware/config.txt with 'dtparam=spi=on'
Updated /boot/firmware/config.txt with 'camera_auto_detect=0'

Enter the camera model (e.g., ov5647 or imx219):
```

OV5647



Imx219



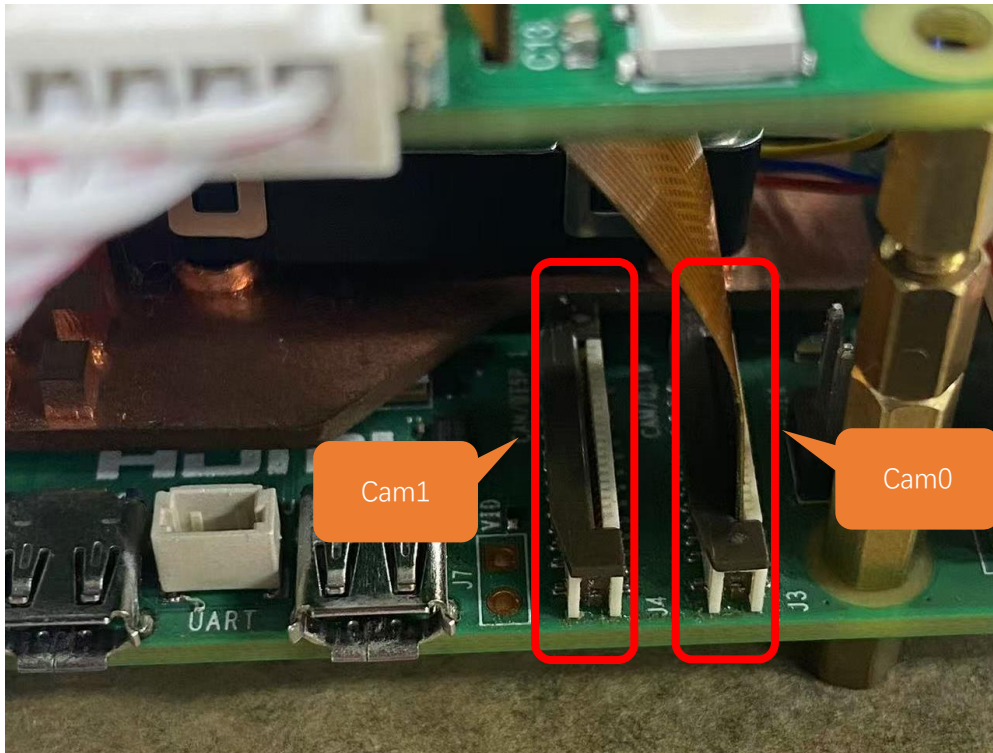
If your Rpi is a Pi 5, it will ask you which camera interface to use. You can type in **cam0** or **cam1** based on the interface you want to use.

**Note:** Make sure you connect the camera cable to the corresponding interface.

If it is not Raspberry Pi, this setting is not needed, as there is only camera interface available.

```
Backup of /boot/firmware/config.txt created at /boot/firmware/config.txt.bak
Updated /boot/firmware/config.txt with 'dtparam=spi=on'
Updated /boot/firmware/config.txt with 'camera_auto_detect=0'

Enter the camera model (e.g., ov5647 or imx219): ov5647
Setting up for Raspberry Pi 5
You have a Raspberry Pi 5. Which camera port is the camera connected to? cam0 or cam1:
```



Reboot your Raspberry Pi after installing the libraries.

```
sudo reboot
```

```
You have a Raspberry Pi 5. Which camera port is the camera connected to? cam0 or cam1: cam0
Updated /boot/firmware/config.txt with 'dtoverlay=ov5647,cam0'
Please reboot your Raspberry Pi to complete the installation.
pi@raspberrypi:~/Freenove_Tank_Robot_Kit_for_Raspberry_Pi/Code $ sudo reboot
```

**The reboot of the pi takes some time, please wait with patience.**

If the installation fails, please rerun setup.py. After the installation is completed, restart the Raspberry Pi. Most installation failures are caused by network reasons.

```
sudo python setup.py
```

## Chapter 2 Assembly of the Robot (Important)

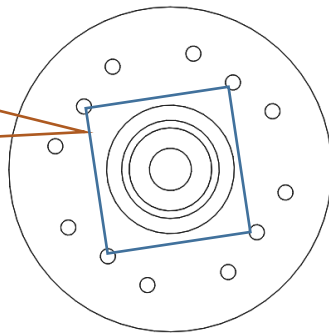
You can watch this video while reading this chapter: <https://youtu.be/l4R5f8EvvPo>

We will help you with any concerns via [support@freenove.com](mailto:support@freenove.com)

### Step 1 Installing Servo Base

Take out the disc servo arms from the servo package.

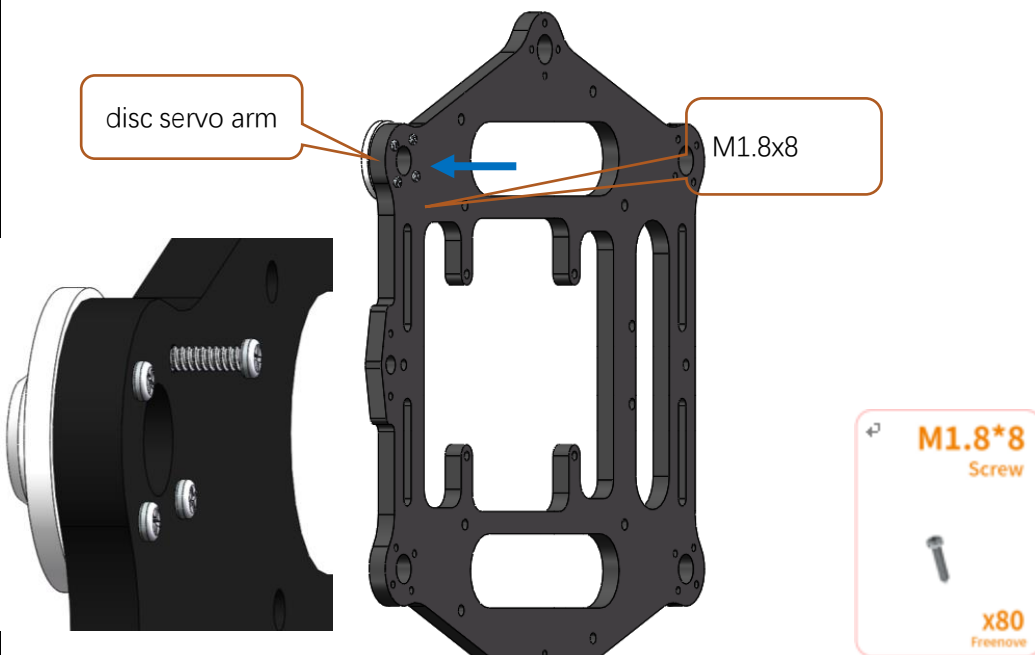
Use the middle hole of each group of holes



Fix each disc servo arm to the joints.

disc servo arm

M1.8x8



M1.8x8

disc servo arm



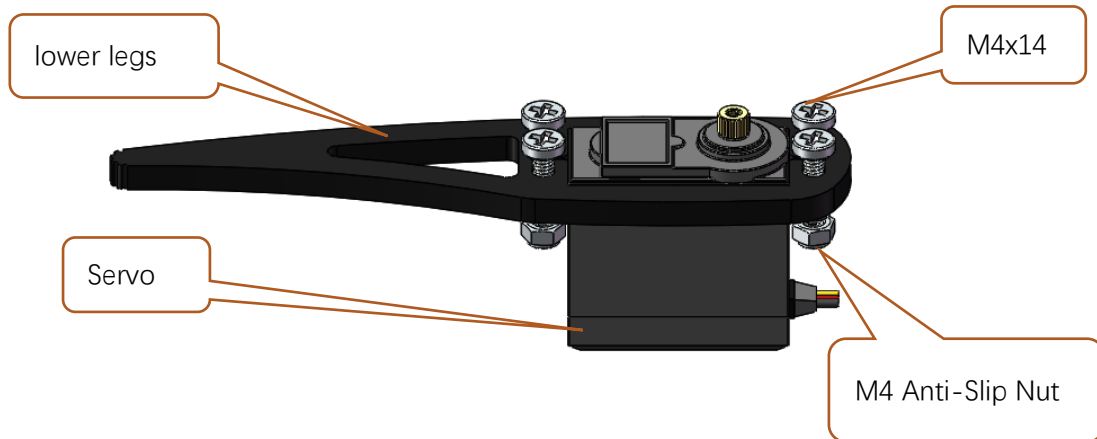
After all disc servo arms are installed successfully, it should look as follows:



## Step 2 Installing Servos

### Installing Servos on the Legs

Install servos on the lower legs

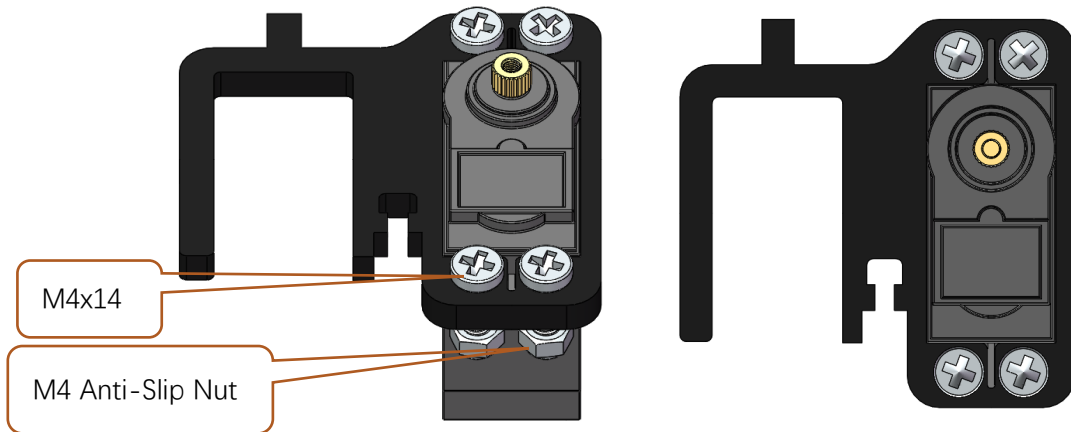


The other 5 leg parts are installed in the same way.

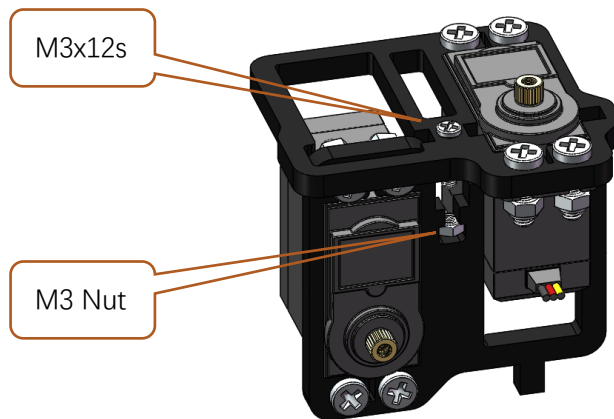
Note: The robot has six legs and its left lower legs and right lower legs are in different directions.



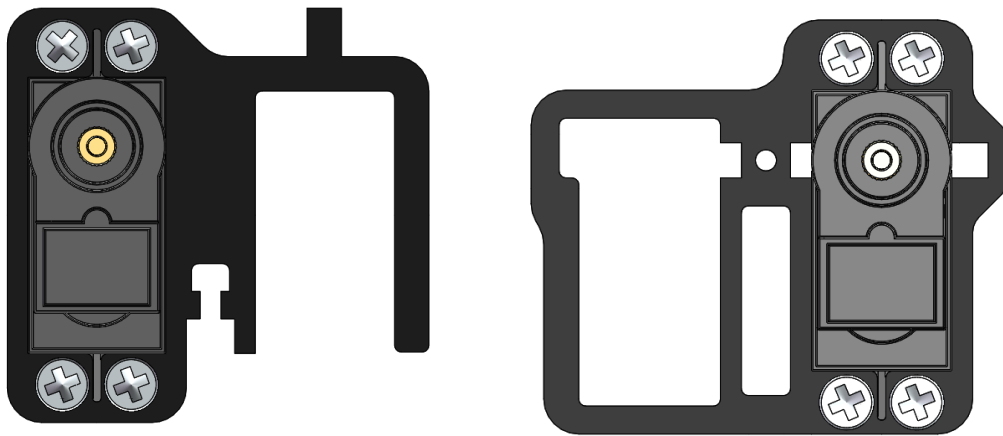
Installation of servos on the hip joints of the three left legs.



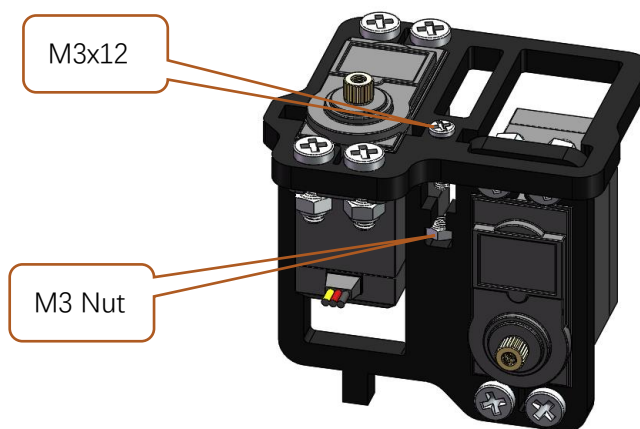
Assembly



Installation of servos on the hip joints of the three right legs.



Assembly

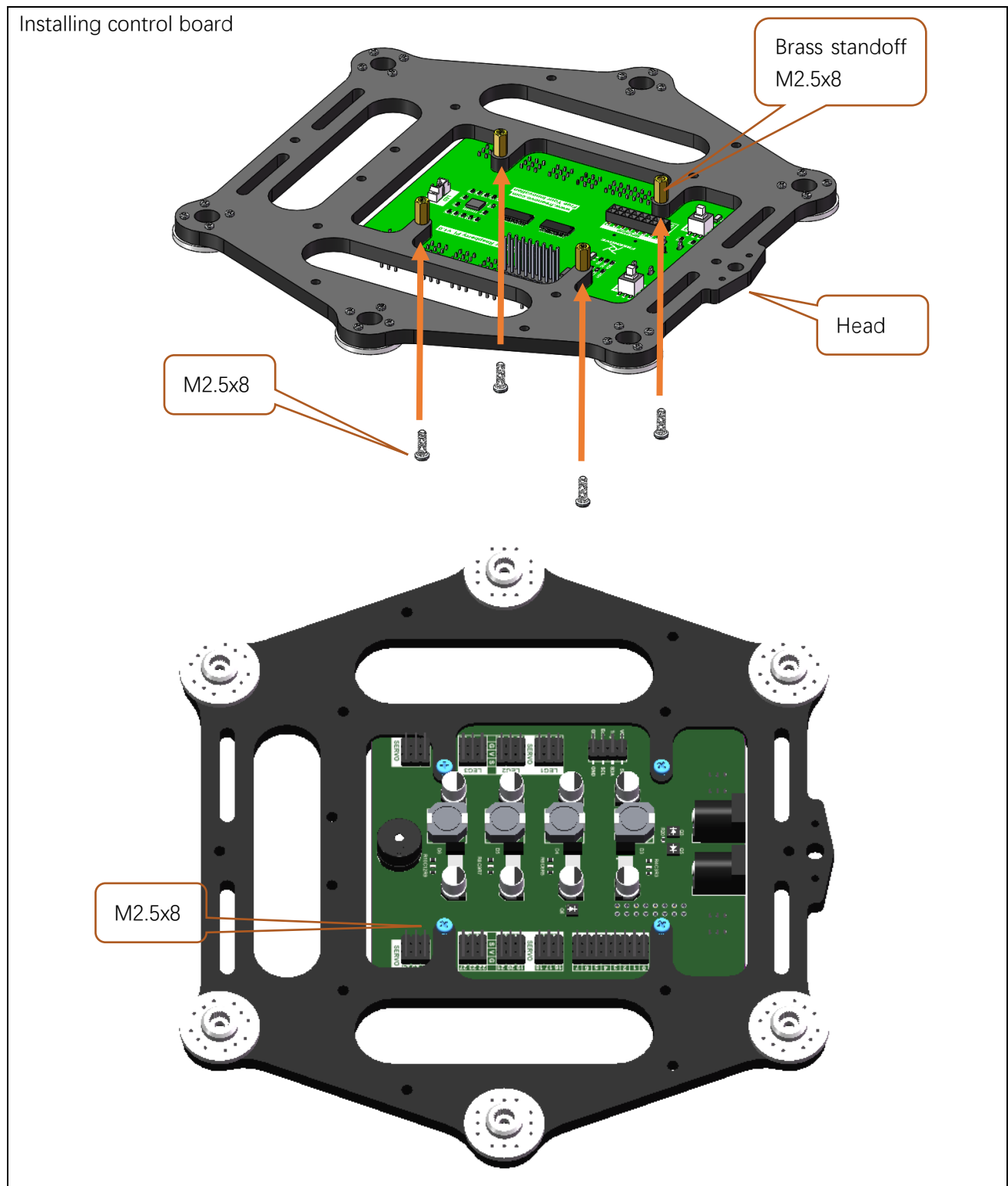


Servos at left and right hip joints (Note: the direction of servos on each side is different.)

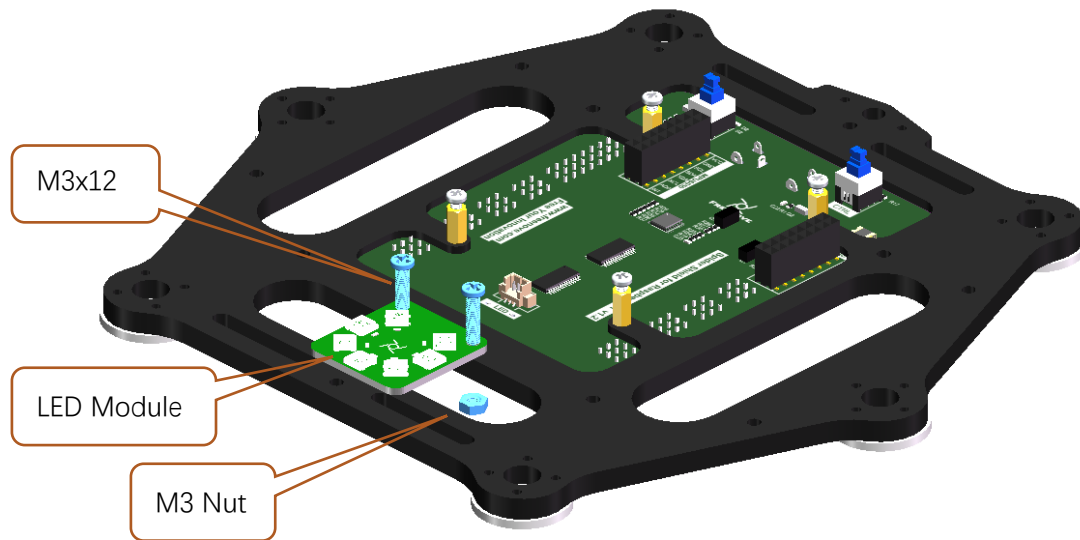


## Step 3 Assembling the center of the body

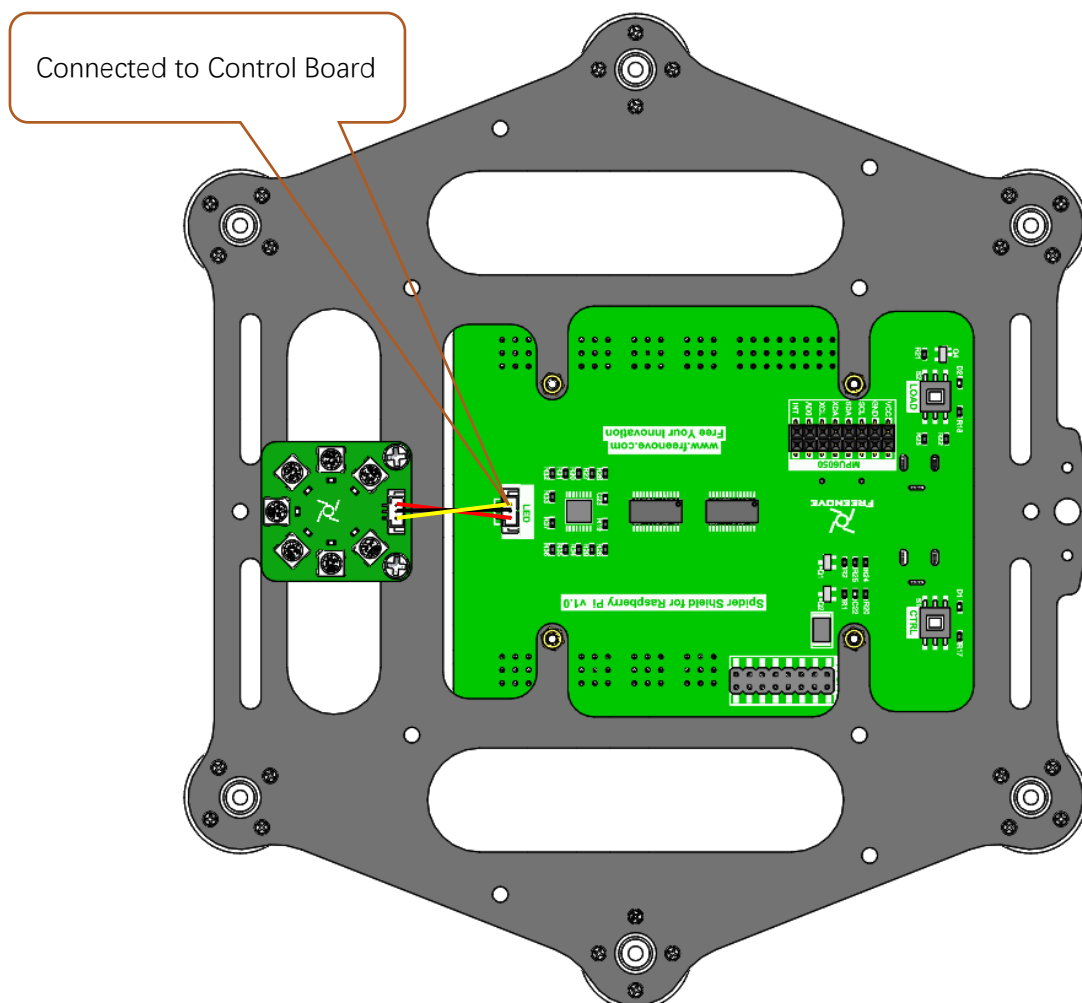
### Installing Control Board



## Installing LED Module

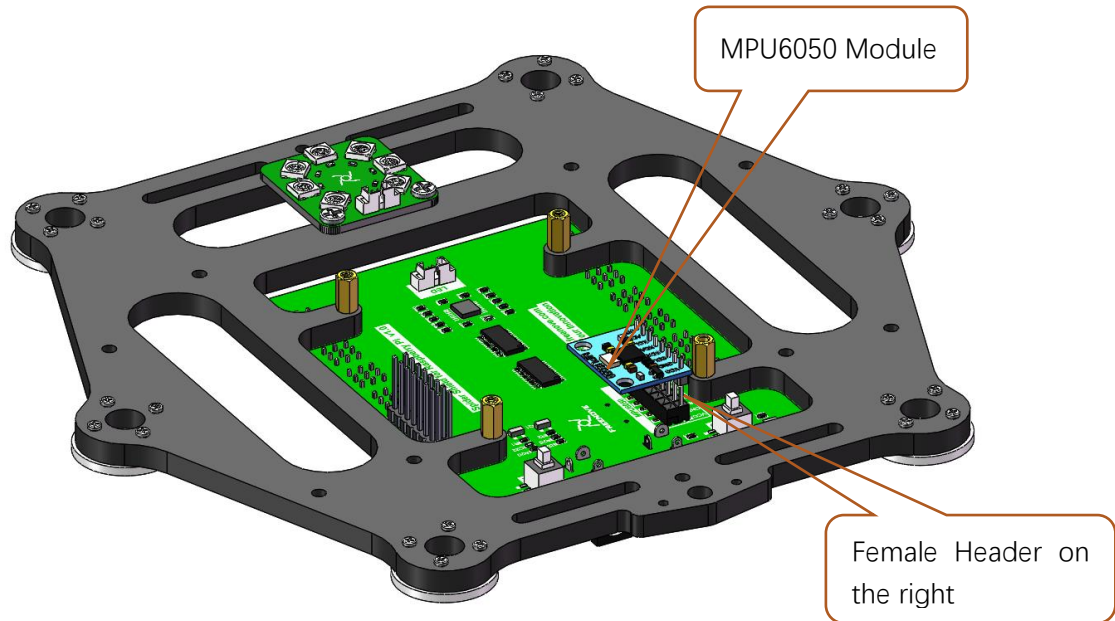


### Wiring

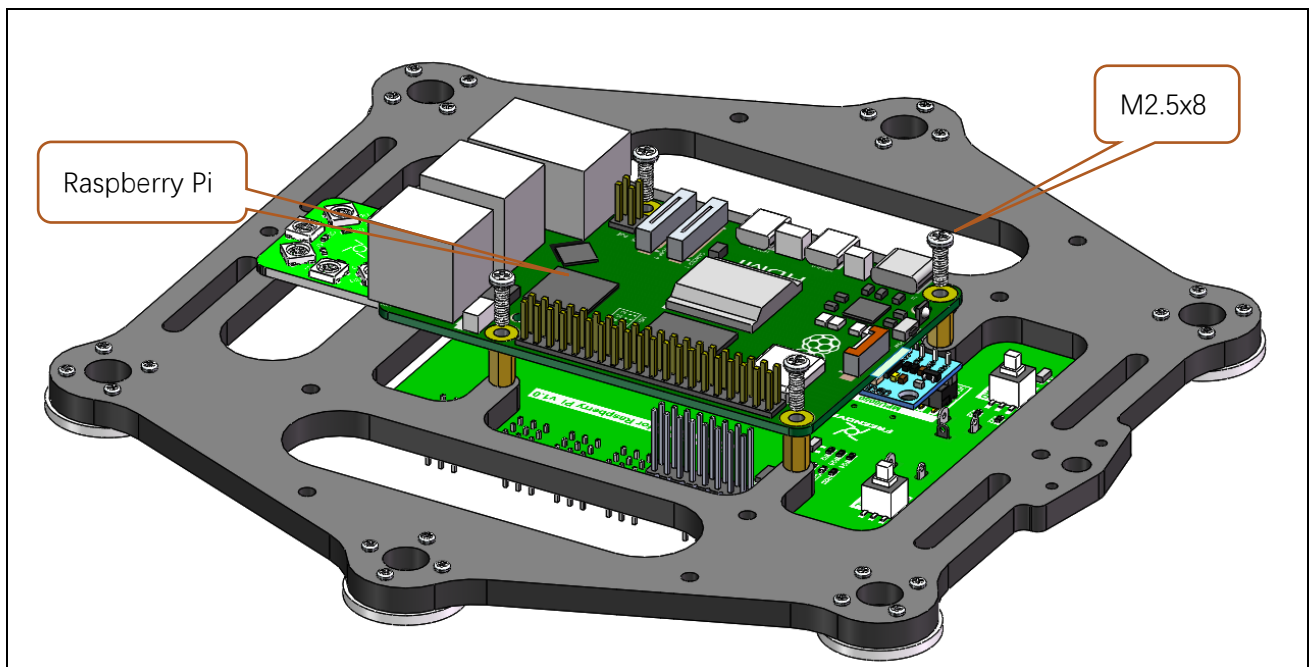


## Installing MPU6050 Module

Note: There are two rows of female header, please plug the module into the right.

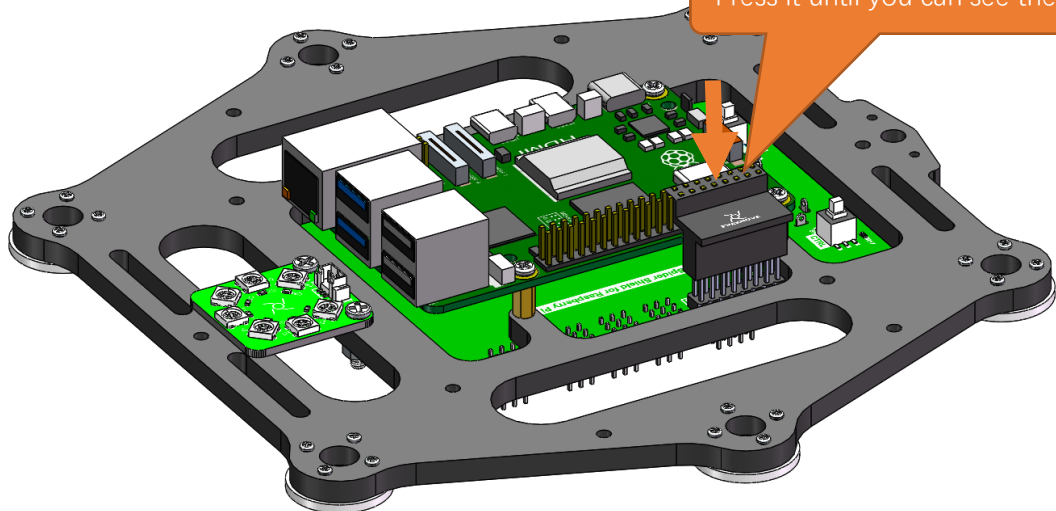


## Installing Raspberry Pi



## Installing Connector

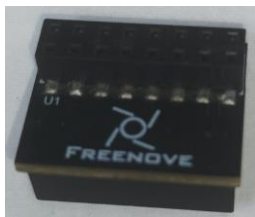
Install the connection board



We have two types of connection boards, but they are installed in the same way.

See [here](#) for more information.

Connection board (PCB\_V1.0)



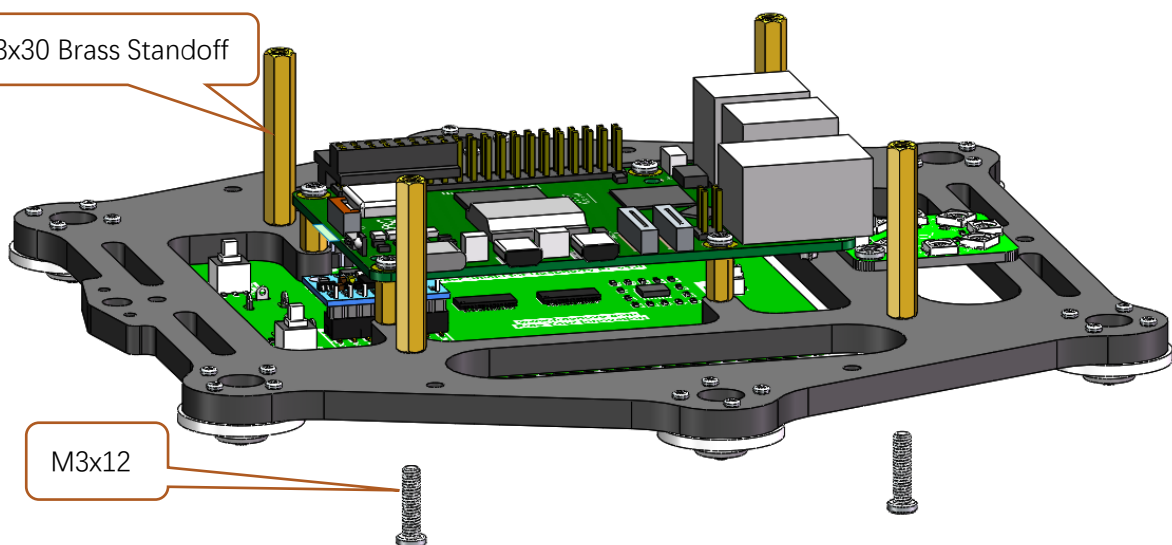
Connection board (PCB\_V2.0)



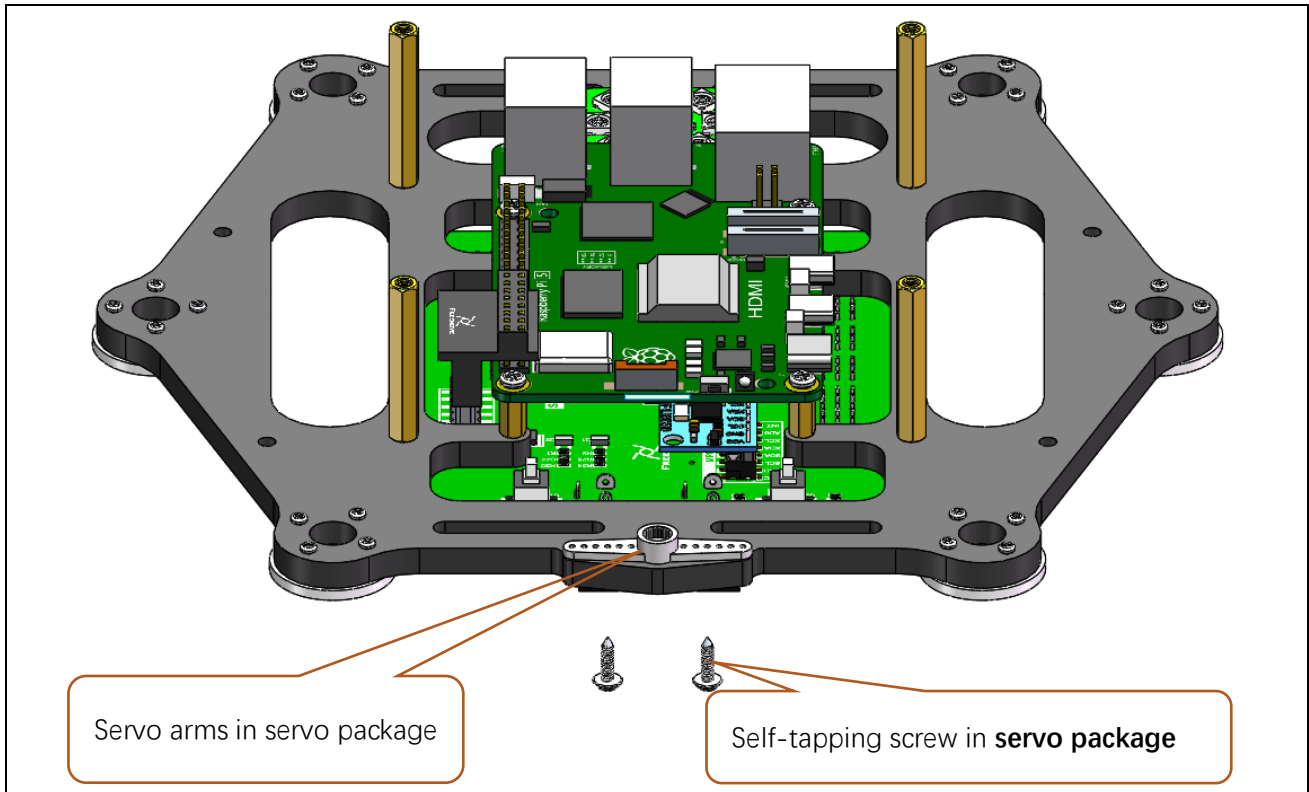
## Installing Brass Standoff

M3x30 Brass Standoff

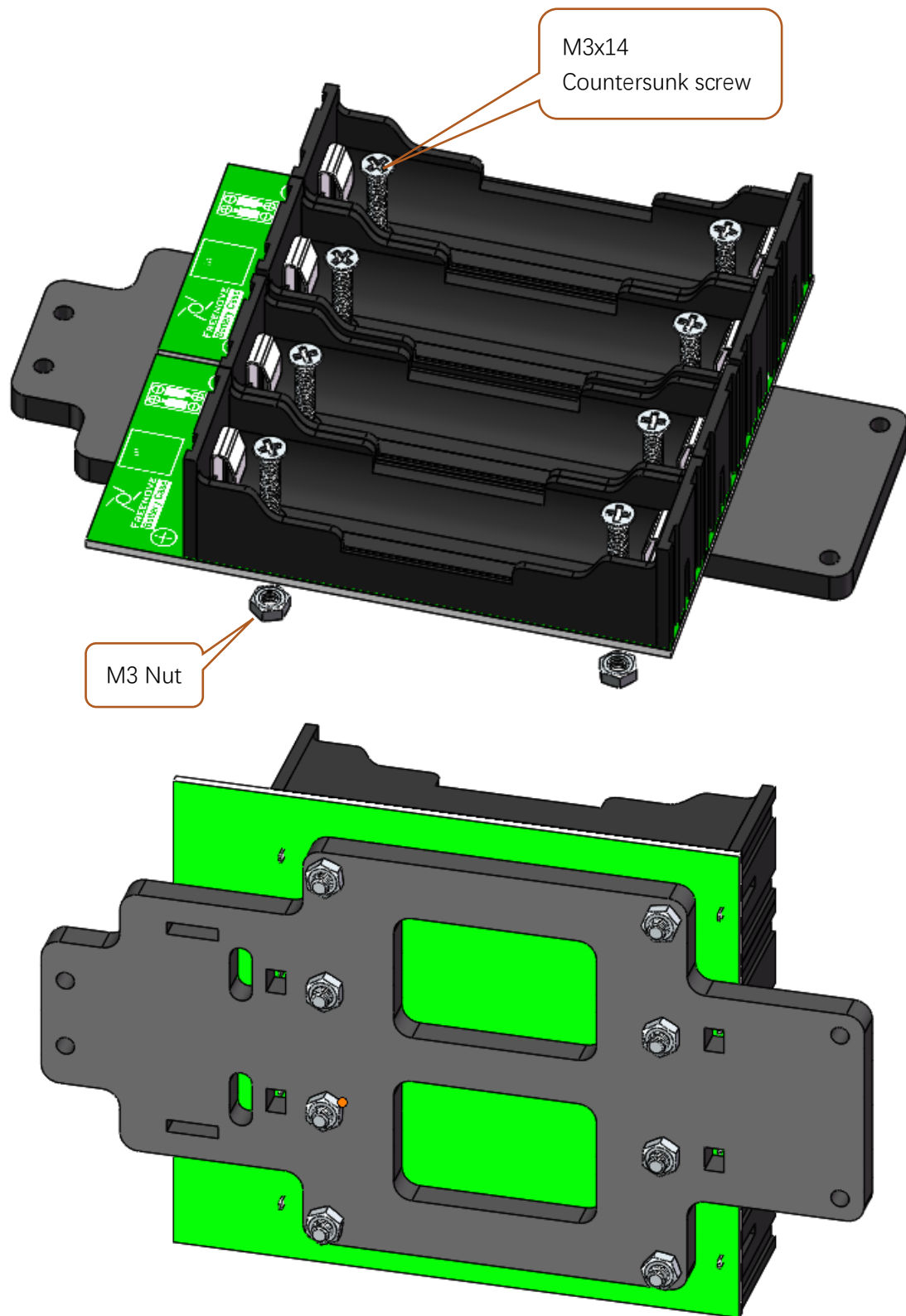
M3x12

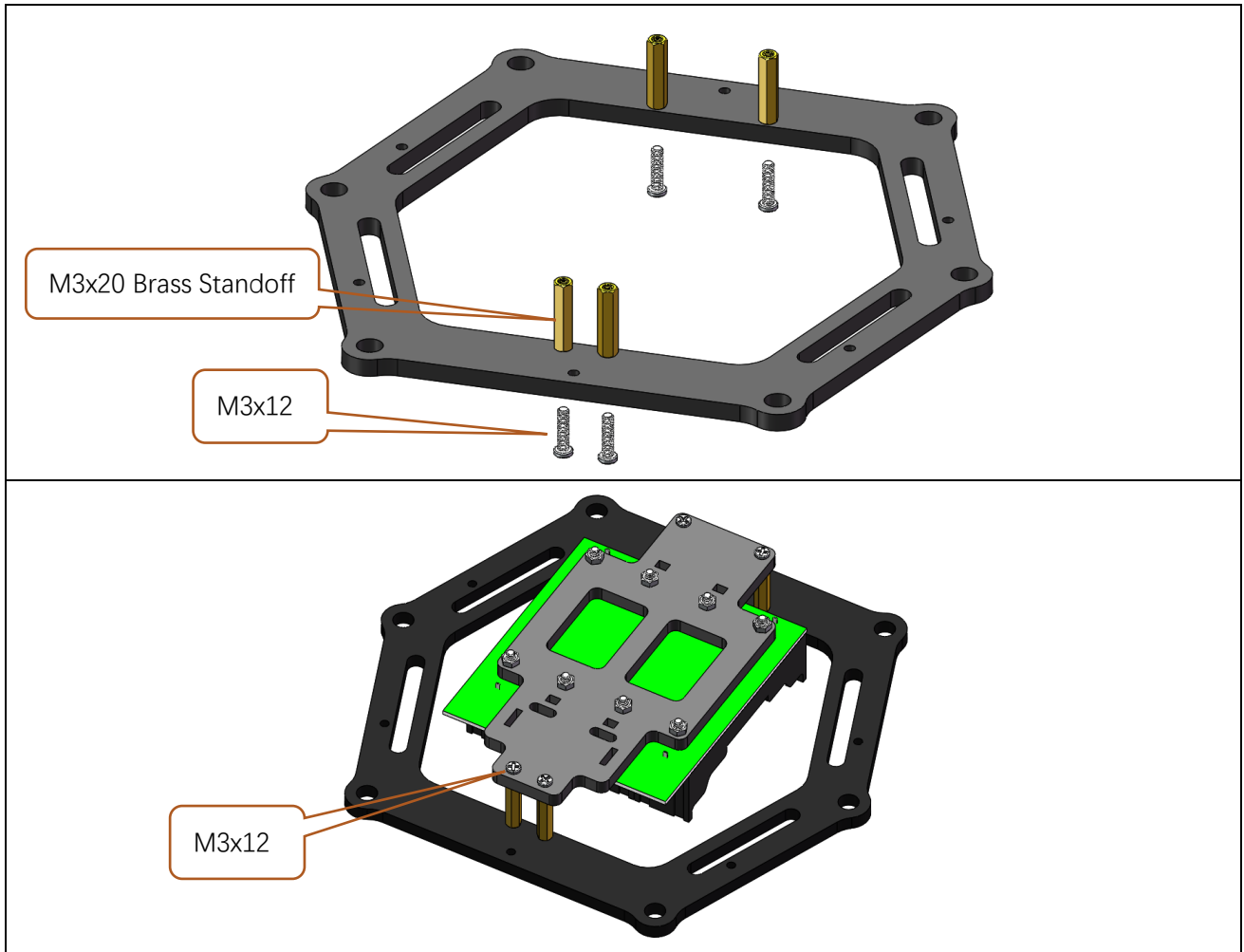


## Installing Servo Arms



## Step 4 Installing Battery Holder

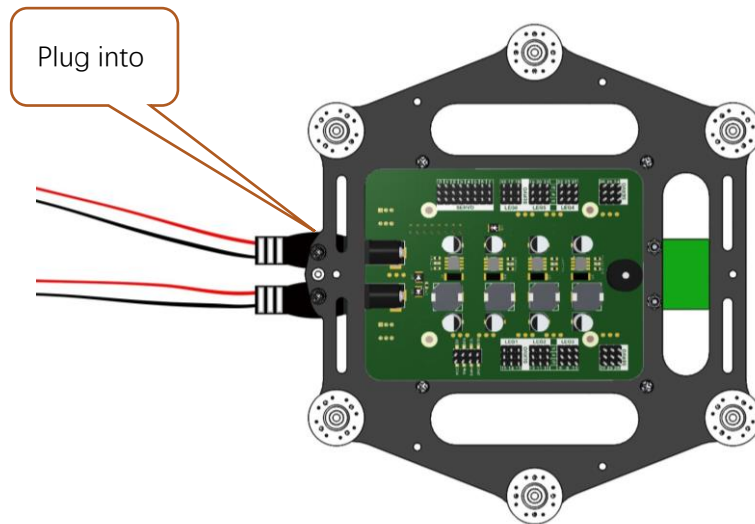




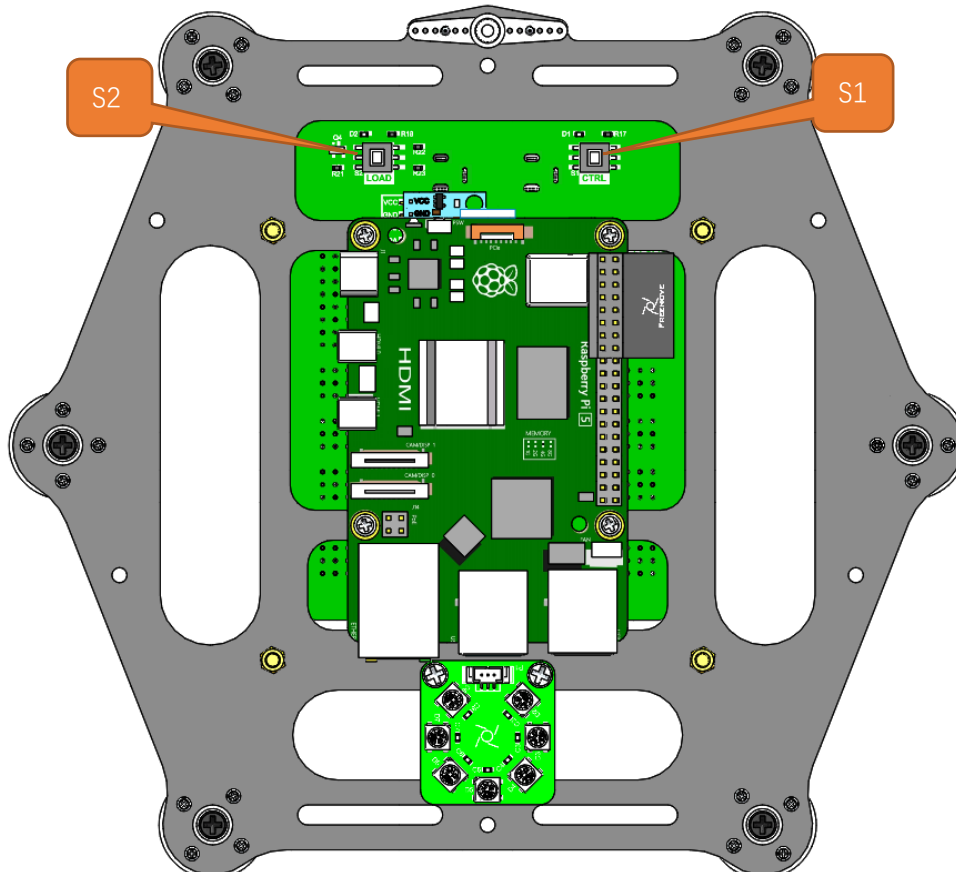
## Step 5 Assembling Legs and Head to the Body

### Run the servo installation program

Before installing, fix the servos at the specified angle. At this point you need to run the servo program. Plug batteries into the battery holder. And then plug the battery holder to control board.



Press S1 and S2.



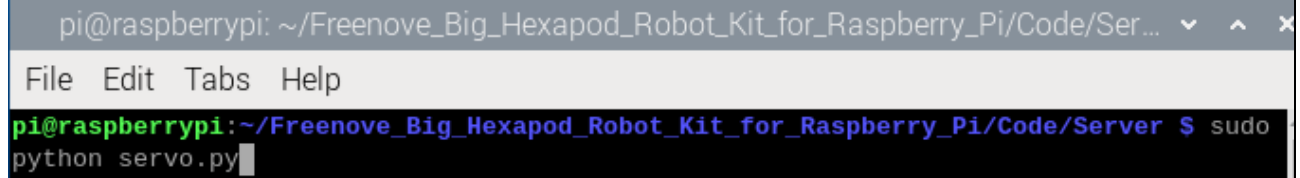
Open Raspberry Pi. (All contents in Chapter 1 should be executed correctly first. If not, please return Chapter 1 to finish it first.)

Enter the following command to enter the directory where the servo program locates.

```
cd ~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server
```

b. Execute the following command to run "Servo.py".

```
sudo python servo.py
```



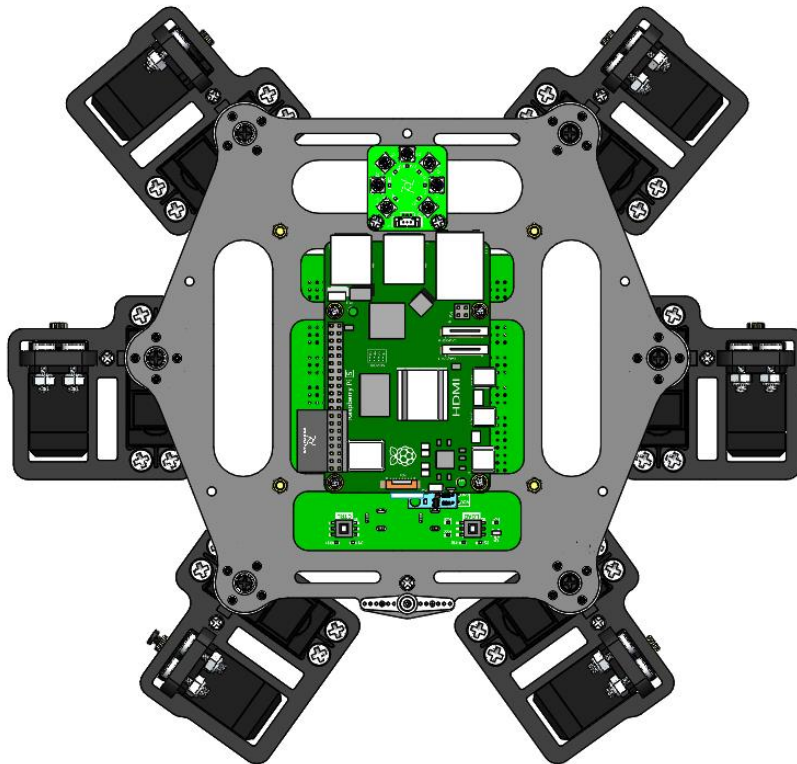
Next, fix each servo to the corresponding disc servo arm. **Do NOT pull the servo wires** during the installation process.

## Installing Hip Joint

Connect the servo wires to corresponding servo port.

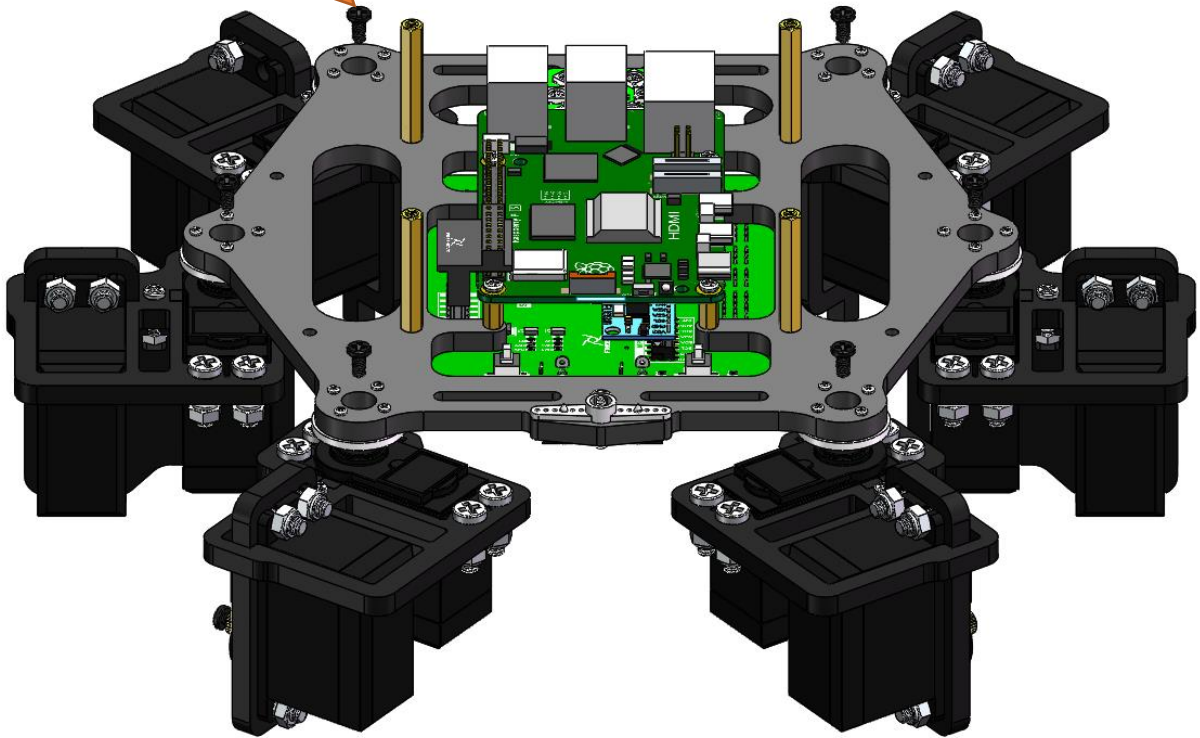


Bottom



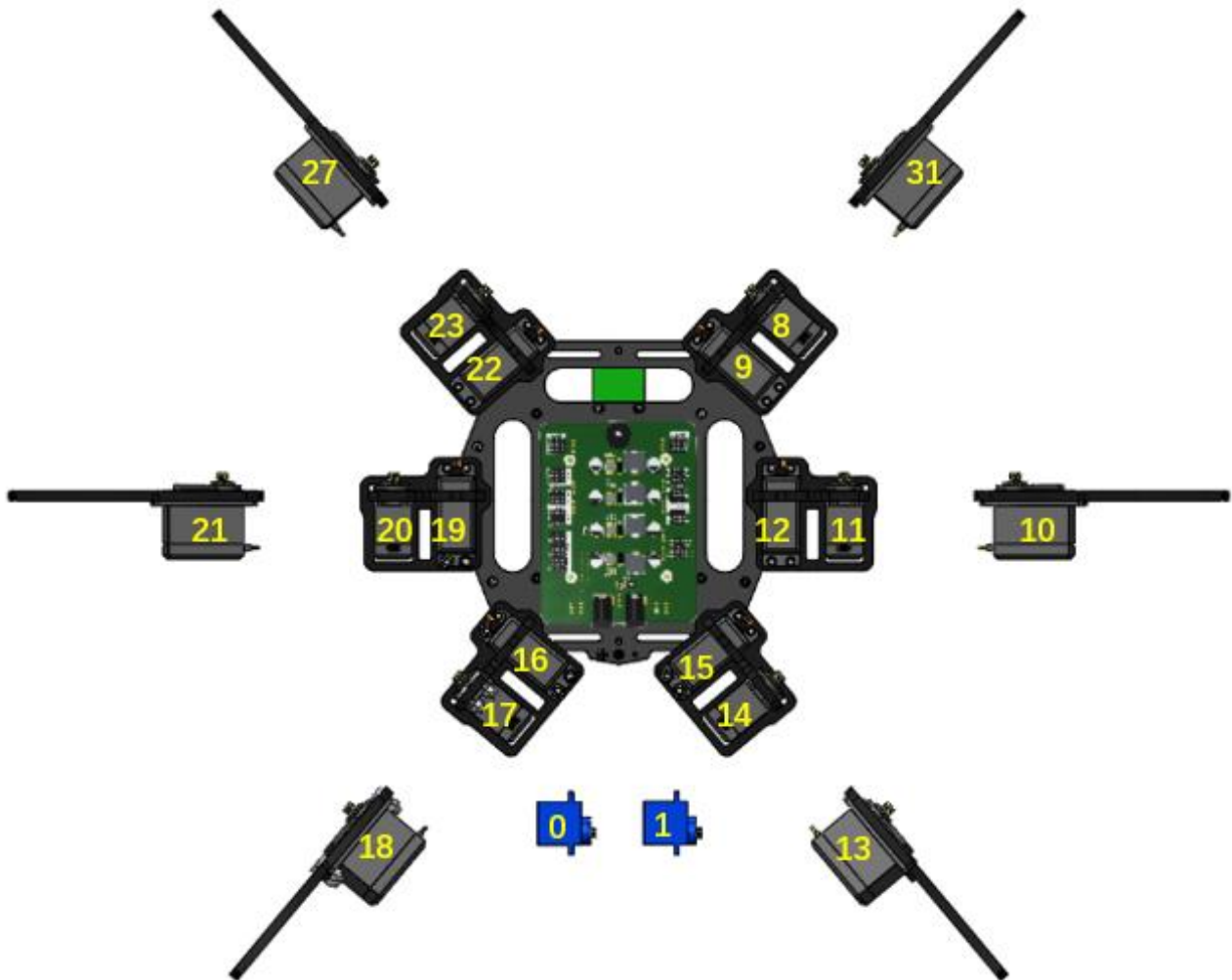
Top

Black screws in the servo package



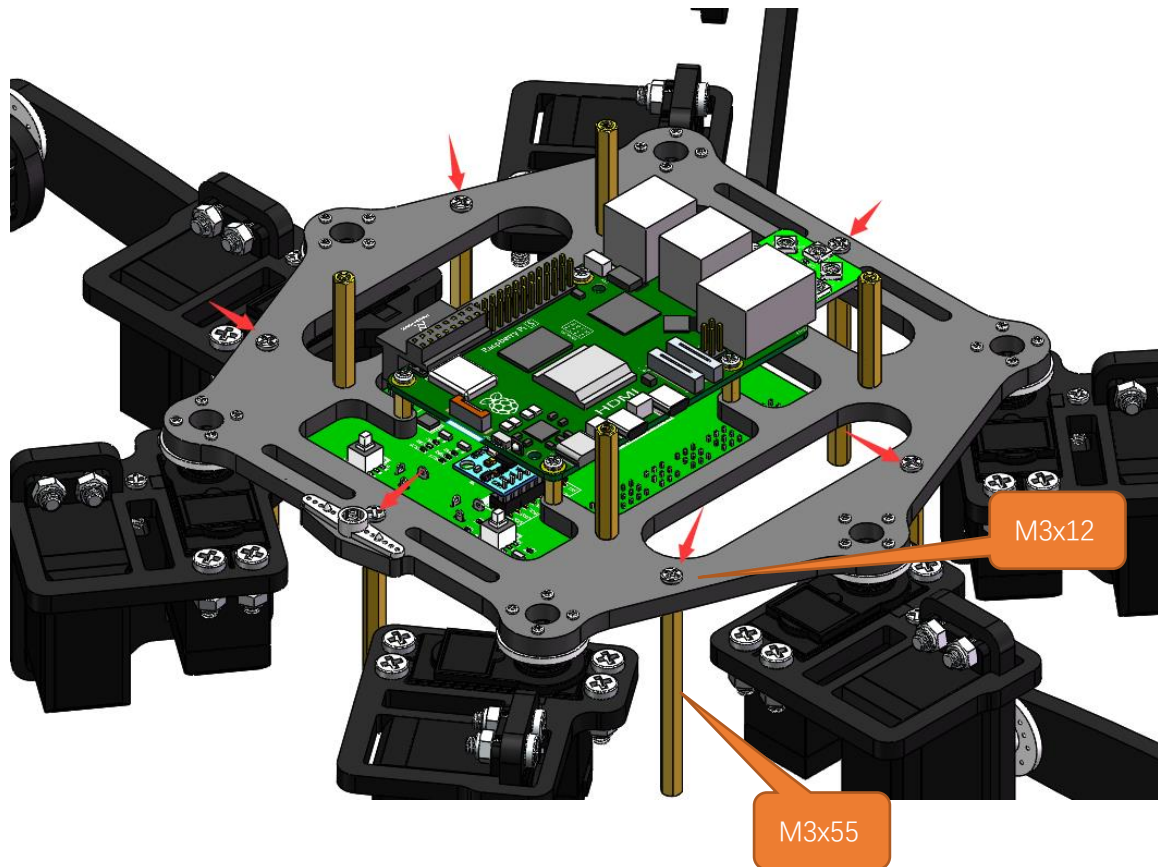
## Installing Lower Leg and the Head

Connect the servo wires to corresponding servo port.

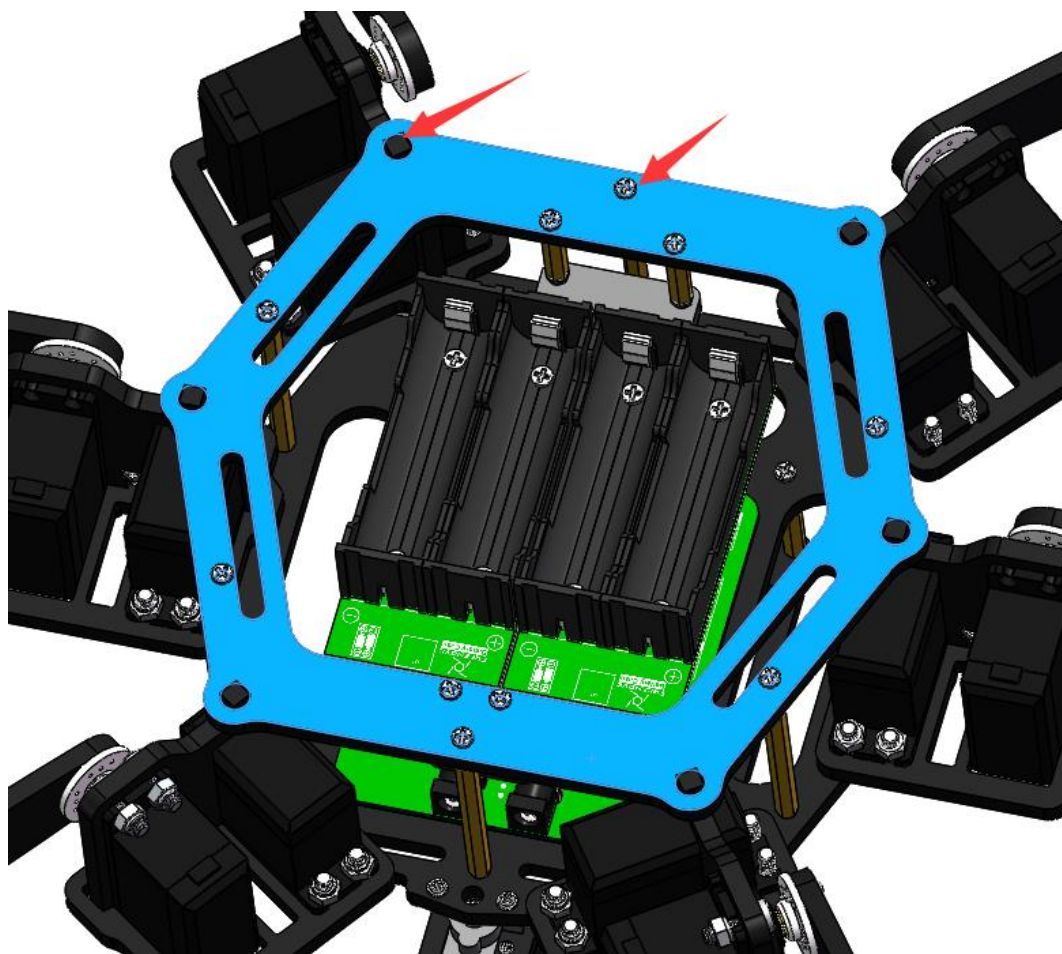
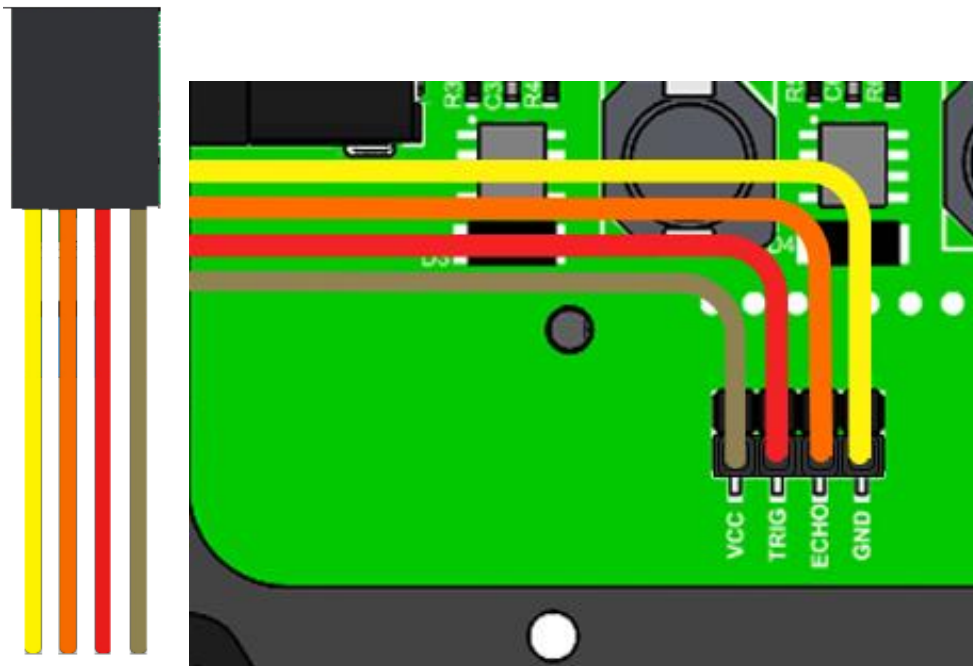




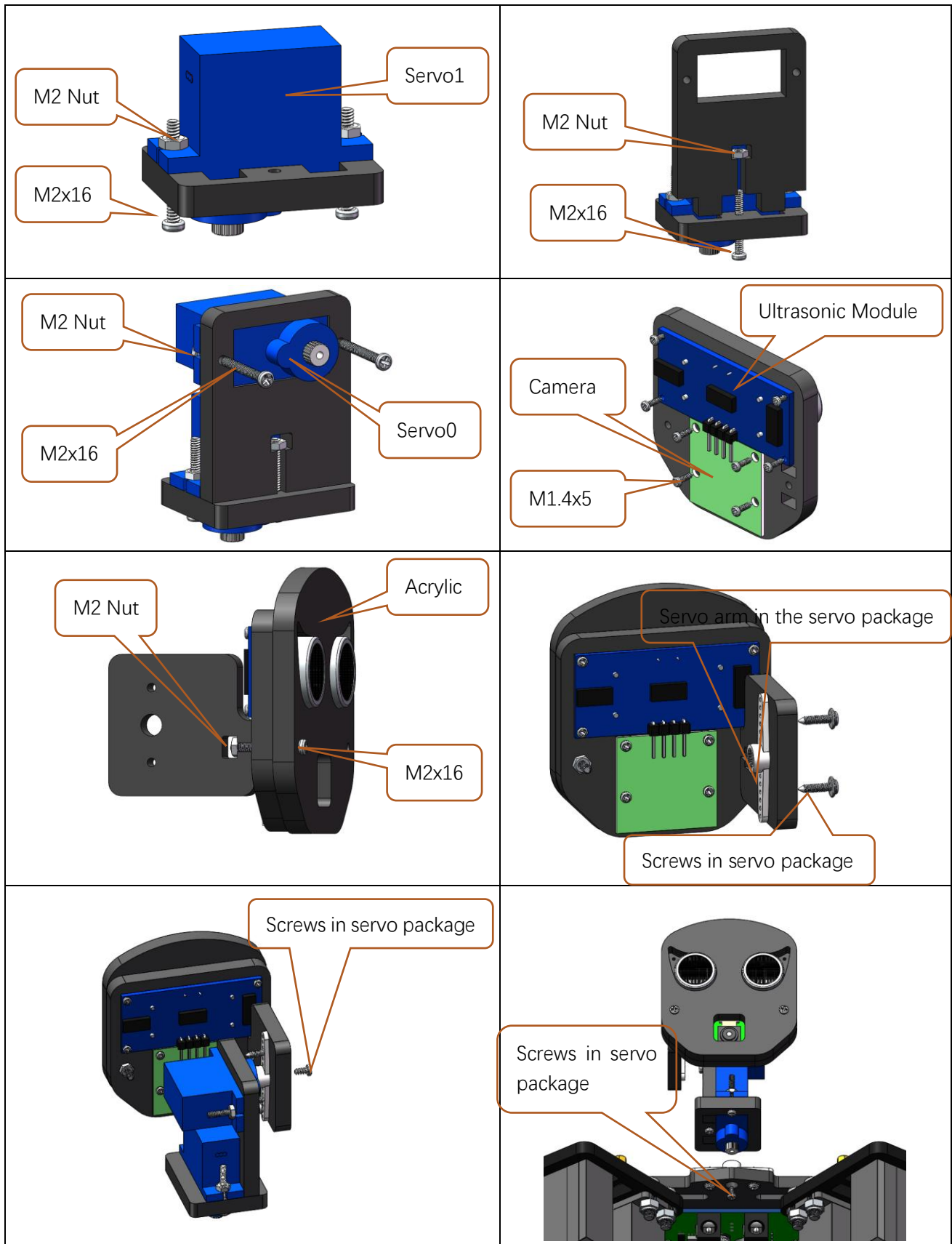
Assemble 6 standoff



Connect a 4P cable which will be used later.

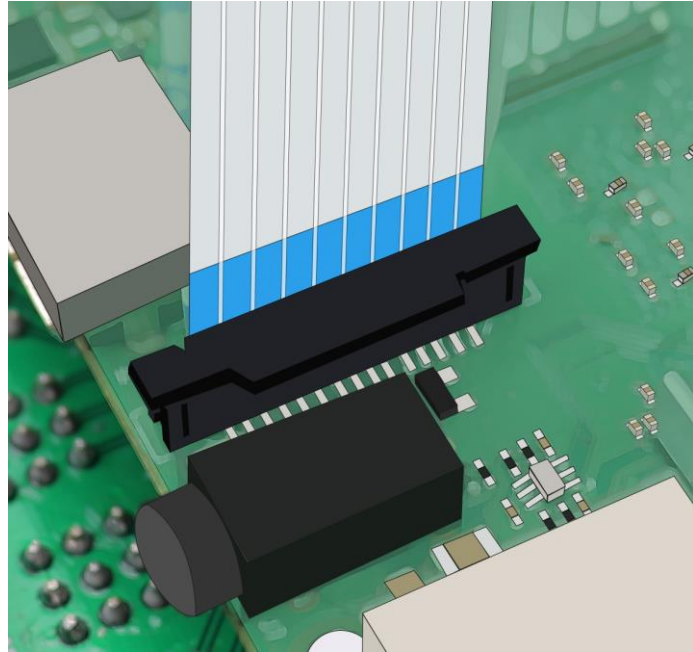
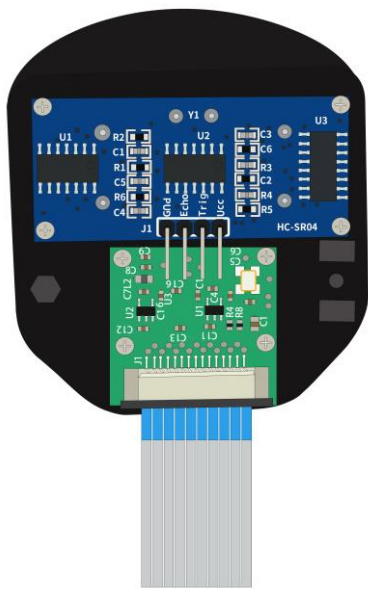


## Installing head



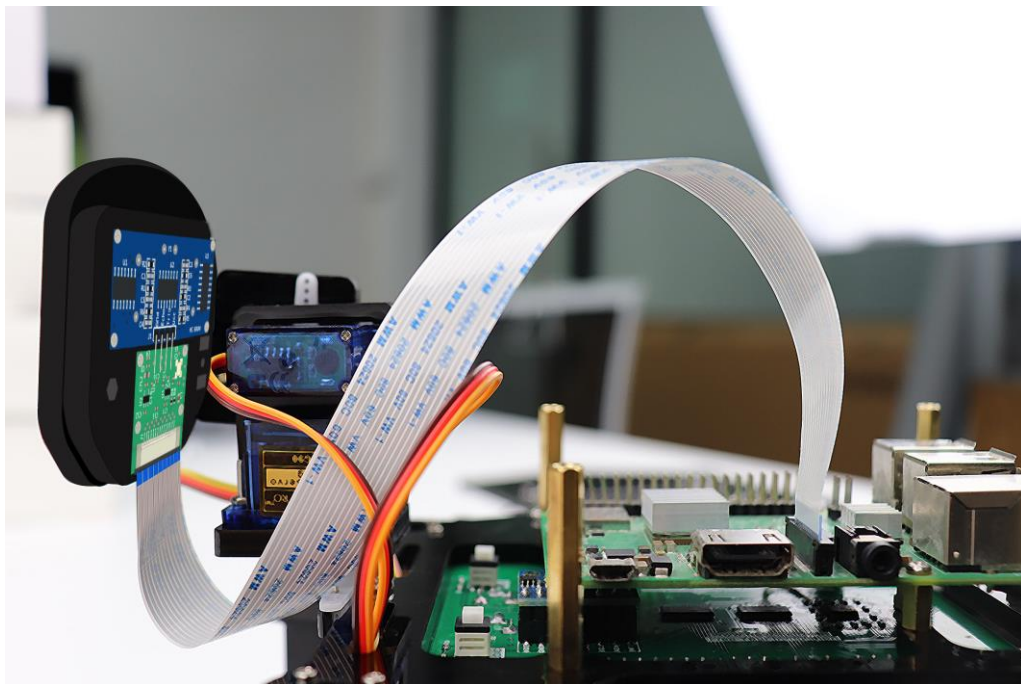
## Connecting Wires of Camera and Ultrasonic Module

Connect camera wire



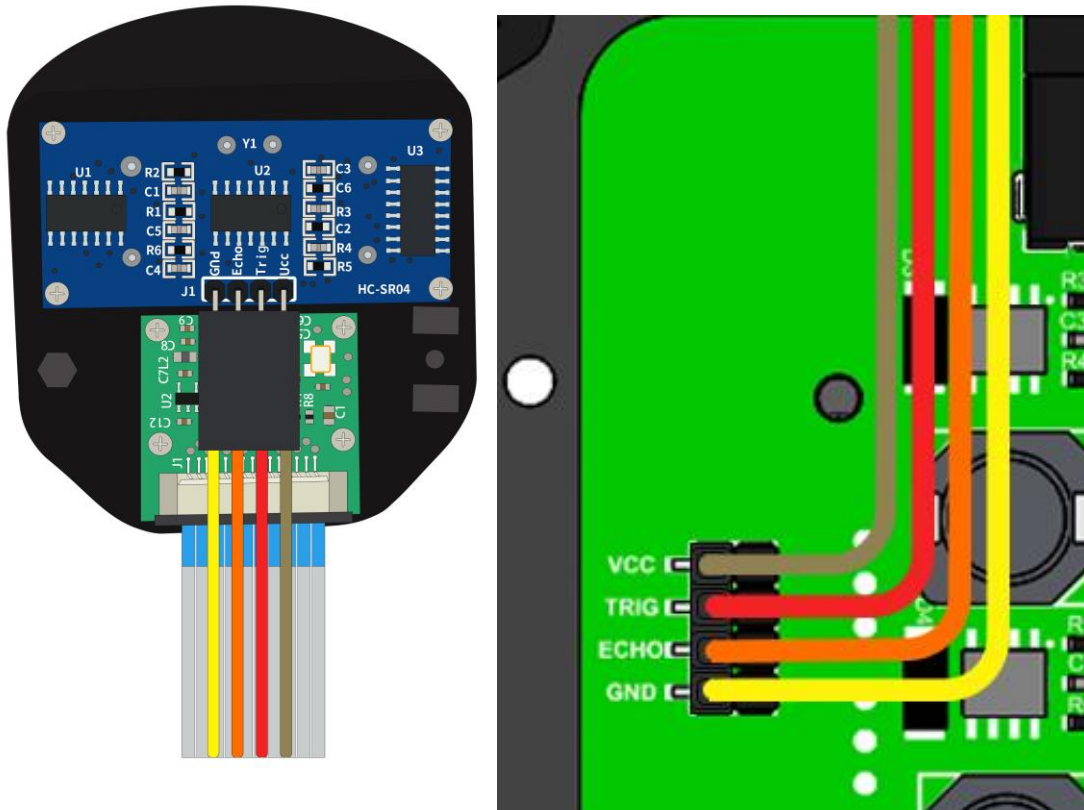
You need power off Raspberry Pi when wire camera.

After connected, it should be as follows

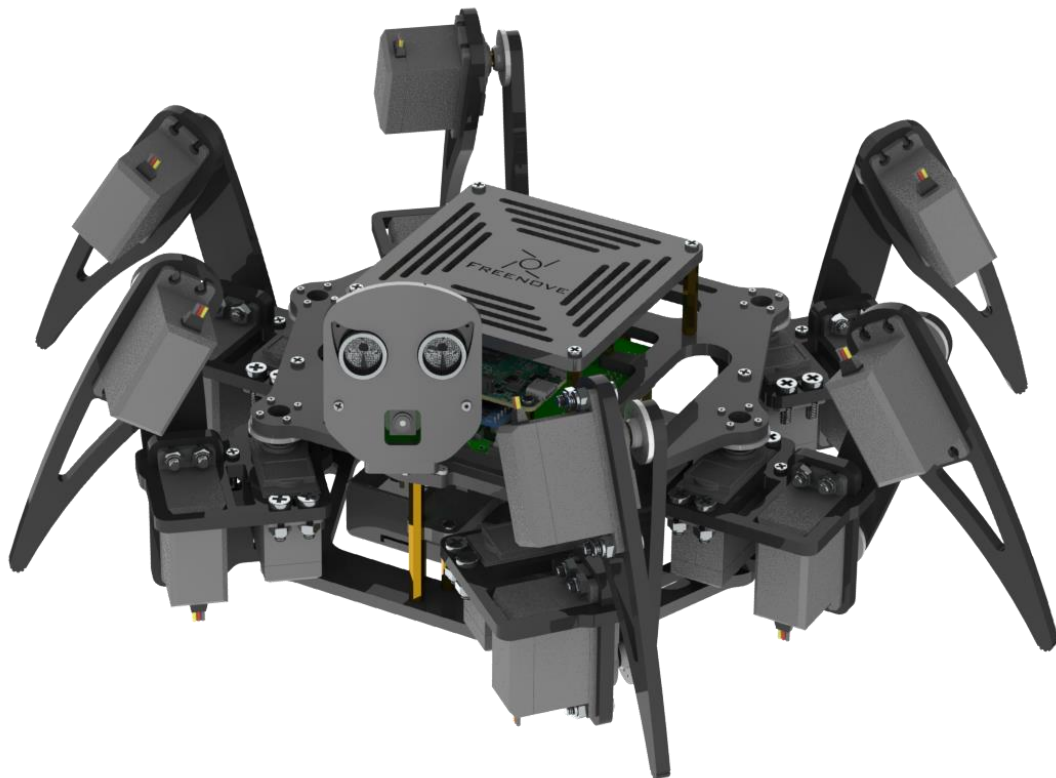
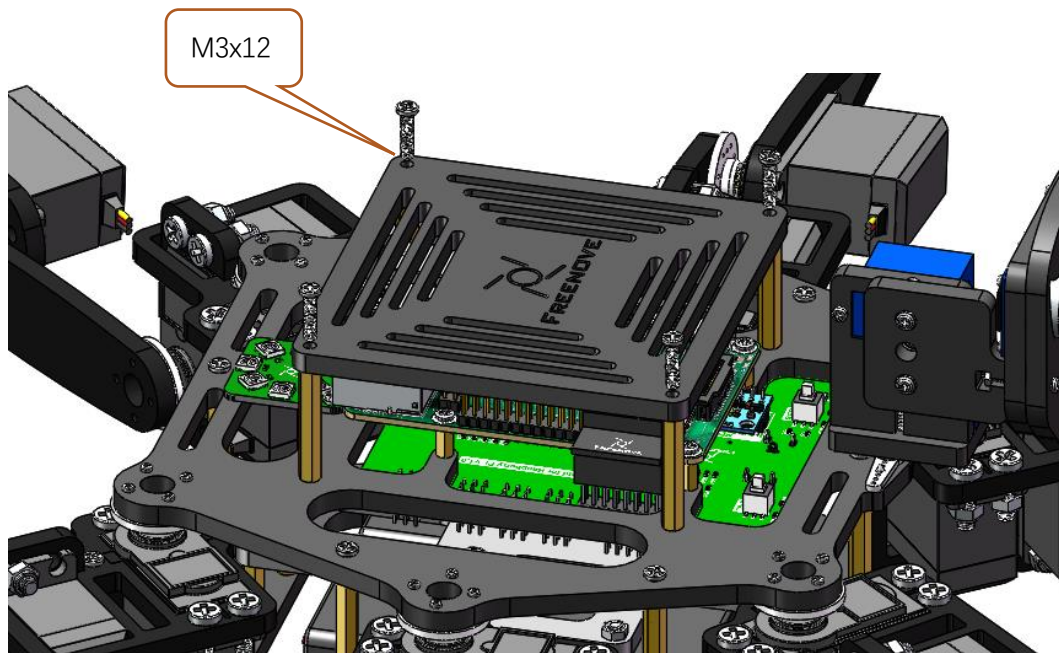


### Connecting Ultrasonic Module

Note: VCC, Echo, Trig, GND are connected to VCC, Echo, Trig and GND on the connection board respectively. Do NOT connect them wrongly. Otherwise, it may burn out the Ultrasonic module.



## Step 6 Installing Base and Cover Plates



Congratulations! You have finished the assembly of the robot.

## Chapter 3 Module Test (necessary)

In the previous chapter, we have assembled the robot. In this chapter, we will test all modules of the robot to see whether they work.

Before starting, please plug in the batteries, and switch ON S1 and S2. Place the assembled robot on a horizontal table and make sure that the wiring of each servo is correct.

If you have never learned python before, you can learn some basic knowledge via the link below:  
<https://python.swaroopch.com/basics.html>

You can watch this video while reading this chapter:<https://youtu.be/D00LdVqXC5E>

We will help you with any concerns via [support@freenove.com](mailto:support@freenove.com)

### Servo

Enter the following command to check I2C.

```
i2cdetect -y 1
```

As shown in the following illustration, the address of two PCA9685 chips are 0x40 and 0x41 respectively, the address of ADS7830 chip is 0x48 and the address of MPU6050 module is 0x68.

```

pi@raspberrypi:~ $ i2cdetect -y 1
    0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
00:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
10:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
20:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
30:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
40: 40 41 -- -- -- -- -- 48 -- -- -- -- -- -- --
50:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
60:  --  --  --  --  --  --  -- 68 -- -- -- -- -- --

```

### Run program

Enter following commands in the terminal to test servos.

If the terminal displays the directory as below (where test.py is located). You can **directly** execute the test.py command.

```
pi@raspberrypi:~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server $
```

1. If not, execute the cd command:

```
cd ~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server
```

2. Execute test.py command:

```
sudo python test.py Servo
```



The code is as below:

```

1  def test_Servo():
2      from servo import Servo
3      servo = Servo()
4      try:
5          for i in range(50):
6              servo.set_servo_angle(15, 90+i)
7              servo.set_servo_angle(12, 90+i)
8              servo.set_servo_angle(9, 90+i)
9              servo.set_servo_angle(16, 90+i)
10             servo.set_servo_angle(19, 90+i)
11             servo.set_servo_angle(22, 90+i)
12             time.sleep(0.005)
13         for i in range(60):
14             servo.set_servo_angle(14, 90+i)
15             servo.set_servo_angle(11, 90+i)
16             servo.set_servo_angle(8, 90+i)
17             servo.set_servo_angle(17, 90-i)
18             servo.set_servo_angle(20, 90-i)
19             servo.set_servo_angle(23, 90-i)
20             time.sleep(0.005)
21         for i in range(120):
22             servo.set_servo_angle(13, i)
23             servo.set_servo_angle(10, i)
24             servo.set_servo_angle(31, i)
25             servo.set_servo_angle(18, 180-i)
26             servo.set_servo_angle(21, 180-i)
27             servo.set_servo_angle(27, 180-i)
28             time.sleep(0.005)
29         print ("\nEnd of program")
30         servo.relax()
31     except KeyboardInterrupt:
32         servo.relax()
33         print ("\nEnd of program")

```

## Reference

### set\_servo\_angle(channel, angle)

There are two parameters in this function to rotate the servo of specified channel to a specified angle.

The first parameter indicates the servo channel

The second parameter indicates the rotation angle

For example:

set\_servo\_angle(0,60)      Rotate servo of channel 0 to 60 degrees

set\_servo\_angle(15,160)    Rotate servo of channel 15 to 160 degrees

## ADC Module

### Run program

Enter following commands to test ADC module.

If the terminal displays the directory as below (where test.py is located). You can **directly** execute the test.py command.

```
pi@raspberrypi:~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server $
```

1.If not, execute the cd command:

```
cd ~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server
```

2.Execute test.py command:

```
sudo python test.py ADC
```

```
pi@raspberrypi: ~/Freenove_Big_Hex...ot_Kit_for_Raspberry_Pi/Code/Server  ▾  ^  x
File Edit Tabs Help
pi@raspberrypi:~ $ cd ~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server
pi@raspberrypi:~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server $ sudo
python test.py ADC
Program is starting ...
The battery voltage is (8.18, 7.53)

The battery voltage is (8.18, 7.53)

The battery voltage is (8.18, 7.47)

The battery voltage is (8.18, 7.53)
```

### Result:

Print out the voltage value of two sets of batteries every one second,You can press "Ctrl + C" to end program.  
The code is as below:

```
1  def test_Adc():
2      from adc import ADC
3      adc = ADC()
4      try:
5          while True:
6              Power=adc.read_battery_voltage()
7              print ("The battery voltage is "+str(Power)+'\n')
8              time.sleep(1)
9      except KeyboardInterrupt:
10         print ("\nEnd of program")
```

### Reference

read\_battery\_voltage()

Return the voltage value of the battery, retaining two decimal places.

## Ultrasonic module

### Run program

Enter the following command in the terminal to test the ultrasonic module:

If the terminal displays the directory as below (where test.py is located). You can **directly** execute the test.py command.

```
pi@raspberrypi:~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server $
```

1. If not, execute the cd command:

```
cd ~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server
```

2. Execute test.py command:

```
sudo python test.py Ultrasonic
```

```
pi@raspberrypi: ~/Freenove_Big_Hex...ot_Kit_for_Raspberry_Pi/Code/Server
File Edit Tabs Help
pi@raspberrypi:~ $ cd ~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server
pi@raspberrypi:~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server $ sudo
python test.py Ultrasonic
Program is starting ...
Obstacle distance is 32CM
Obstacle distance is 32CM
Obstacle distance is 9CM
Obstacle distance is 15CM
Obstacle distance is 14CM
Obstacle distance is 7CM
Obstacle distance is 31CM
^C
End of program
pi@raspberrypi:~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server $
```

### Result:

Every 1s, the distance between the obstacle and the ultrasonic module will be printed in, and you can press "Ctrl + C" to end the program.

The code is as below:

```
1 def test_Ultrasonic():
2     from ultrasonic import Ultrasonic
3     ultrasonic=Ultrasonic()
4     try:
5         while True:
6             data=ultrasonic.get_distance() #Get the value
7             print ("Obstacle distance is "+str(data)+"CM")
8             time.sleep(0.01)
9     except KeyboardInterrupt:
10        print ("\nEnd of program")
```

### Reference

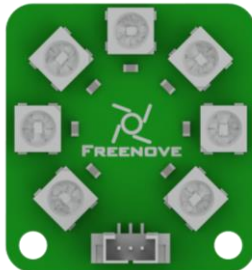
get\_distance()

This function is used to obtain the distance of ultrasonic module and front obstacle, with unit CM.

## Led

**Note: Raspberry PI 5 is temporarily unavailable because the Library rpi-ws281x-python does not support Raspberry PI 5. Please skip this section.**

There is an LED module on the robot, as shown below:



### Run program

Enter the following commands to test LED.

If the terminal displays the directory as below (where test.py is located). You can **directly** execute the test.py command.

```
pi@raspberrypi:~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server $
```

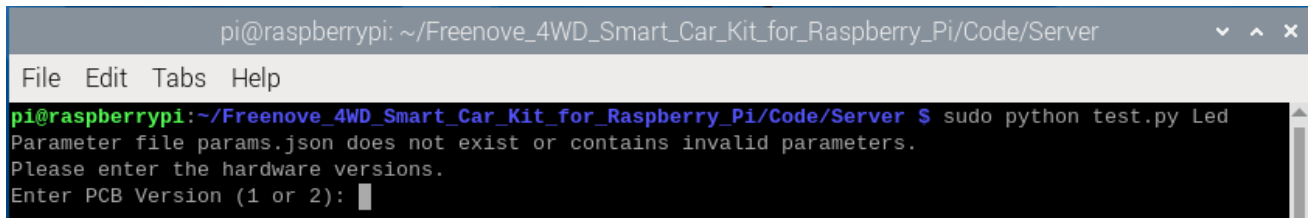
1.If not, execute the cd command:

```
cd ~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server
```

2.Execute test.py command:

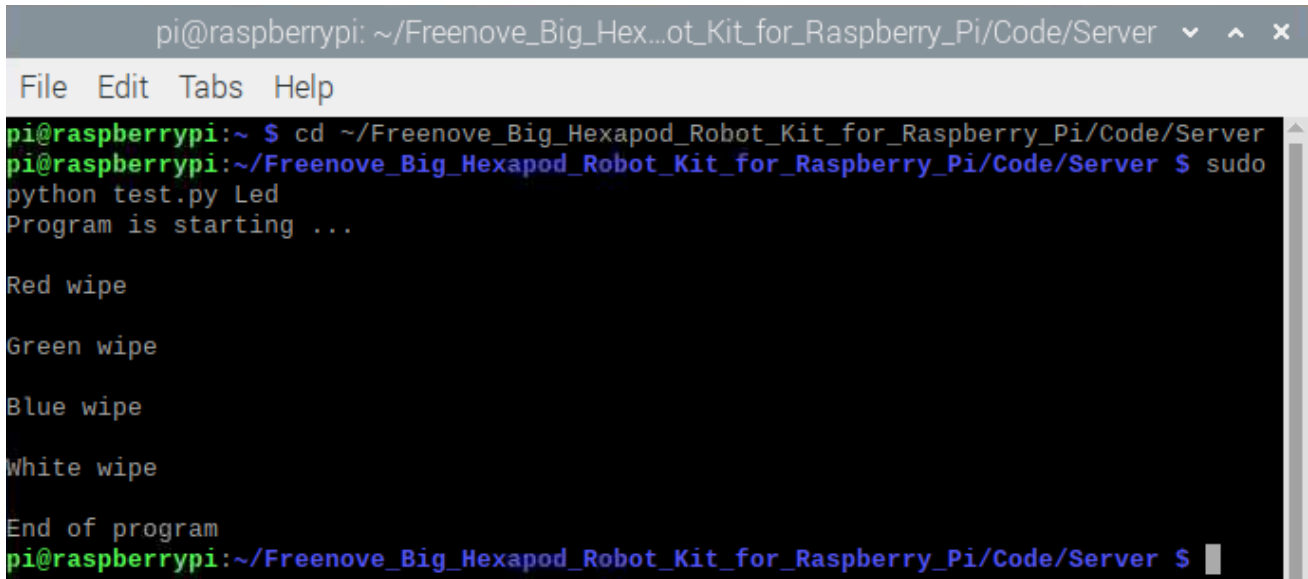
```
sudo python test.py Led
```

If you are using the Led feature for the first time, it will ask for the PCB version of the connection board.



Please enter 1 or 2 according to the version number of your connection board.

|                                                                                                                 |                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Connection board (PCB_V1.0)  | Connection board (PCB_V2.0)  |
|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|



```
pi@raspberrypi: ~/Freenove_Big_Hex...ot_Kit_for_Raspberry_Pi/Code/Server
File Edit Tabs Help
pi@raspberrypi:~ $ cd ~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server
pi@raspberrypi:~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server $ sudo
python test.py Led
Program is starting ...

Red wipe

Green wipe

Blue wipe

White wipe

End of program
pi@raspberrypi:~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server $
```

If your connection board version is V2.0 and you do not have the SPI feature turned on, the LedPixel will not show the color. Refer [here](#) to turn on the SPI

#### Result:

The LED lights up with the color of red, green, blue and white in turn and then it turns OFF. You can end the program ahead of time by pressing "ctrl+c".

If the LED color display order is not correct, open the "**Led.py**" file in the current directory and modify the value of the "self.ORDER" variable on line 16.



## Buzzer

### Run the program

Enter following command in the terminal to test buzzer.

If the terminal displays the directory as below (where test.py is located). You can **directly** execute the test.py command.

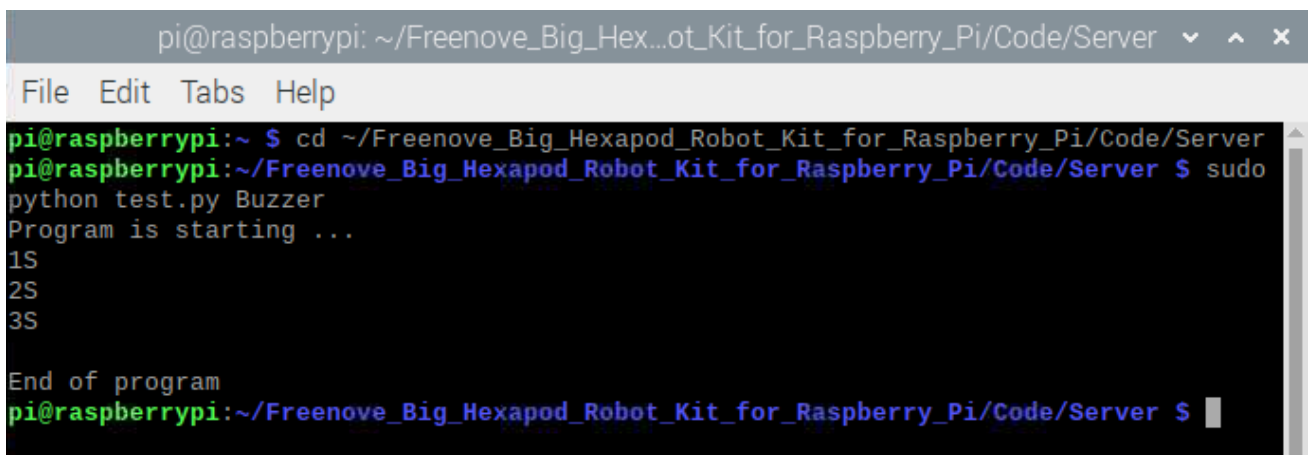
```
pi@raspberrypi:~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server $
```

1.If not, execute the cd command:

```
cd ~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server
```

2.Execute test.py command:

```
sudo python test.py Buzzer
```



```
pi@raspberrypi: ~/Freenove_Big_Hex...ot_Kit_for_Raspberry_Pi/Code/Server
File Edit Tabs Help
pi@raspberrypi:~ $ cd ~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server
pi@raspberrypi:~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server $ sudo
python test.py Buzzer
Program is starting ...
1S
2S
3S

End of program
pi@raspberrypi:~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server $
```

### Result:

The buzzer will be turn on lasting for 3s. Then the program will automatically end or you can press "Ctrl + C" to end the program.



## Camera

### Run program

1. execute the cd command:

```
cd ~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server
```

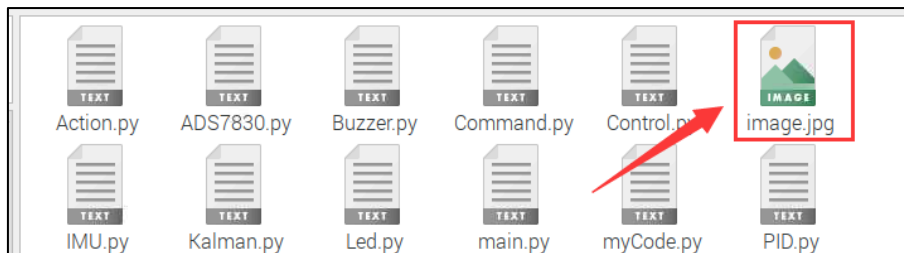
2. Execute test.py command:

```
python camera.py
```

Then please open and check the generated image.jpg under  
/Freenove\_Big\_Hexapod\_Robot\_Kit\_for\_Raspberry\_Pi/Code/Server.

```
[0:43:46.576778300] [2970] INFO RPI pisp.cpp:1450 Sensor: /base/axi/pcie@120000/rp1/i2c@88000/ov5647@36 - Selected sensor format: 640x480-SGBRG10_1X10 - Selected CFE format: 640x480-PC1g
QStandardPaths: wrong permissions on runtime directory /run/user/1000, 0770 instead of 0700
[0:43:48.531984920] [2974] WARN V4L2 v4l2_pixelformat.cpp:344 Unsupported V4L2 pixel format RPBP
[0:43:48.532711766] [2974] INFO Camera camera.cpp:1183 configuring streams: (0) 2592x1944-BGR888 (1) 2592x1944-GBRG_PISP_COMP1
[0:43:48.536032796] [2970] INFO RPI pisp.cpp:1450 Sensor: /base/axi/pcie@120000/rp1/i2c@88000/ov5647@36 - Selected sensor format: 2592x1944-SGBRG10_1X10 - Selected CFE format: 2592x1944-PC1g
[0:43:48.945554937] [2974] WARN V4L2 v4l2_pixelformat.cpp:344 Unsupported V4L2 pixel format RPBP
[0:43:48.946127747] [2974] INFO Camera camera.cpp:1183 configuring streams: (0) 640x480-XBGR8888 (1) 640x480-GBRG_PISP_COMP1
[0:43:48.949655072] [2970] INFO RPI pisp.cpp:1450 Sensor: /base/axi/pcie@120000/rp1/i2c@88000/ov5647@36 - Selected sensor format: 640x480-SGBRG10_1X10 - Selected CFE format: 640x480-PC1g
```

Then please open and check the generated image.jpg under  
/Freenove\_Big\_Hexapod\_Robot\_Kit\_for\_Raspberry\_Pi/Code/Server.



## Chapter 4 Hexapod Robot

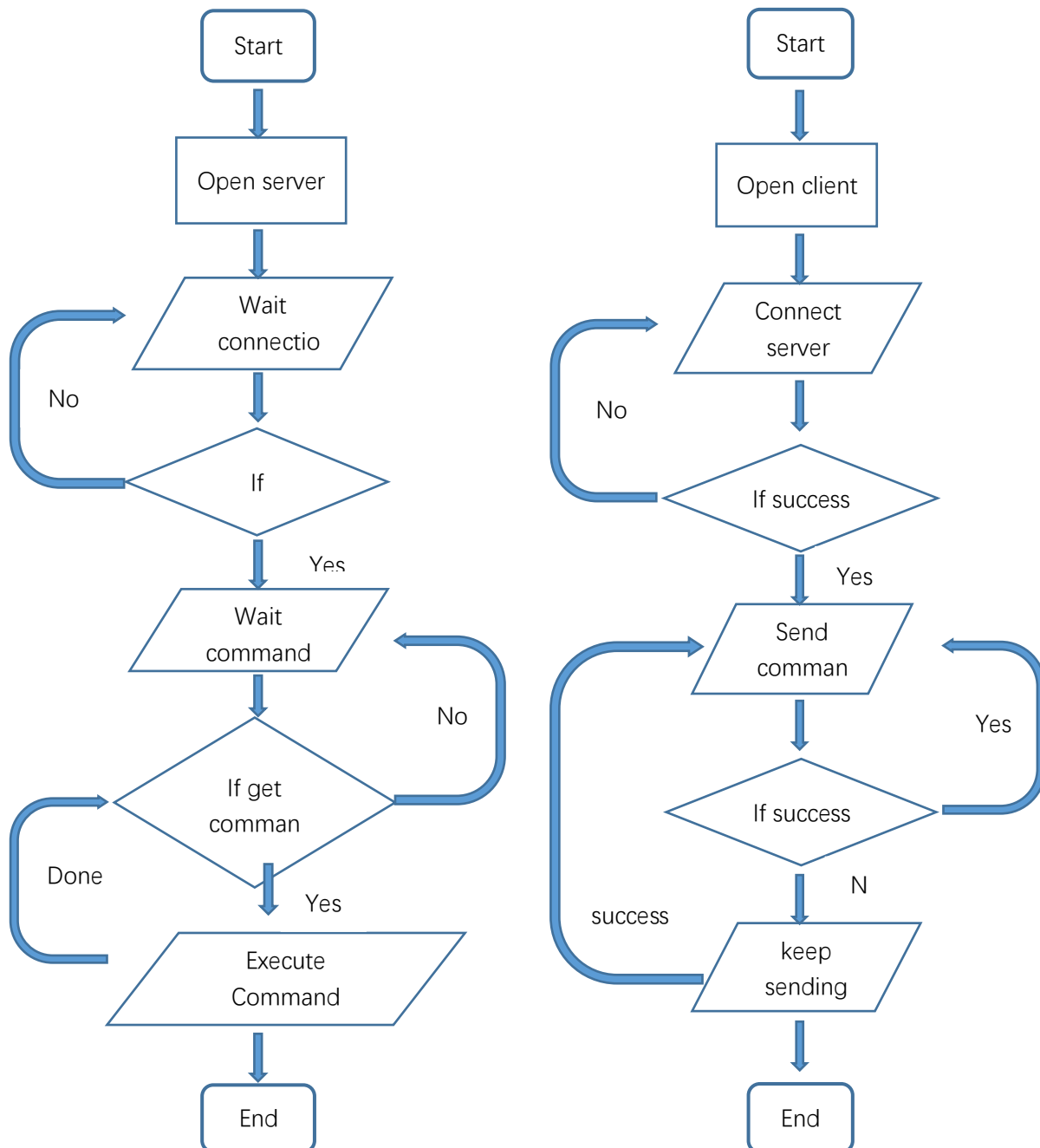
You can refer to these videos to open client and calibrate the robot:

Client control: <https://youtu.be/KlxxgnW6uyo>

Calibration: <https://youtu.be/NdXG3wmdufI>

The robot features with live video, ultrasonic ranging and other functions. Based on Python3 and PyQt5, it has also been built with server and client, which communicate with each other through TCP/IP protocol and can be controlled remotely with the same LAN.

We will help you with any concerns via [support@freenove.com](mailto:support@freenove.com)



## Server

The server runs on the Raspberry Pi. It sends the camera data and ultrasonic module data to the client, and receive commands from client.

**The code in the Server folder is used as an example.**

Part of the server code is as follows:

```

1  def get_interface_ip(self):
2      s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
3      return socket.inet_ntoa(fcntl.ioctl(s.fileno(),
4                                          0x8915,
5                                          struct.pack('256s', b'wlan0'[:15])
6                                          )[20:24])
7
8  def start_server(self):
9      #ip address
10     HOST=self.get_interface_ip()
11     #Port 8002 for video transmission
12     self.server_socket = socket.socket()
13     self.server_socket.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEPORT, 1)
14     self.server_socket.bind((HOST, 8002))
15     self.server_socket.listen(1)
16     #Port 5002 is used for instruction sending and receiving
17     self.server_socket1 = socket.socket()
18     self.server_socket1.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEPORT, 1)
19     self.server_socket1.bind((HOST, 5002))
20     self.server_socket1.listen(1)
21     print('Server address: ' +HOST)
22
23 def stop_server(self):
24     try:
25         self.connection.close()
26         self.connection1.close()
27     except :
28         print ('\n'+ "No client connection")
29
30 def reset_server(self):
31     self.turn_off_server()
32     self.turn_on_server()
33     self.video=threading.Thread(target=self.transmission_video)
34     self.instruction=threading.Thread(target=self.receive_instruction)
35     self.video.start()
36     self.instruction.start()

```

## Reference

For more code, please check "Server.py" in the Server directory.

|                                 |                                                     |
|---------------------------------|-----------------------------------------------------|
| <code>get_interface_ip()</code> | To obtain the IP address of RPi's WLAN0             |
| <code>start_server()</code>     | To turn ON TCP to wait for the connection of Client |
| <code>stop_server()</code>      | To turn OFF TCP                                     |
| <code>send_data()</code>        | To send commands to client                          |
| <code>reset_server ()</code>    | To restart TCP                                      |
| <code>transmit_video()</code>   | To send video data to client                        |
| <code>receive_commands()</code> | To receive commands from client                     |

## Open Server

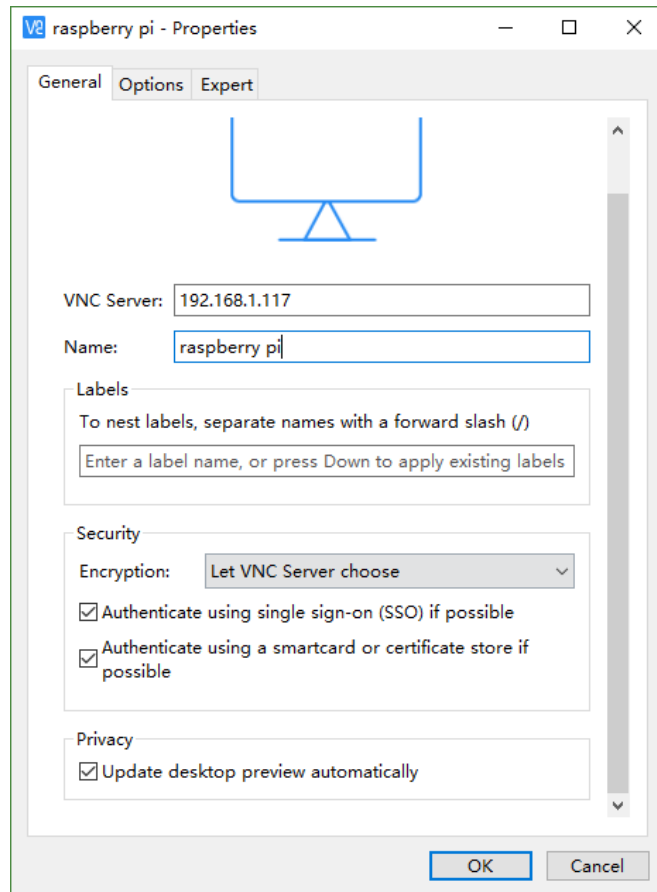
### Step 1 Login Raspberry Pi via VNC viewer

**Because server and client use GUI. You need use VNC viewer as remote desktop way.**

Download and install VNC Viewer according to your computer system by clicking following link:

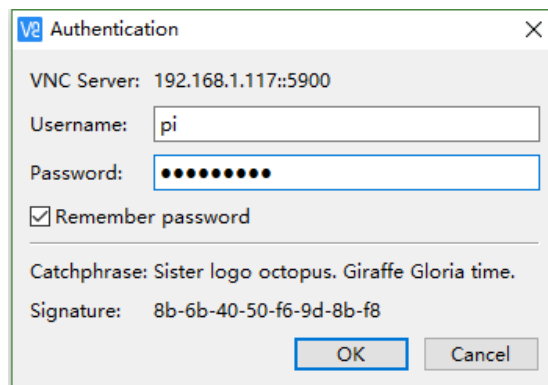
<https://www.realvnc.com/en/connect/download/viewer/>

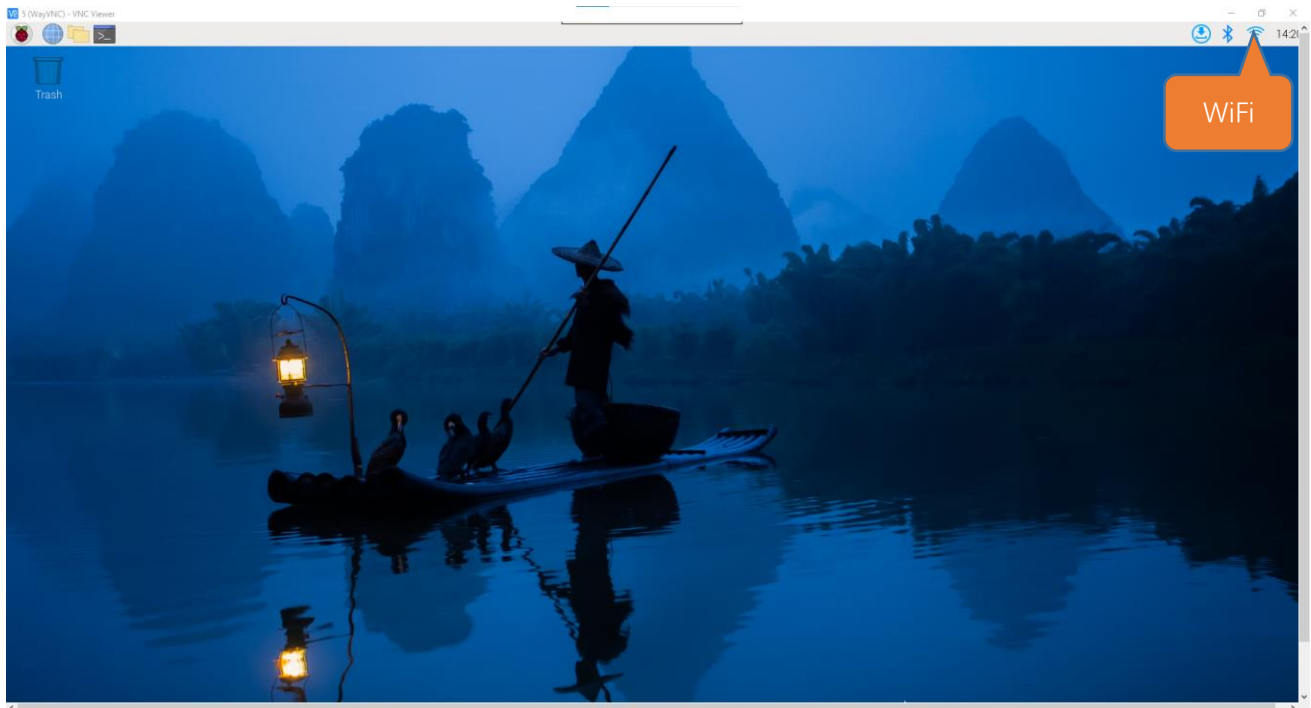
After installation is completed, open VNC Viewer. And click File → New Connection. Then the interface is shown below.



Enter IP address of your Raspberry Pi and fill in a Name. And click OK.

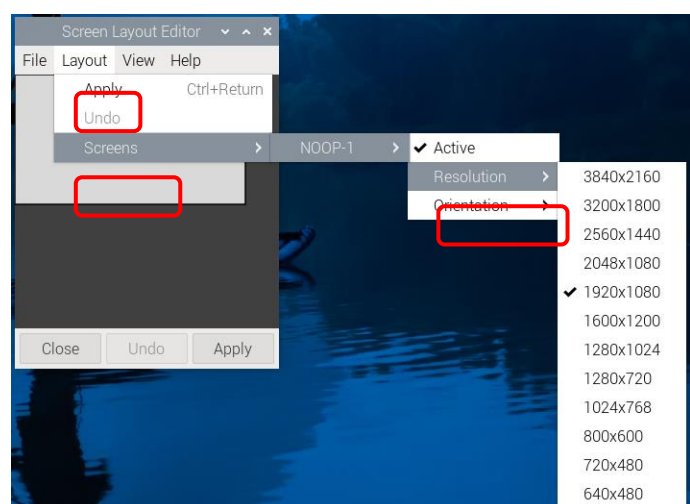
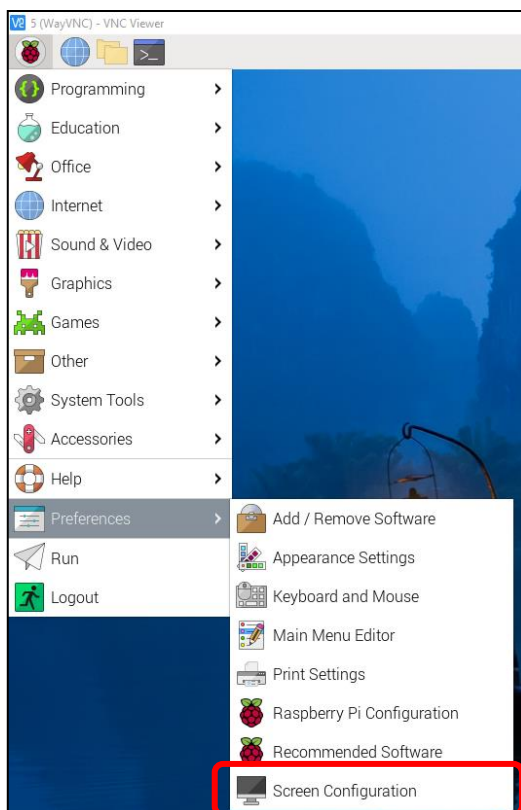
Then on the VNC Viewer panel, double-click new connection you just created, and the following dialog box pops up. Enter username: **pi** and Password: **raspberry**. And click OK.





If the resolution ratio is not great or there is just a **little window**, you can set a proper resolution ratio via steps below.

Select Screen Configuration. Select the appropriate resolution in the new window. Click Apply.



In addition, your VNC Viewer window may zoom your Raspberry Pi desktop. You can change it. On your VNC View control panel, click right key. And select Properties->Options label->Scaling. Then set proper scaling.

### Step 2 Run commands

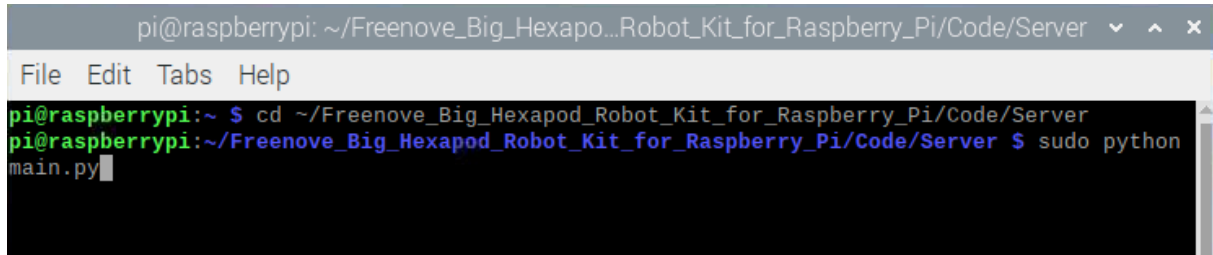
Enter following command in the terminal.

1. Use cd command to enter directory where main.py is located:

```
cd ~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server
```

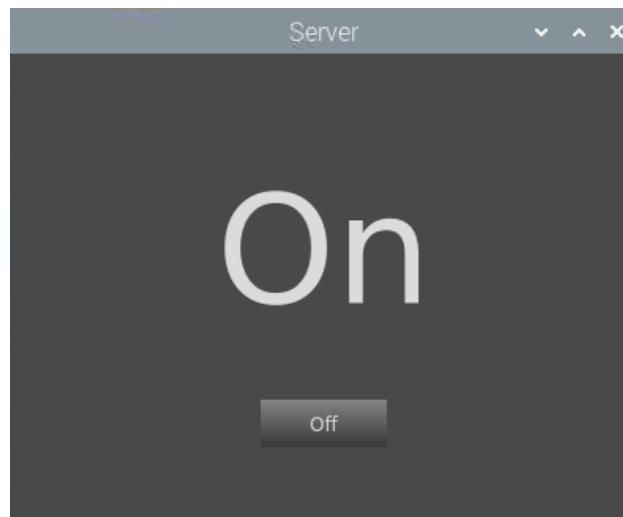
2. Run main.py:

```
sudo python main.py
```



```
pi@raspberrypi: ~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server
File Edit Tabs Help
pi@raspberrypi:~ $ cd ~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server
pi@raspberrypi:~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server $ sudo python
main.py
```

The interface is as below:



If you don't like the interface, you can also enter the commands to open the server. It is more convenient.

1. Use cd command to enter directory where main.py is located:

```
cd ~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server
```

2. Run main.py:

```
sudo python main.py -t -n
```

or Run main.py with following command:

```
sudo python main.py -tn
```

"-t" means open TCP communication. "-n" means don't show interface.

### Sever Auto Start

1. Open the terminal and execute the following two commands respectively to create a "start.sh" file.

```
cd ~
sudo touch start.sh
```

2. Open "start.sh".

```
sudo nano start.sh
```

3. Add the following contents to "start.sh" file.

```
#!/bin/sh
cd "/home/pi/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server"
pwd
sleep 10
sudo cp point.txt /home/pi
sudo python main.py
```

Press Ctrl + O and then press Enter to save it. Press Ctrl+X to exit.

4. Modify permissions.

```
sudo chmod 777 start.sh
```

5. Enter the following command to create a directory.

```
mkdir ~/.config/autostart/
```

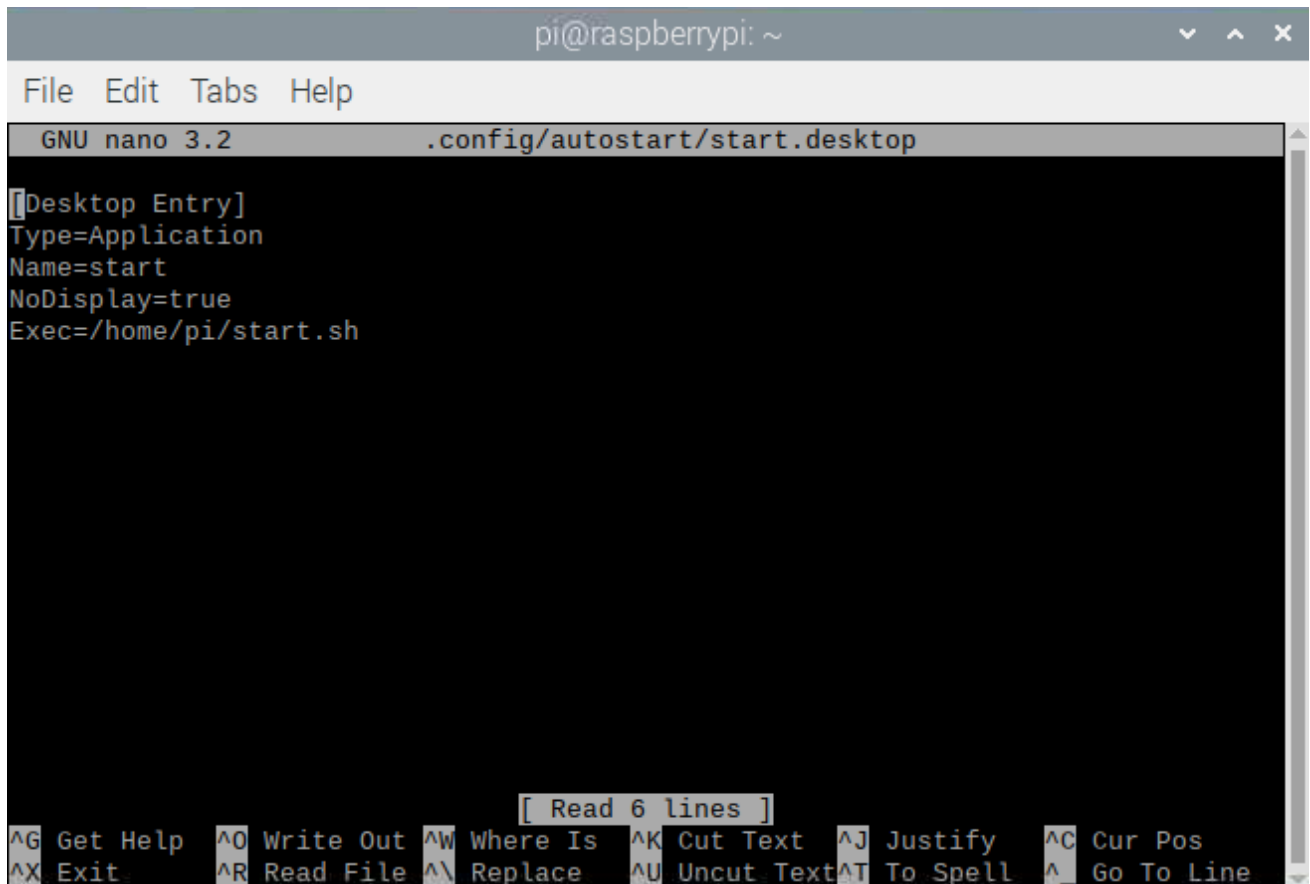
6. create and open "start.desktop" file

```
sudo nano .config/autostart/start.desktop
```

- 7 Add the following content to “start.desktop” file.

```
[Desktop Entry]
Type=Application
Name=start
NoDisplay=true
Exec=/home/pi/start.sh
```

Press Ctrl + O and then press Enter to save it. Press Ctrl+X to exit.



```
pi@raspberrypi: ~
File Edit Tabs Help
GNU nano 3.2 .config/autostart/start.desktop
[Desktop Entry]
Type=Application
Name=start
NoDisplay=true
Exec=/home/pi/start.sh
[ Read 6 lines ]
^G Get Help  ^O Write Out ^W Where Is  ^K Cut Text  ^J Justify   ^C Cur Pos
^X Exit      ^R Read File ^\ Replace   ^U Uncut Text ^T To Spell  ^_ Go To Line
```

- 8 Modify permissions.

```
sudo chmod +x .config/autostart/start.desktop
```

- 9 Finally enter the following content to reboot Raspberry Pi.

```
sudo reboot
```

Note: To cancel auto start, please delete the files “start.sh” and “start.desktop” created above.

## Client

Client can receive video data and commands from the server as well as send commands to server. It can also be run in different systems including Windows and macOS as long as you install the related software and libraries.

Part of the client code is as follows:

```

1  def turn_on_client(self, ip):
2      self.client_socket1 = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
3      self.client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
4      print (ip)
5  def turn_off_client(self):
6      try:
7          self.client_socket.shutdown(2)
8          self.client_socket1.shutdown(2)
9          self.client_socket.close()
10         self.client_socket1.close()
11     except Exception as e:
12         print(e)
13  def receiving_video(self, ip):
14      try:
15         self.client_socket.connect((ip, 8002))
16         self.connection = self.client_socket.makefile('rb')
17     except:
18         #print ("command port connect failed")
19         pass
20     while True:
21         try:
22             stream_bytes= self.connection.read(4)
23             leng=struct.unpack('<L', stream_bytes[:4])
24             jpg=self.connection.read(leng[0])
25             if self.is_valid_image_4_bytes(jpg):
26                 if self.video_flag:
27                     self.image = cv2.imdecode(np.frombuffer(jpg, dtype=np.uint8),
28 cv2.IMREAD_COLOR)
29                     if self.fece_id == False and self.fece_recognition_flag:
30                         self.face.face_detect(self.image)
31                         self.video_flag=False
32         except BaseException as e:
33             print (e)
34             break
35  def send_data(self, data):
36      if self.tcp_flag:

```

```

37         try:
38             self.client_socket1.send(data.encode('utf-8'))
39         except Exception as e:
40             print(e)
41     def receive_data(self):
42         data=""
43         data=self.client_socket1.recv(1024).decode('utf-8')
44         return data

```

### Reference

For more code, please check "Client.py" in the Client director.

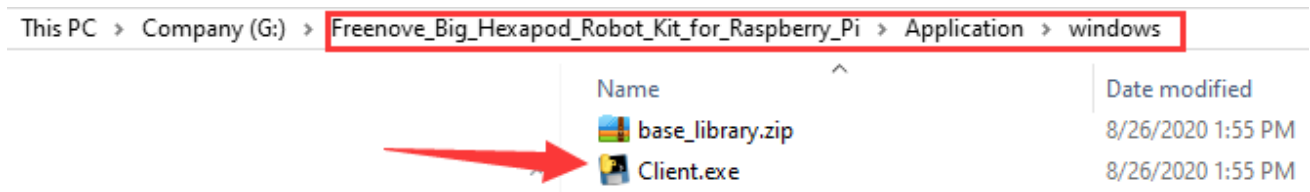
|                                                    |  |
|----------------------------------------------------|--|
| turn_on_client ()                                  |  |
| To connect to the server                           |  |
| turn_off_client ()                                 |  |
| To disconnect server。                              |  |
| receiving_video()                                  |  |
| To receive video data from server                  |  |
| is_valid_image_4_bytes ()                          |  |
| To check the integrity of each frame of video data |  |
| send_data ()                                       |  |
| To send commands to server                         |  |
| receive_data ()                                    |  |
| To receive commands from server                    |  |

## Run Client on Windows system

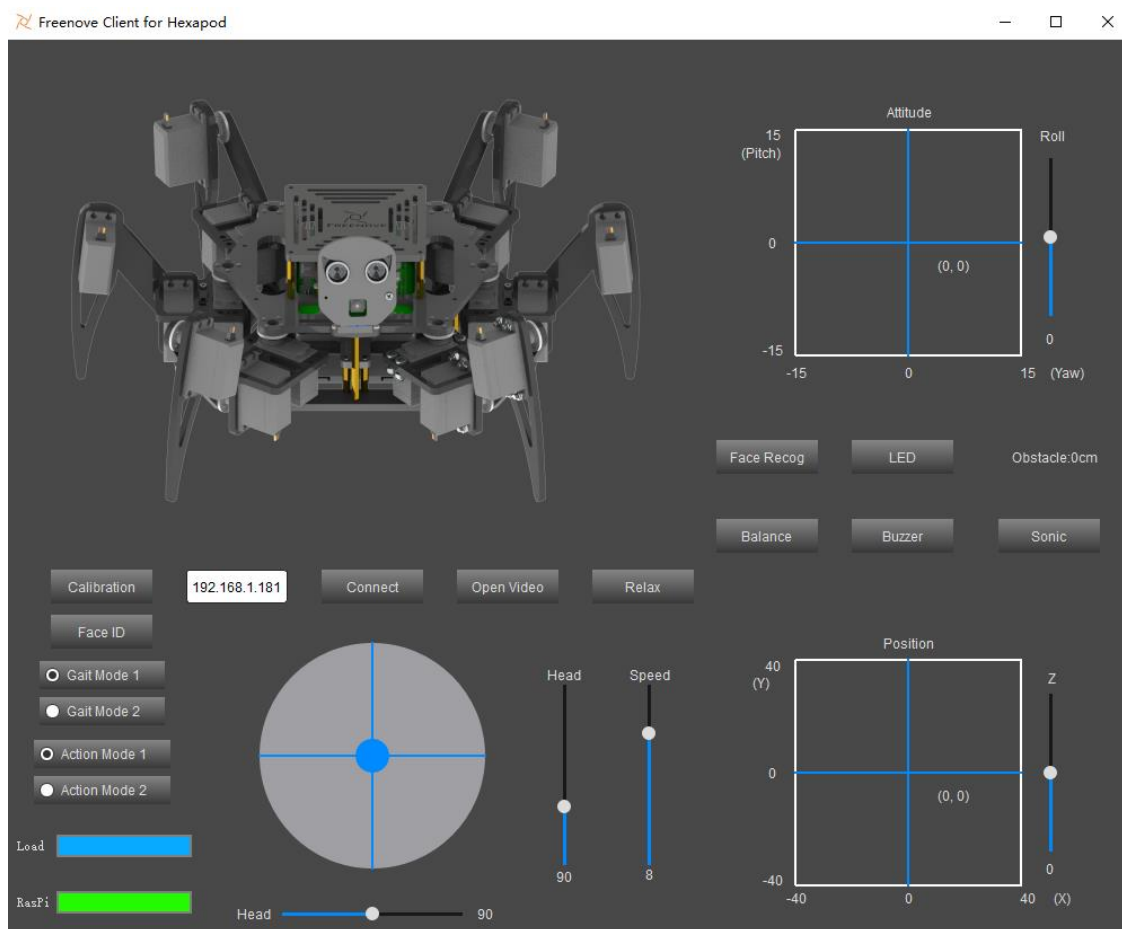
There are two ways to run Client on Windows.

### 1. Running executable file directly

Find the "Client.exe" file in the specified directory, double click it and the Client is opened.



The client interface is shown as below:



When the Client is opened successfully, you need to turn on Raspberry Pi and open the server. Enter Raspberry Pi's IP address in the white IP editor and click "Connect" to connect the robot with RPi. After connecting successfully, you need to calibrate the robot's six legs before control it to move.

**Note: when Raspberry Pi is shut down, server will be closed. You need open server again the next time.**

2. Install python3 and some related python libraries and execute the program through python3 editor.

This section will be completed in your **computer with windows system**, **not Raspberry Pi**.

Running client on Windows system requires the installation of some software and libraries, which take some time. At this point, it does not require to run server and use the Raspberry Pi, so you can turn OFF RPi temporarily. After installing successfully, turn ON the RPi and run the Server.

#### Install python3

Download the installation file via below:

<https://www.python.org/downloads/windows/>



Click Latest Python Release - Python 3.8.1

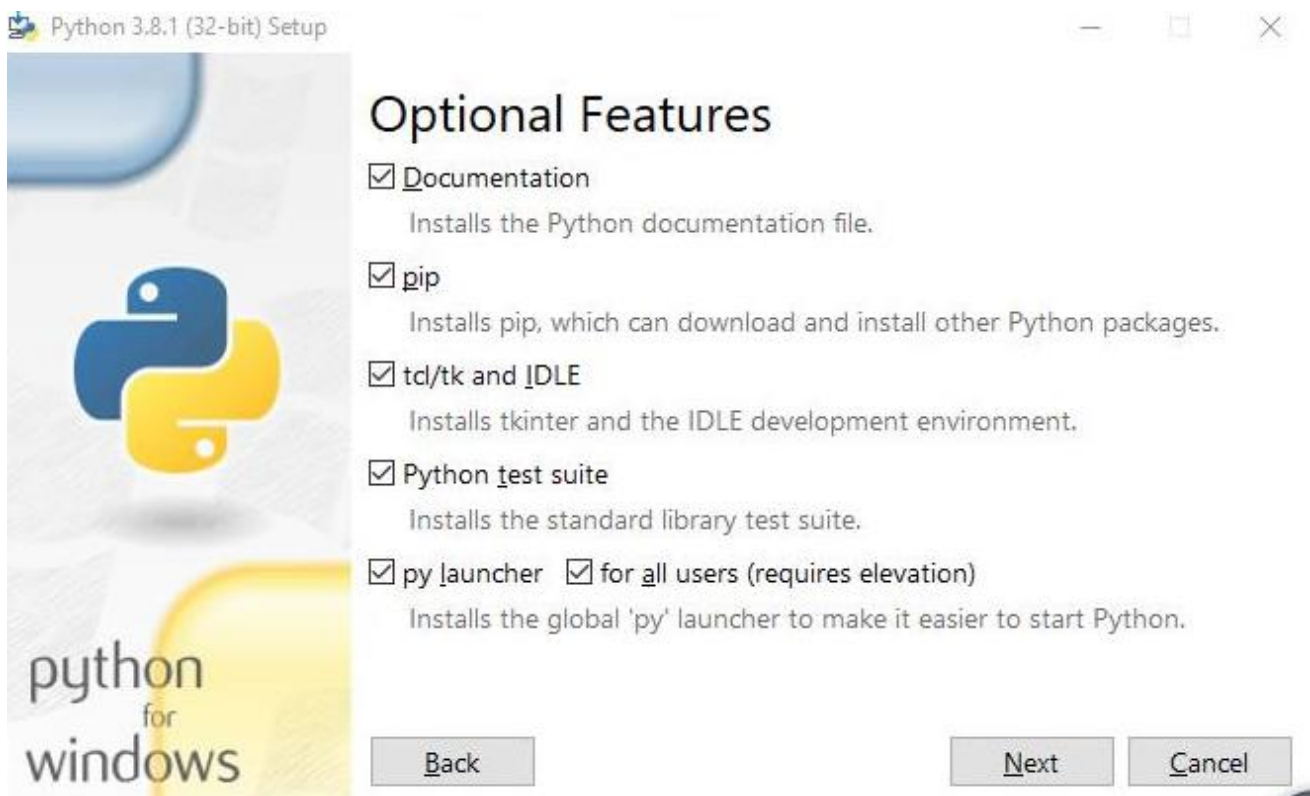
| Version                                             | Operating System | Description             |
|-----------------------------------------------------|------------------|-------------------------|
| <a href="#">Gzipped source tarball</a>              | Source release   |                         |
| <a href="#">XZ compressed source tarball</a>        | Source release   |                         |
| <a href="#">macOS 64-bit installer</a>              | Mac OS X         | for OS X 10.9 and later |
| <a href="#">Windows help file</a>                   | Windows          |                         |
| <a href="#">Windows x86-64 embeddable zip file</a>  | Windows          | for AMD64/EM64T/x64     |
| <a href="#">Windows x86-64 executable installer</a> | Windows          | for AMD64/EM64T/x64     |
| <a href="#">Windows x86-64 web-based installer</a>  | Windows          | for AMD64/EM64T/x64     |
| <a href="#">Windows x86 embeddable zip file</a>     | Windows          |                         |
| <a href="#">Windows x86 executable installer</a>    | Windows          |                         |
| <a href="#">Windows x86 web-based installer</a>     | Windows          |                         |

Select "Windows x86 executable installer". Execute the installer when completes downloading.

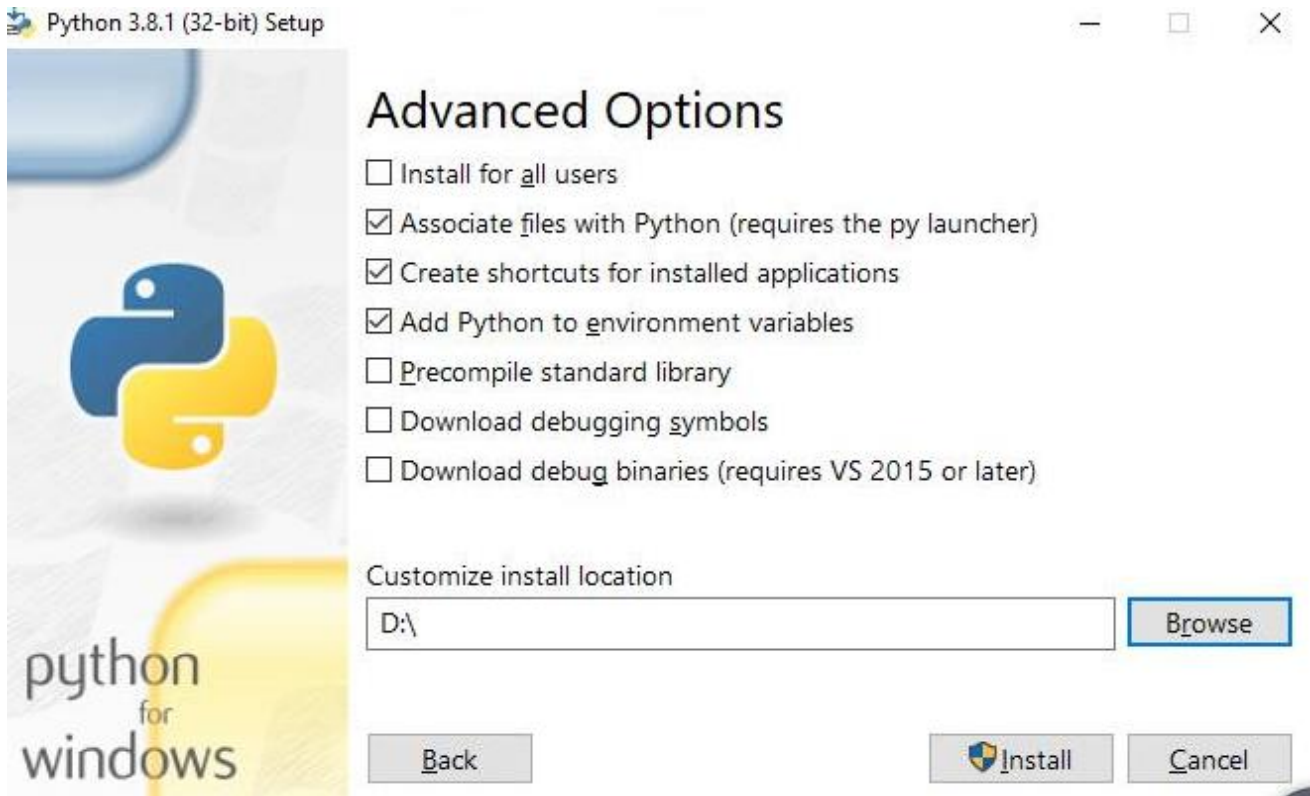
 [support@freenove.com](mailto:support@freenove.com)



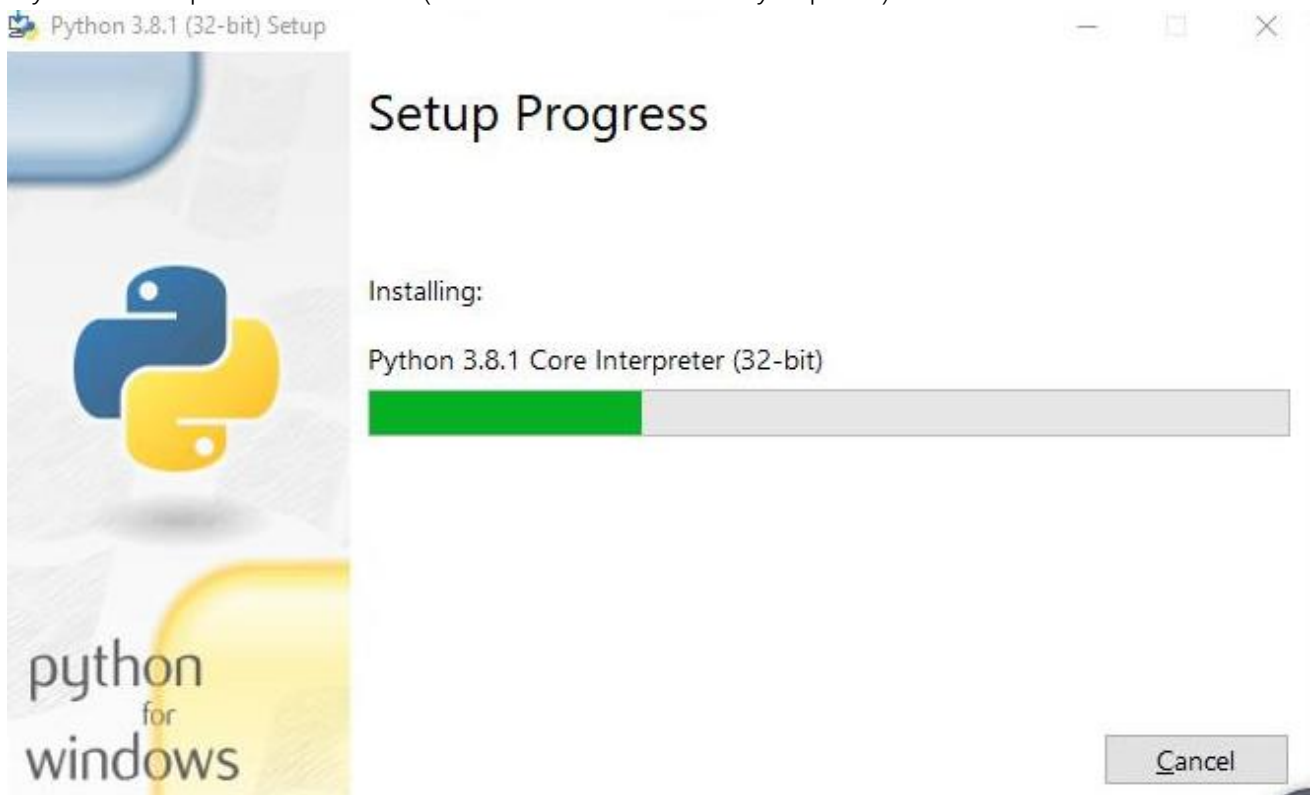
Check “Add Python 3.8 to PATH” and install according to your needs.



Check all the options and click “Next”.



My installation path here is Disk D (You can choose whichever you prefer). Click Install.



Wait for the installation,



Installation finishes.

Install libraries including PyQt5 library, opencv library, numpy library, etc.

If have not download the zip file, download it via below:

[https://github.com/Freenove/Freenove\\_Big\\_Hexapod\\_Robot\\_Kit\\_for\\_Raspberry\\_Pi/archive/master.zip](https://github.com/Freenove/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/archive/master.zip)

Then unzip it and delete “-master” to rename it to “Freenove\_Big\_Hexapod\_Robot\_Kit\_for\_Raspberry\_Pi”.

Then put it into D disk for example.

You can also place it into other disks (like E), but the path in following command should be modified accordingly (replace D: by E:).

Press “win + R” and enter cmd, and click ok. Then enter following commands.

1.Enter D disk. If you put it into E, it should be E:

```
D:
```

2.Enter directory where setup\_windows.py is located:

```
cd D:\Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi\Code
```

3.Run: setup\_windows.py

```
Python setup_windows.py
```

```
C:\Users\Freenove>D:
```

```
D:\>cd D:\Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi\Code
```

```
D:\Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi\Code>Python setup_windows.py
```

Or you can enter “Freenove\_Big\_Hexapod\_Robot\_Kit\_for\_Raspberry\_Pi\Code\” directory and double click “setup\_windows.py” to execute the installer.

Or open “setup\_windows.py” with python3 and execute it.

**Caution:** If the default python of your Windows system is not python3 but python2, change the command “setup\_windows.py” as follows and replace all “python” in setup\_windows.py to “python3”.

```
Python3 setup_windows.py
```

The installation takes some time, please wait with patience. When all the libraries are installed successfully, you can see the prompt “All libraries installed successfully” on the screen.

```
Package                Version
-----
click                  7.1.2
numpy                  1.18.5
opencv-contrib-python  4.3.0.36
opencv-python          4.3.0.36
Pillow                 7.1.2
pip                    20.2.1
PyInstaller             3.6
PyQt5                  5.15.0
PyQt5-sip              12.8.0
pyqt5-tools            5.15.0.1.7
python-dotenv           0.14.0
setuptools              41.2.0

All libraries installed successfully
```

If there is any library fails to install, it will print "Some libraries have not been installed yet. Please run 'python setup\_windows.py' again" on the screen, then you need to execute the command `python setup_windows.py` again. Most installation fails because of poor network, so you can check your network before installing the libraries.

### Open client

Press “win + R” and enter cmd, and click ok. Then enter following commands.

1.Enter D disk. If you put it into E, it should be E:

```
D:
```

2.Enter directory where Main.py is located:

```
cd D:\Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi\Code\Client
```

3.Run Main.py:

```
Python Main.py
```

```
C:\Users\Freenove>D:
D:\>cd D:\Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi\Code\Client
D:\Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi\Code\Client>Python Main.py_
```

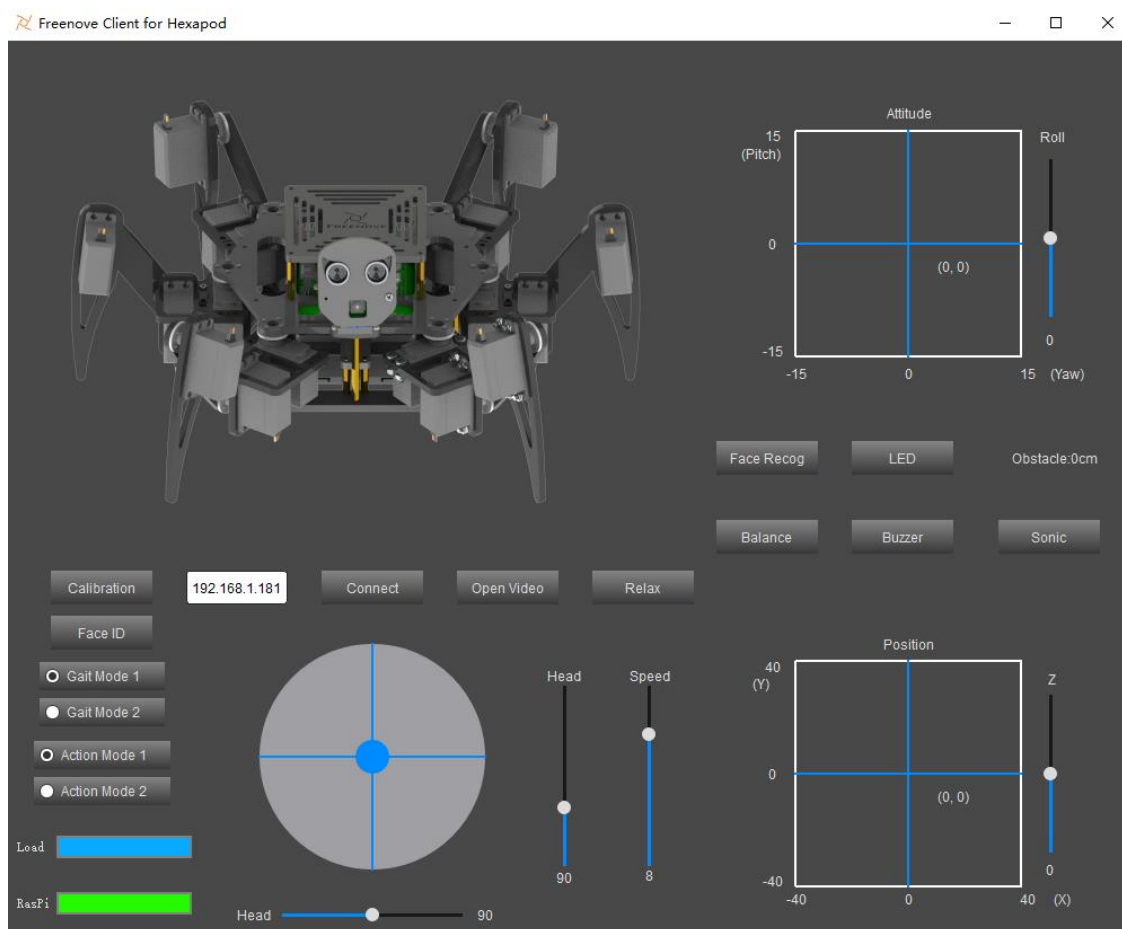
Or you can enter “Freenove\_Big\_Hexapod\_Robot\_Kit\_for\_Raspberry\_Pi\Code\Client” directory and double-click “Main.py”.

Or use python3 to open and execute “Main.py”.

**Note:** If the default python of your Windows system is not Python3 but python2, change the command Main.py as follows:

```
Python3 Main.py
```

The client interface is shown as below:



When the Client is opened successfully, you need to turn on Raspberry Pi and open the server. Enter Raspberry Pi's IP address in the white IP editor and click "Connect" to connect the robot with RPi. After connecting successfully, you need to calibrate the robot's six legs before control it to move.

**Note: when Raspberry Pi is shut down, server will be closed. You need open server again the next time.**

## Run Client on macOS system

Running client on macOS system requires the installation of some software and libraries, which take some time. At this point, it does not require to run server and use the Raspberry Pi, so you can turn OFF RPi temporarily. After installing successfully, turn ON the RPi and run the Server. The system comes with python2 instead of python3, so you need to install python3 first, as all the program of this robot are run with python3.

### Install python3

Download the installation package via: <https://www.python.org/downloads/>

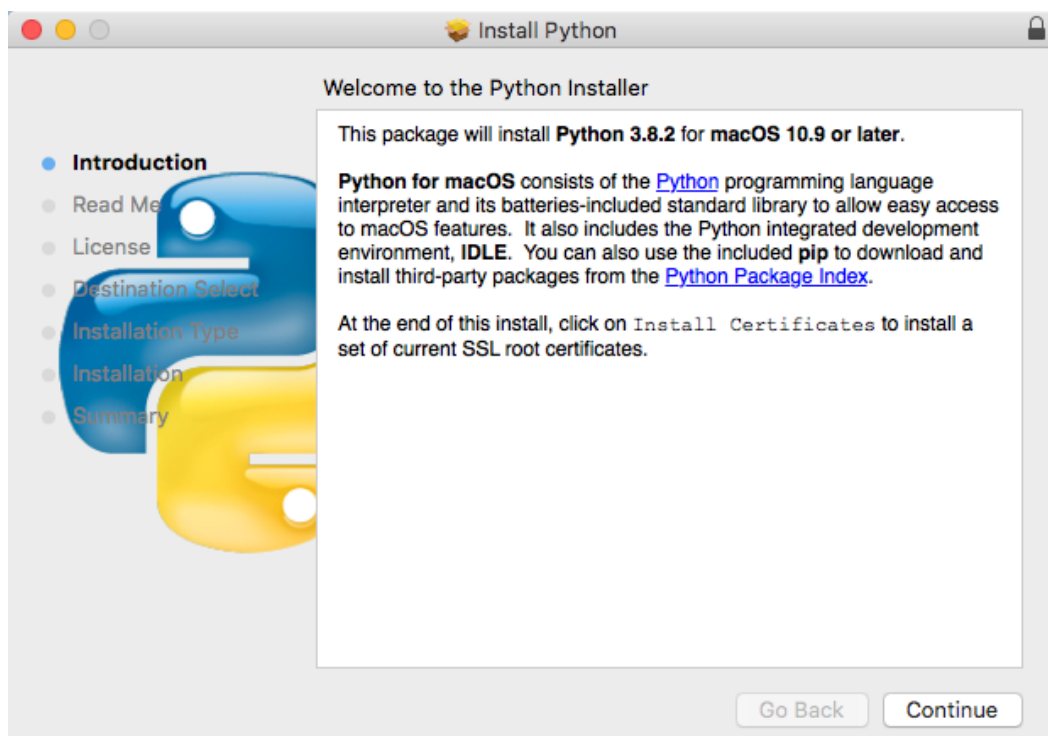
|              |               |                                                                                              |
|--------------|---------------|----------------------------------------------------------------------------------------------|
| Python 3.8.1 | Dec. 18, 2019 |  Download |
| Python 3.7.6 | Dec. 18, 2019 |  Download |

If your macOS is 11. Like 11.0, please install **python 3.9**.

If your macOS is NOT 11, like 10.15, please install **python 3.8**. If you have installed python 3.9. You need uninstall it first.

| Version                                             | Operating System | Description             |
|-----------------------------------------------------|------------------|-------------------------|
| <a href="#">Gzipped source tarball</a>              | Source release   |                         |
| <a href="#">XZ compressed source tarball</a>        | Source release   |                         |
| <a href="#">macOS 64-bit installer</a>              | Mac OS X         | for OS X 10.9 and later |
| <a href="#">Windows help file</a>                   | Windows          |                         |
| <a href="#">Windows x86-64 embeddable zip file</a>  | Windows          | for AMD64/EM64T/x64     |
| <a href="#">Windows x86-64 executable installer</a> | Windows          | for AMD64/EM64T/x64     |
| <a href="#">Windows x86-64 web-based installer</a>  | Windows          | for AMD64/EM64T/x64     |
| <a href="#">Windows x86 embeddable zip file</a>     | Windows          |                         |
| <a href="#">Windows x86 executable installer</a>    | Windows          |                         |
| <a href="#">Windows x86 web-based installer</a>     | Windows          |                         |

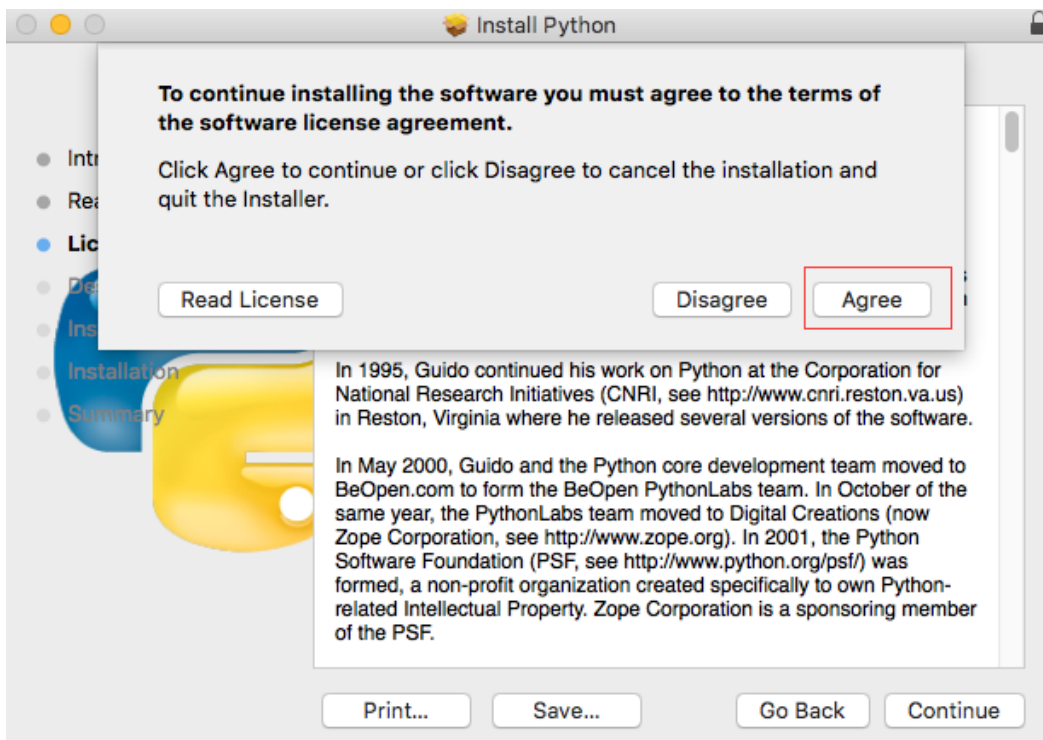
Slide to the bottom of the page and click macOS 64-bit installer.



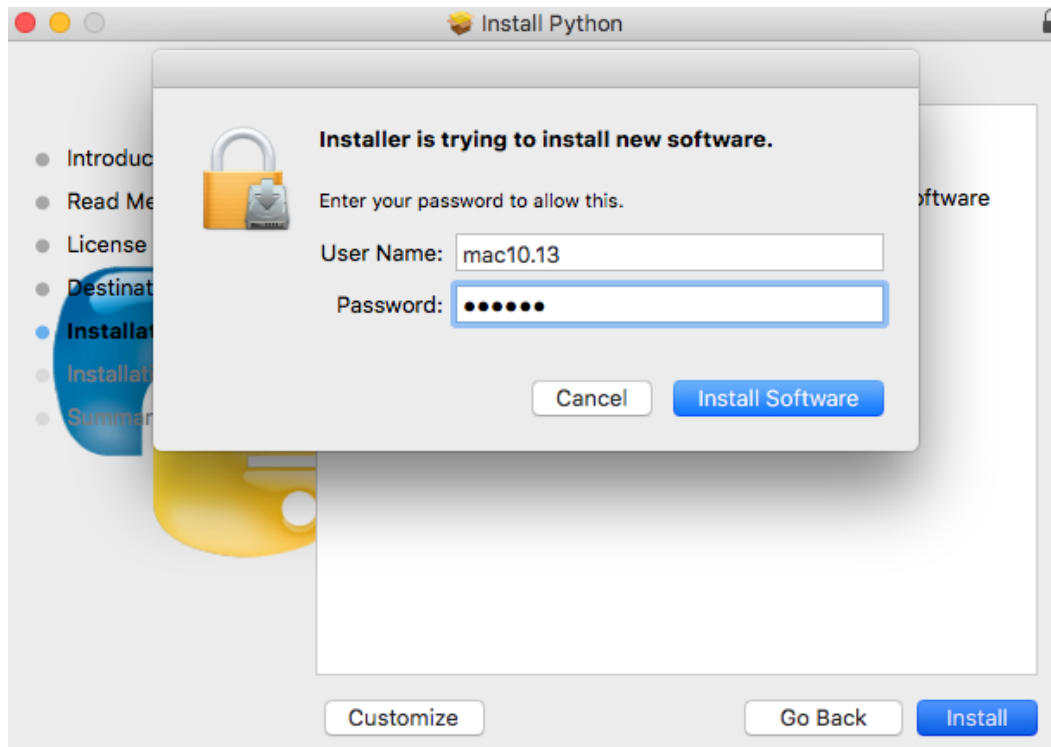
Click Continue



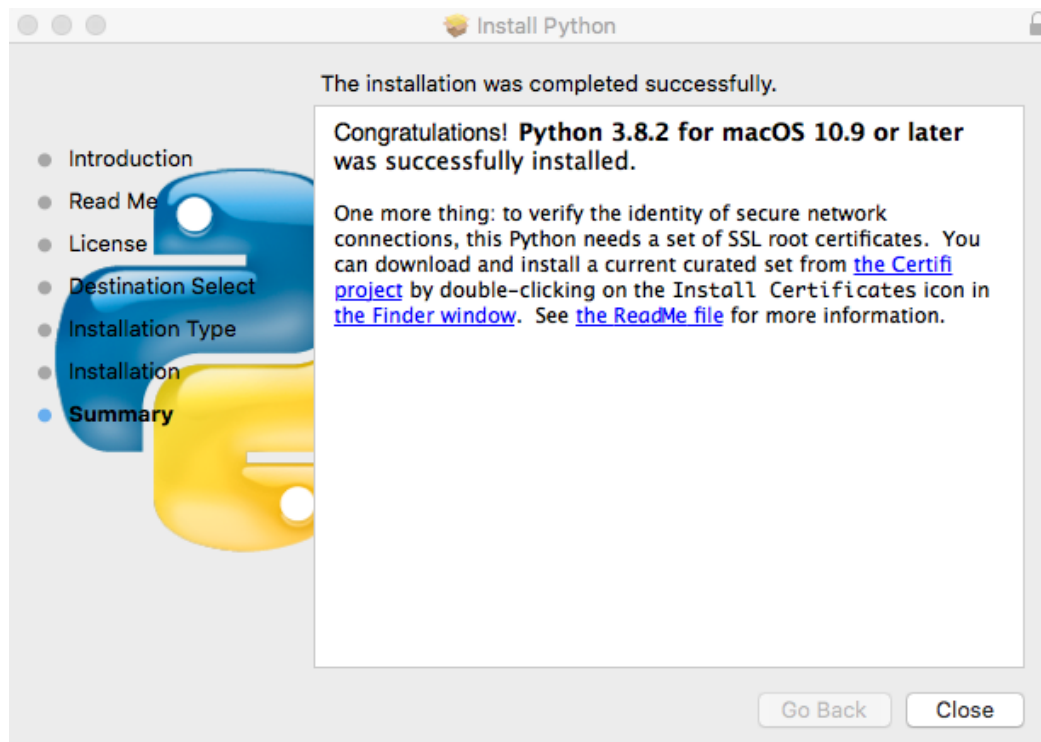
Click Continue



Click Agree



Click “Install”, enter the computer password and click “Install Software”.



Install successfully



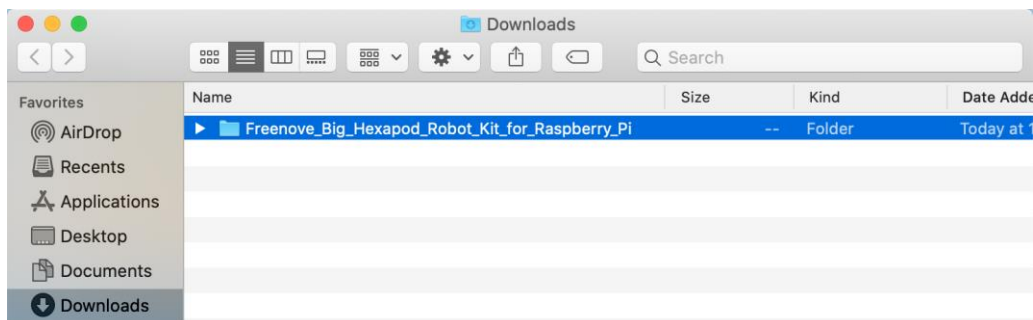
You can find it in your Applications.

Install PyQt5 library, opencv library, numpy library, etc.

If you have not yet downloaded the robot code, you can download it through the following link:

[https://github.com/Freenove\\_Big\\_Hexapod\\_Robot\\_Kit\\_for\\_Raspberry\\_Pi/archive/master.zip](https://github.com/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/archive/master.zip)

After downloading, you can find it in "Downloads".



Open "Terminal"



Enter the following commands in "Terminal"

1. Enter "Downloads" (where you save the robot code. If your path is different, please change it.)

```
cd Downloads
```

2. Enter directory where setup\_macos.py is located:

```
cd Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/
```

3. Run setup\_macos.py:

```
python3 setup_macos.py
```

```
[freenove@PandeMacBook-Air ~ % cd Downloads
freenove@PandeMacBook-Air Downloads % cd Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/
freenove@PandeMacBook-Air Code % python3 setup_macos.py]
```

The installation takes some time, please wait with patience.

| Package                        | Version  |
|--------------------------------|----------|
| numpy                          | 1.19.0   |
| opencv-contrib-python-headless | 4.3.0.36 |
| opencv-python-headless         | 4.3.0.36 |
| Pillow                         | 7.2.0    |
| pip                            | 20.2.1   |
| PyQt5                          | 5.13.2   |
| PyQt5-sip                      | 12.8.0   |
| setuptools                     | 47.1.0   |
| wheel                          | 0.34.2   |

```
All libraries installed successfully
freenove@PandeMacBook-Air Code %
```

When all the libraries are installed successfully, you can see the prompt “All libraries installed successfully” on the screen. If there is any library fails to install, it will print “Some libraries have not been installed yet. Please run 'python3 setup\_macos.py' again” on the screen, then you need to execute the command python3 setup\_macos.py again.

If you are using **macOS under 11.0, like 10.15**. Just skip to “Open client”.

If you are using **macOS 11.0 or later version**. Please run commands below:

```
pip3 uninstall PyQt5
pip3 install PyQt5
```

Most installation fails because of poor network, so you can check your network before installing the libraries.

### Open client

Installing successfully, now you are in the directory where setup\_macos.py locates.

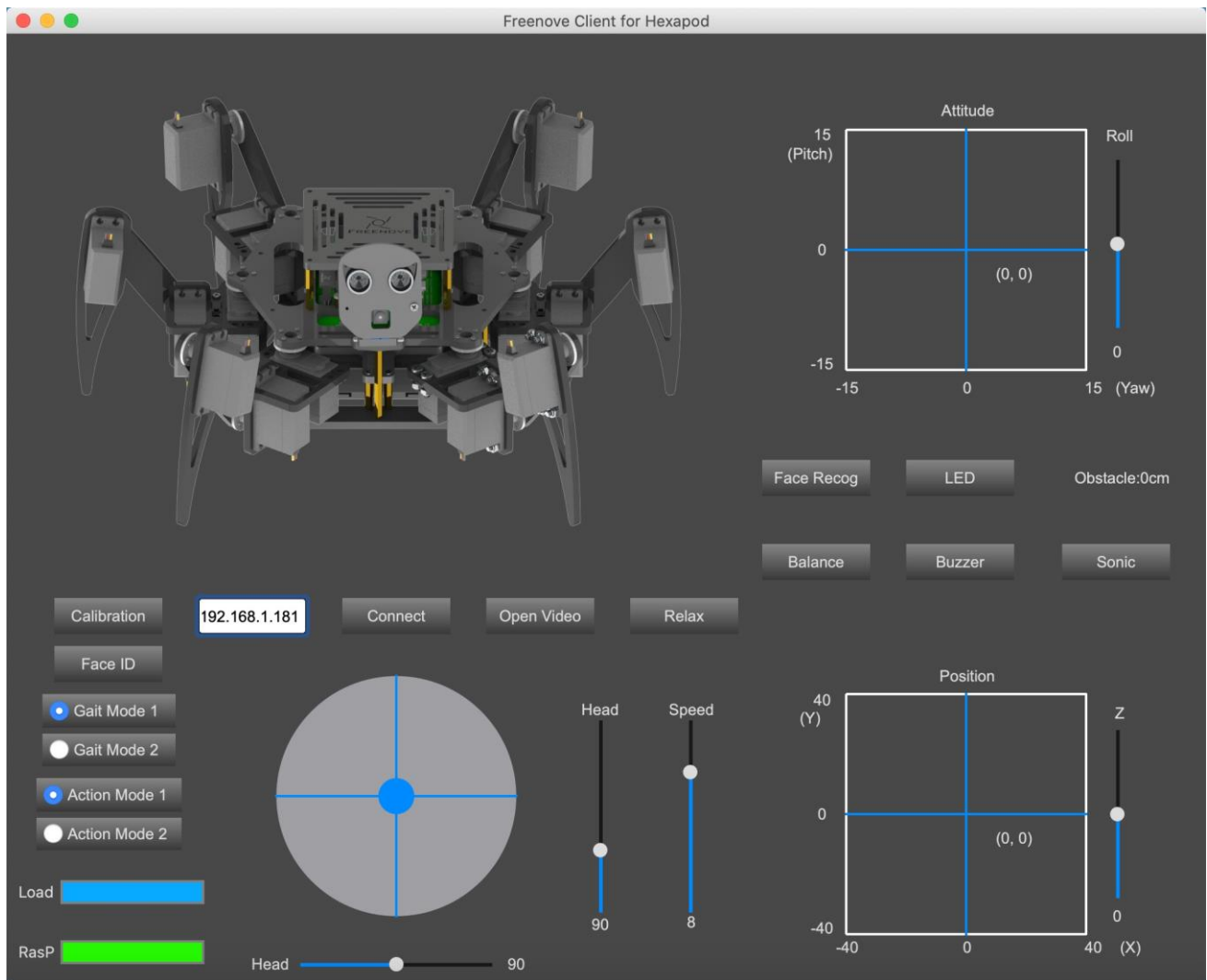
1. Enter the following command to enter the client program folder:

```
cd Client/
```

2. Enter the following command to run the program

```
python3 Main.py
```





When the Client is opened successfully, you need to turn on Raspberry Pi and open the server. Enter Raspberry Pi's IP address in the white IP editor and click "Connect" to connect the robot with RPi. After connecting successfully, you need to calibrate the robot's six legs before control it to move.

**Note: when Raspberry Pi is shut down, server will be closed. You need open server again the next time.**

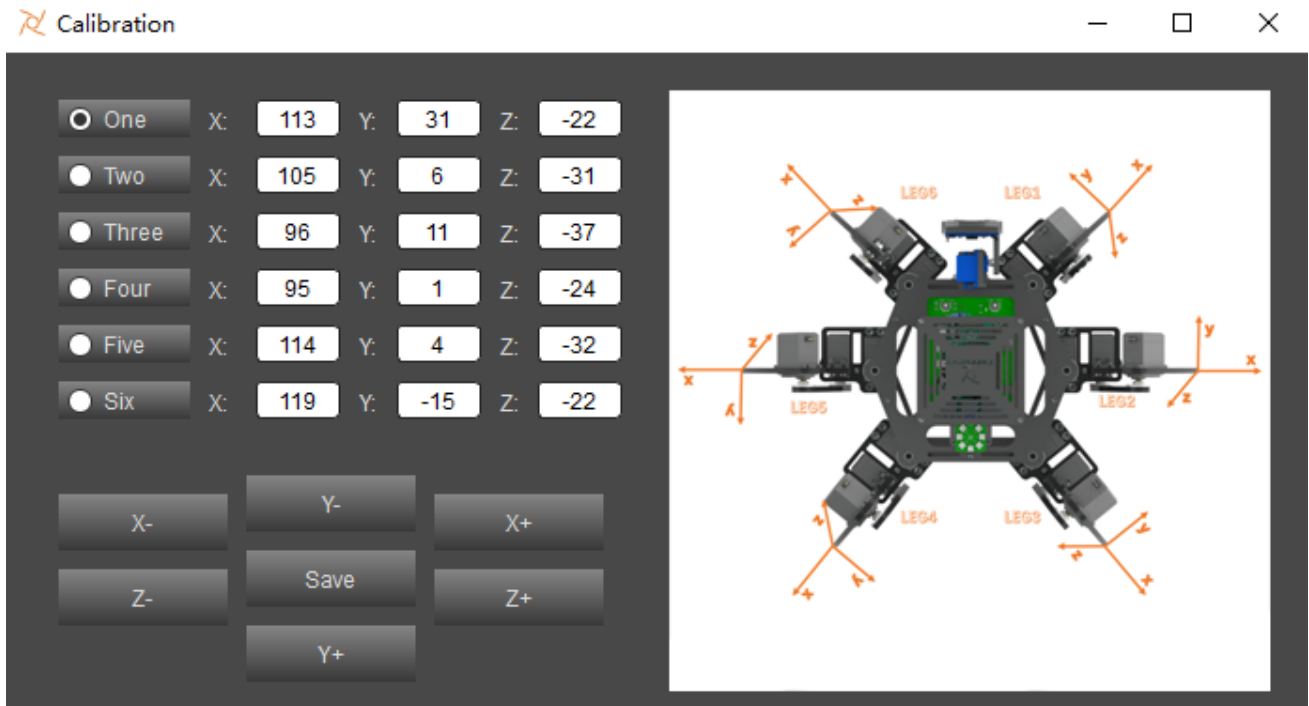
## Calibration

Calibrate the robot's six legs

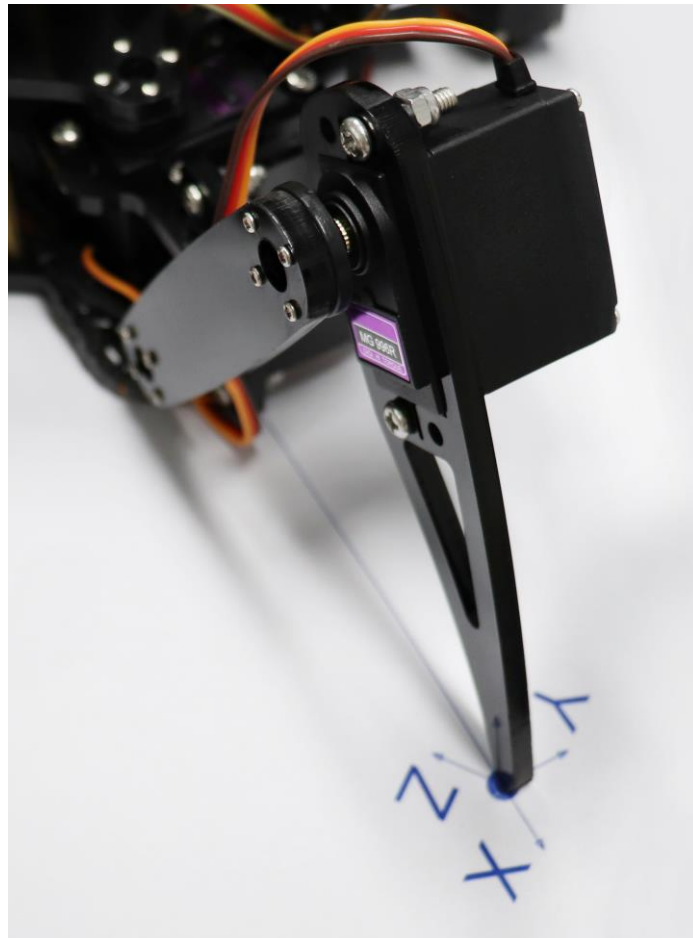
1. Lay the calibration paper on a horizontal table.
2. Place the robot to the corresponding position on the calibration paper as shown below:



3. Connect client with server successfully, click the "Calibration" button on the client, then a calibration window pops up.



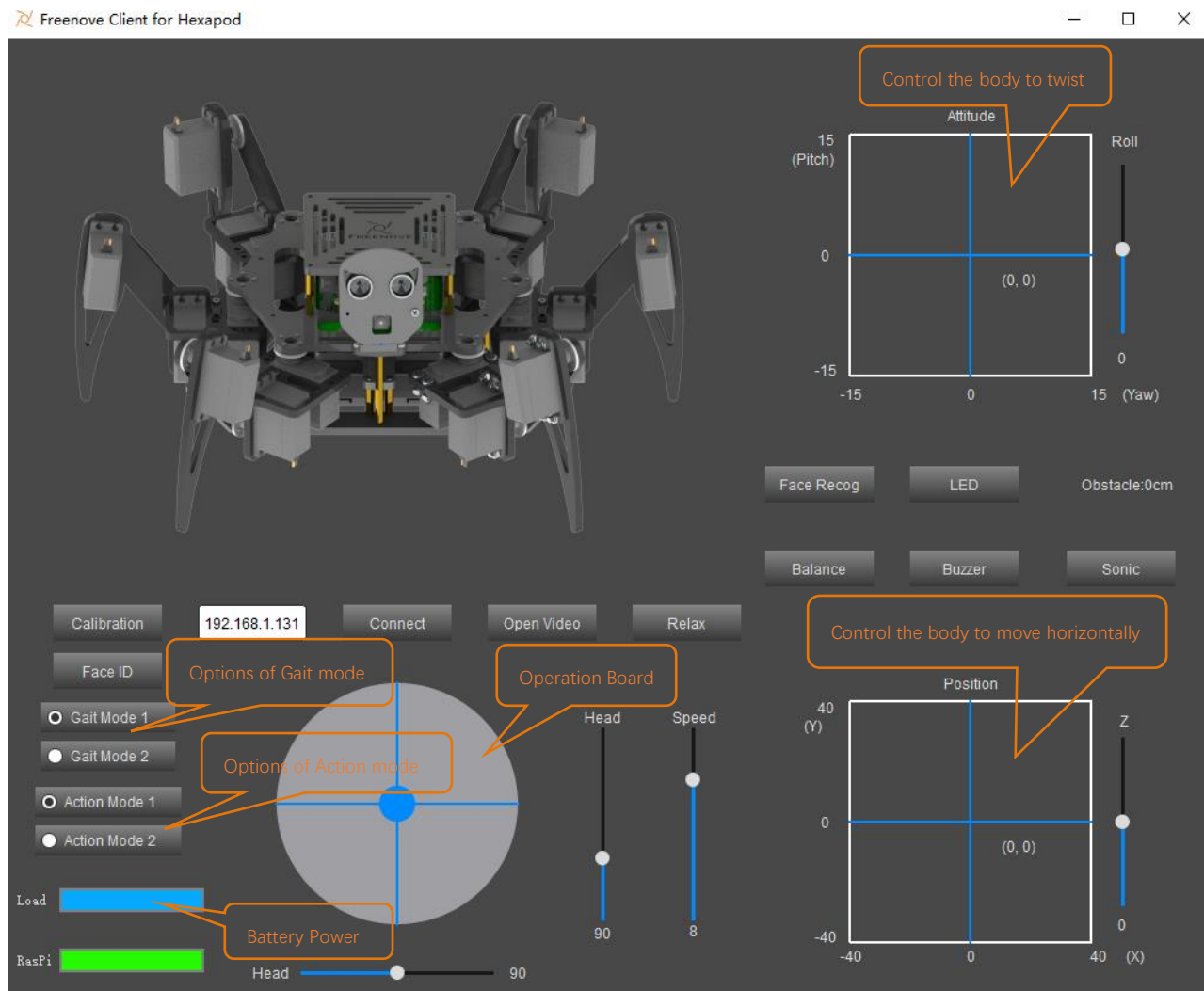
- Calibrate the six legs to make the six foot points fall to the corresponding positions. (Due to the error of the installation and error of the servos themselves, the six foot points may fail to fall to the specified position perfectly, which is reasonable but we should try to make them as accurate as possible.)



5. When all the six foot points fall to the specified positions, click “Save” and you can control the robot to move.

Note: Poor calibration will affect the robot's movement. You can recalibrate it when needed.

## Control



The following is the corresponding operation of the buttons and keyboards.

| Button on Client        | Key | Action                          |
|-------------------------|-----|---------------------------------|
| ForWard                 | W   | Move                            |
| BackWard                | S   | Back off                        |
| Turn Left               | A   | Turn left                       |
| Turn Right              | D   | Turn right                      |
| Connect/ Disconnect     | C   | Connection ON/OFF               |
| Open Video/ Close Video | V   | Video ON/OFF                    |
| Relax                   | R   | Relax/Activated                 |
| LED                     | L   | Open LED control interface      |
| Balance                 | B   | balance mode ON/OFF             |
| Face Recog              | F   | face recognition ON/OFF         |
| Sonic                   | U   | Ultrasonic ranging ON/OFF       |
| Calibration             | T   | Open calibration interface      |
| Face ID                 | I   | Open the face reading interface |
| Buzzer/Noise            | Y   | Buzzer ON/OFF                   |

The function of SliderBar is below:

| SliderBar | Function                  |
|-----------|---------------------------|
| Head      | Rotate the robot's head   |
| Speed     | Control the robot's speed |

## Free your innovation

If you have any concerns, please feel free to contact us via [support@freenove.com](mailto:support@freenove.com)

If you want to write your own program to control the robot dog, you can follow this chapter.

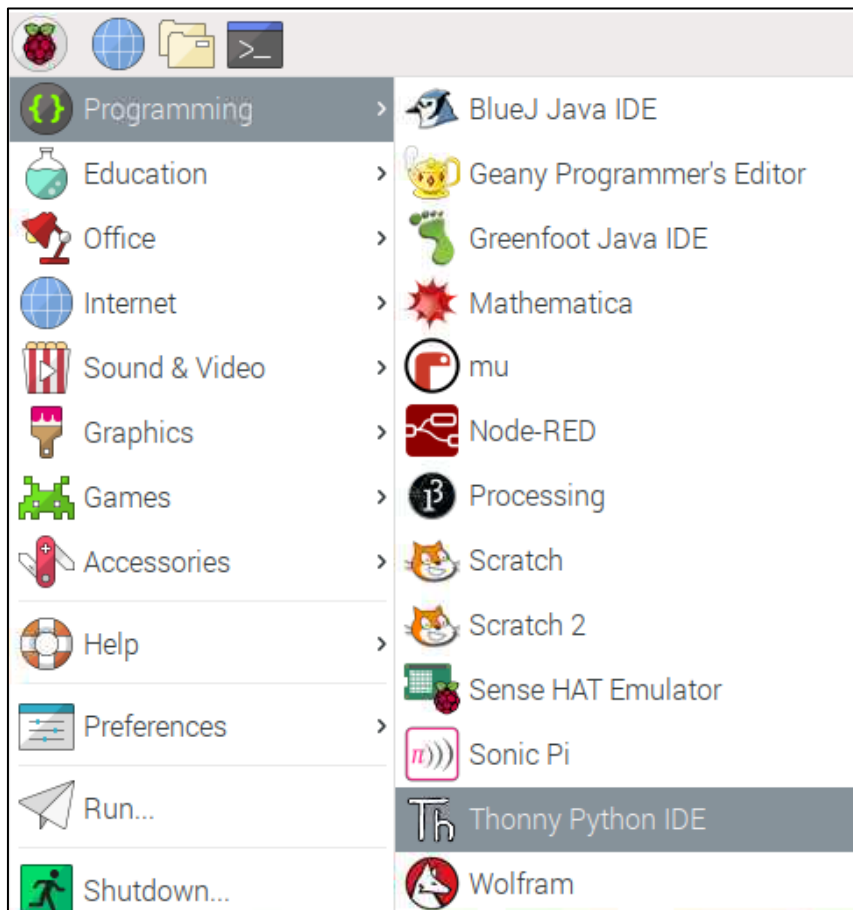
The robot program is based on python3. If your python is python2 by default, please change to python3.

If you have never learned python before, you can learn some basics through the following links:

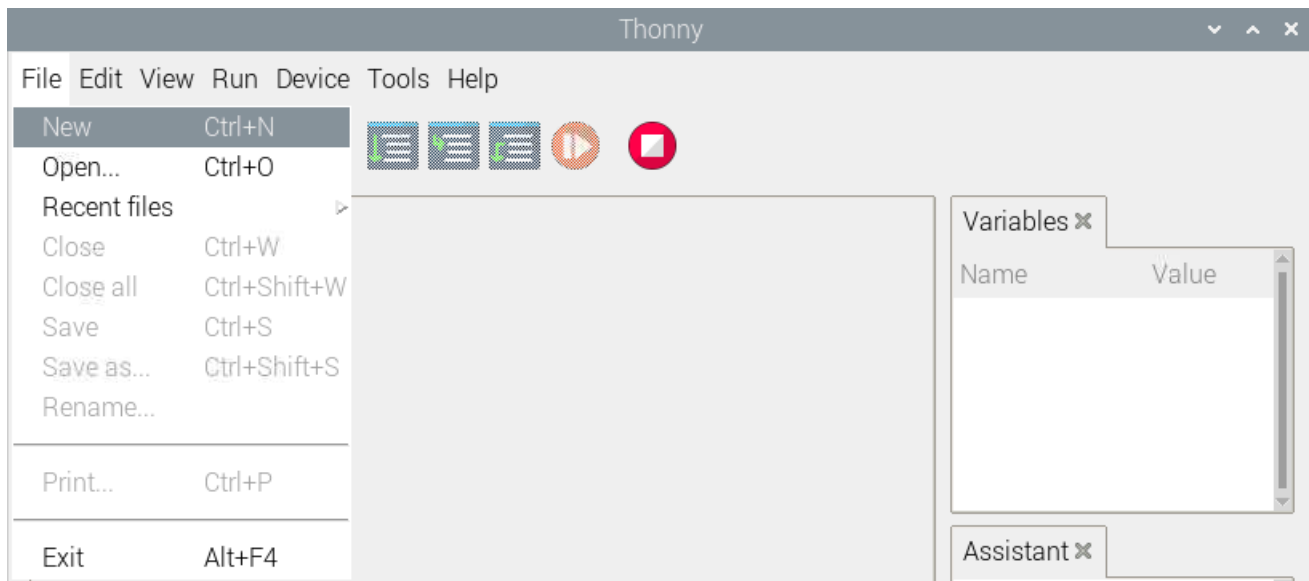
<https://python.swaroopch.com/basics.html>

## Program

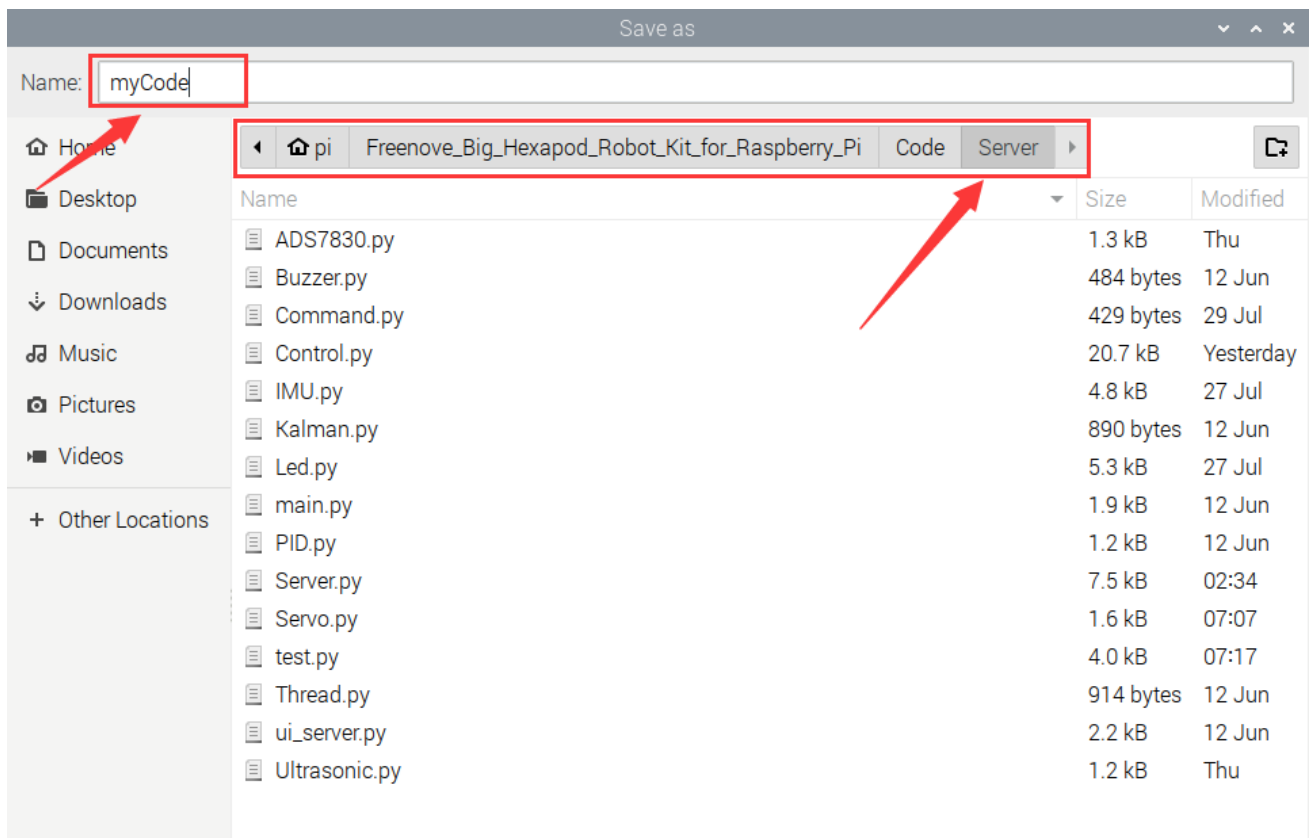
First, open Thonny Python IDE which is easy to use for beginners.



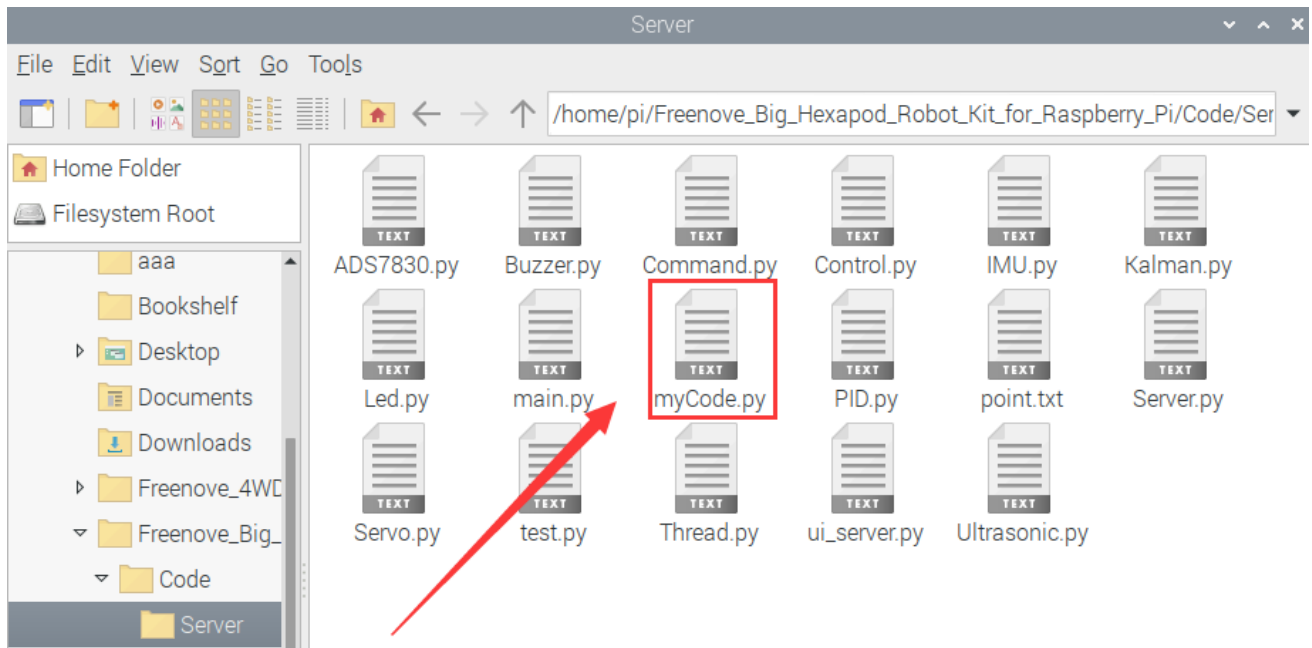
Create a new file.



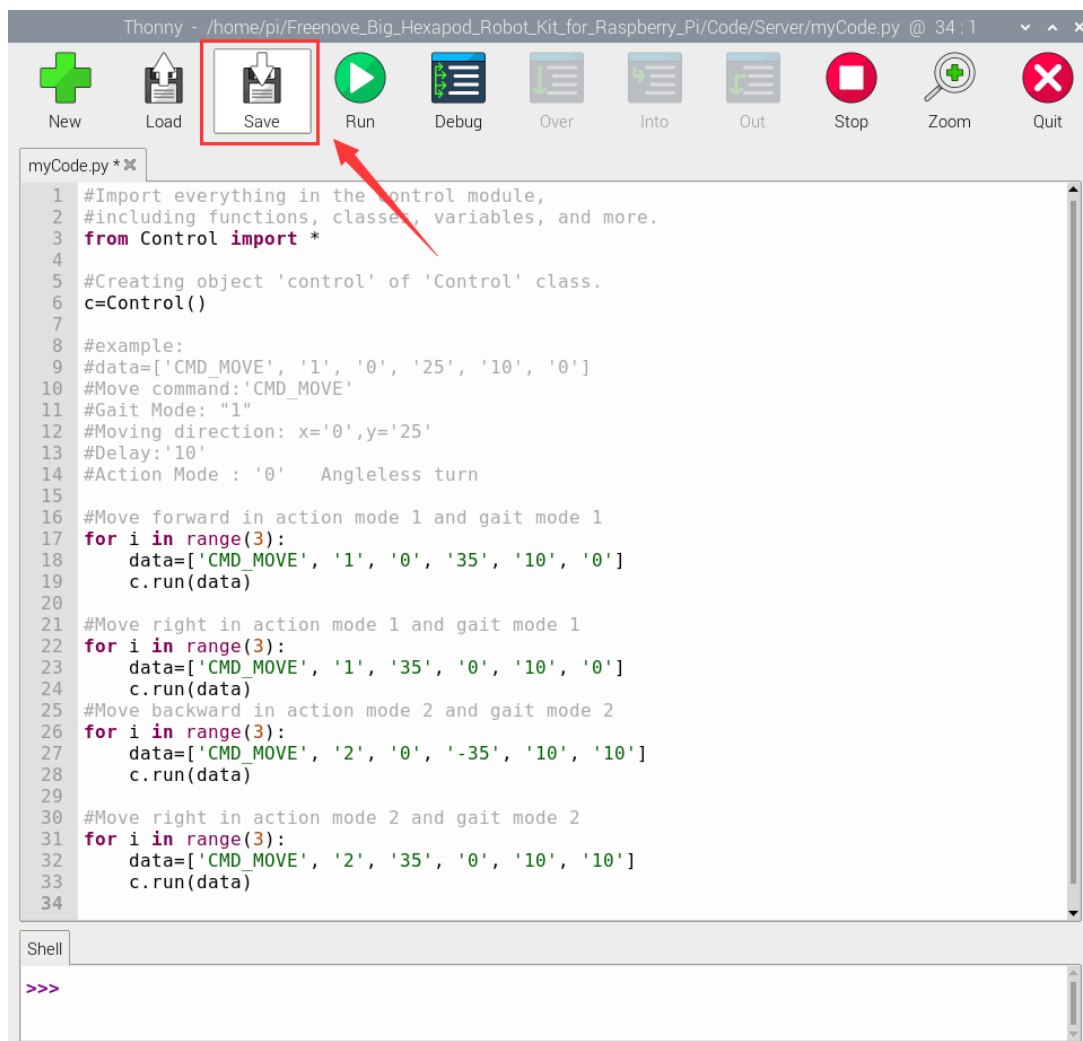
Name it myCode and save it in Server folder of robot code folder that you have downloaded.



Open the Server folder of the robot code and you can see the file you created.



Write the code in myCode.py and click save after finishing, as shown below.



Type the following command to enter the directory where myCode.py is located.

```
cd ~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server
```

Run myCode.py

```
sudo python myCode.py
```

```
pi@raspberrypi: ~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server
File Edit Tabs Help
pi@raspberrypi:~ $ cd ~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server
pi@raspberrypi:~/Freenove_Big_Hexapod_Robot_Kit_for_Raspberry_Pi/Code/Server $ sudo
python myCode.py
```

### Result:

The robot moves straight forward first and then walks to the right (Gait Mode1 and Action Mode1), and then it moves back and turn right.( Gait Mode2 and Action Mode2)

## Related py Files

Some important functions included in the py files are listed below. For more detailed code, please refer to the corresponding py file.

### ADS7830.py

| Function         | Description                                            |
|------------------|--------------------------------------------------------|
| voltage(channel) | Returns the voltage value of corresponding channel     |
| batteryPower()   | Returns the voltage value of the two sets of batteries |

### Buzzer.py

| Function     | Description                                                     |
|--------------|-----------------------------------------------------------------|
| run(command) | Enter 0, the buzzer will not sound; Enter 1, the buzzer sounds. |

### Contorl.py

| Function                   | Description                                                                          |
|----------------------------|--------------------------------------------------------------------------------------|
| coordinateToAngle(x,y,z)   | Enter the foot coordinate and it returns the rotation angle of servos on each joint. |
| angleToCoordinate(a,b,c)   | Enter the joint angles and it returns the foot coordinate.                           |
| Condition()                | Process the action command                                                           |
| coordinateTransformation() | Transformation of coordinate system                                                  |
| map()                      | Mapping a value from one range to another                                            |
| Position(x,y,z)            | Control the in-situ movement of the body in x,y,z directions                         |
| postureBalance(r,p,y)      | Control in-situ twist of the body(deflection angle, roll Angle and course angle)     |
| run()                      | Control the movement of the robot with two gait modes and two                        |

|  |               |
|--|---------------|
|  | action modes. |
|--|---------------|

## IMU.py

| Function    | Description                                                                   |
|-------------|-------------------------------------------------------------------------------|
| imuUpdate() | Update the current deflection angle, roll angle and course angle of the robot |

## Led.py

| Function                            | Description                                                                                                                                                                                                                                                       |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ledIndex( Index, R, G, B)           | Turn ON a single LED and set its color                                                                                                                                                                                                                            |
| wheel(pos)                          | Generate RGB value of different colors                                                                                                                                                                                                                            |
| LED_TYPR(order,R_G_B)               | Change the order in which the LED color data is transmitted. When the value of the order parameter is "RGB", the order of data transmission should be: R-G-B; when the value of the order parameter is "GBR", and the order of data transmission should be: G-B-R |
| theaterChaseRainbow(strip, wait_ms) | The function is used to make 7 LEDs show one color at the same time, and change various colors to blink.<br>The blinking interval is wait_ms, and the default value is 50ms                                                                                       |
| rainbow(strip, wait_ms)             | This function achieves the effect of rainbow breathing. It makes 7 LEDs display same color at the same time, and then change all various colors like breathing. The interval is wait_ms. The default value is 20ms.                                               |
| rainbowCycle(strip, wait_ms)        | This function also achieves the effect of rainbow breathing. but unlike rainbow(), it makes eight LEDs to display different colors at the same time, and then change various color separately. The interval is wait_ms.<br>The default value is 20ms              |

## Servo.py

| Function                      | Description                                                |
|-------------------------------|------------------------------------------------------------|
| setServoAngle(channel, angle) | Rotate the servo of specified channel to a specified angle |

## Ultrasonic.py

| Function      | Description                                                                       |
|---------------|-----------------------------------------------------------------------------------|
| getDistance() | Obtain the distance between the ultrasonic module and the barrier in front of it. |

## Android and iOS app

You can download and install the Freenove Android app from below:

On Google play:

<https://play.google.com/store/apps/details?id=com.freenove.suhayl.Freenovez>

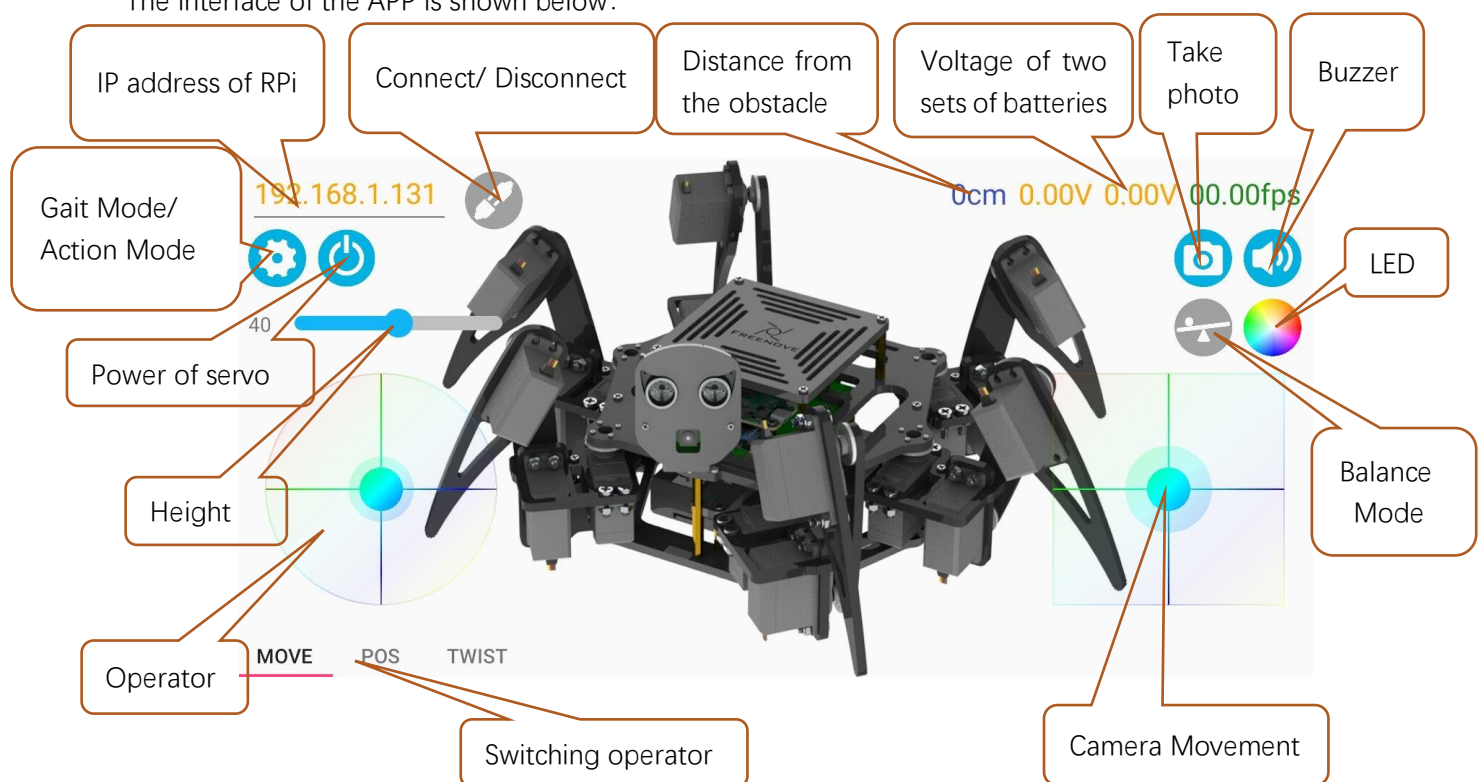
On GitHub:

[https://github.com/Freenove/Freenove\\_App\\_for\\_Android](https://github.com/Freenove/Freenove_App_for_Android)

In this github repository, you can find the App instruction (Tutorial.pdf).

You can download and install the Freenove **iPhone ios app** by searching **freenove** in app store.

The interface of the APP is shown below:



The servos and the Raspberry Pi are powered by two separate, independent power supplies. When the servos are powered OFF, the power supply of RPi remains its original state.

## What's Next?

THANK YOU for participating in this learning experience!

We have reached the end of this Tutorial. If you find errors, omissions or you have suggestions and/or questions about the Tutorial or component contents of this Kit, please feel free to contact us: [support@freenove.com](mailto:support@freenove.com)

We will make every effort to make changes and correct errors as soon as feasibly possible and publish a revised version.

After completing the contents in this book, you can try to reform this robot by purchasing and installing other Freenove electronic modules, or improving the code to achieve different functions. We will also try our best to add more new functions and update the code on our github (<https://github.com/freenove>).

If you want to learn more about Arduino, Raspberry Pi, Smart Cars, Robotics and other interesting products in science and technology, please continue to visit our website. We will continue to launch fun, cost-effective, innovative and exciting products.

<http://www.freenove.com/>

Thank you again for choosing Freenove products.